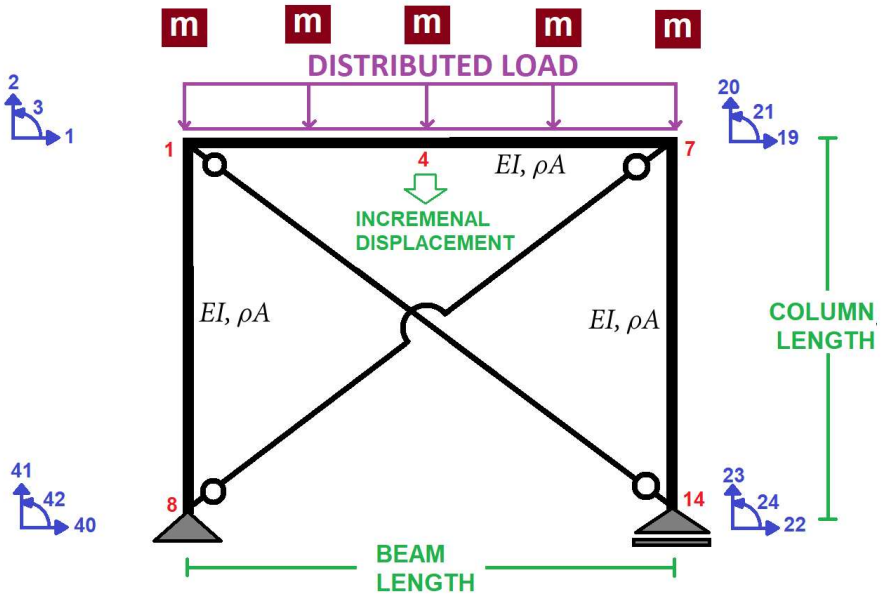
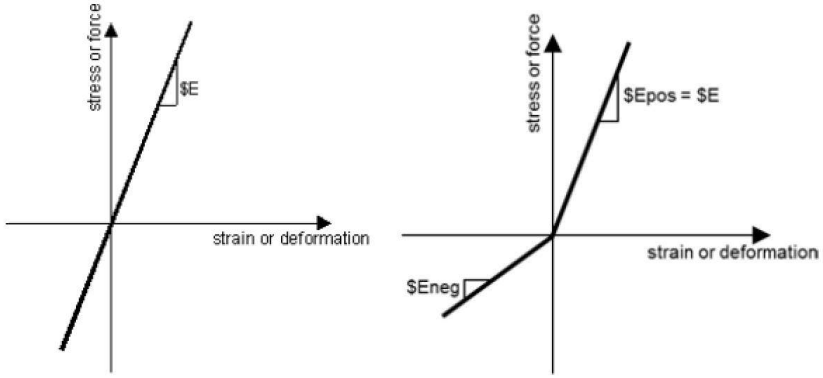
>> IN THE NAME OF ALLAH, THE MOST GRACIOUS, THE MOST MERCIFUL <<

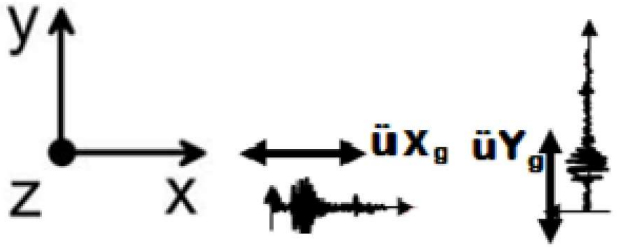
COMPREHENSIVE NONLINEAR SEISMIC ASSESSMENT OF A STEEL BARCED FRAME: AN OPENSEES FRAMEWORK FOR STATIC PUSHOVER, CYCLIC DEGRADATION, STATIC TIME-HISTORY AND DYNAMIC TIME-HISTORY ANALYSIS

THIS PROGRAM IS WRITTEN BY SALAR DELAVAR GHASHGHAEI (QASHQAI)





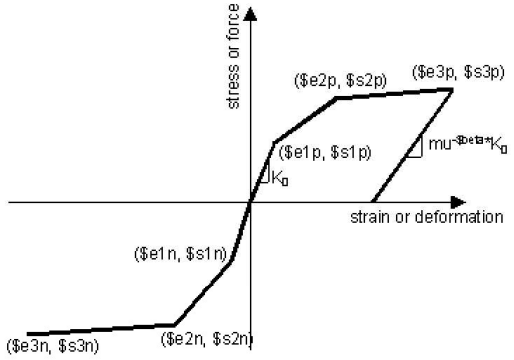
ELEMENT ELASTIC MATERIAL



$$W(x,t) = P(t) = W_0 e^{-0.05\bar{\omega}t} sin(\bar{\omega}t)$$

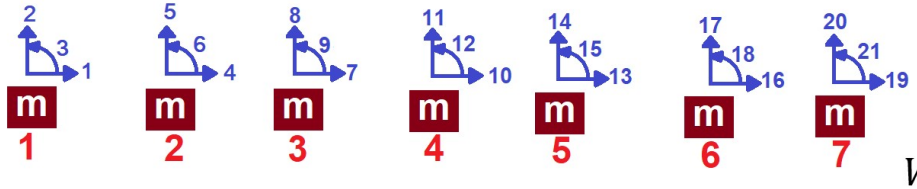
$$m\ddot{u} + c\dot{u} + ku = P(t)$$

$$Structure\ Ductility\ Damage\ Index = \frac{\Delta_d - \Delta_y}{\Delta_u - \Delta_y}$$

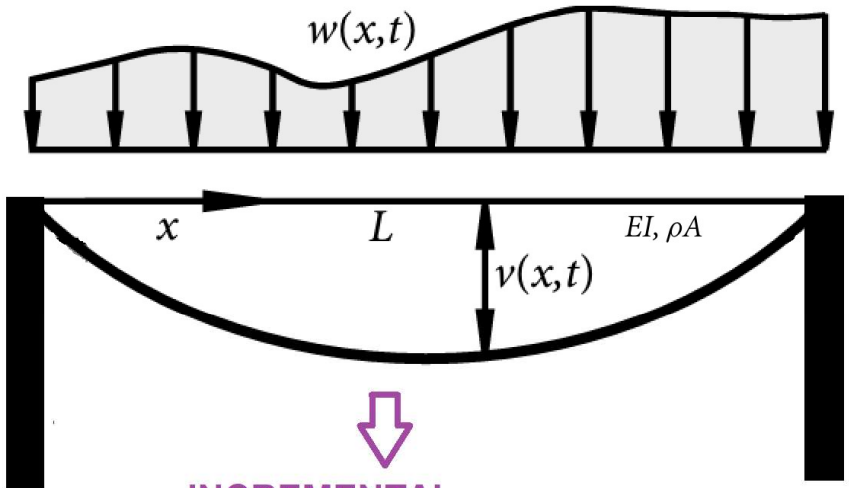


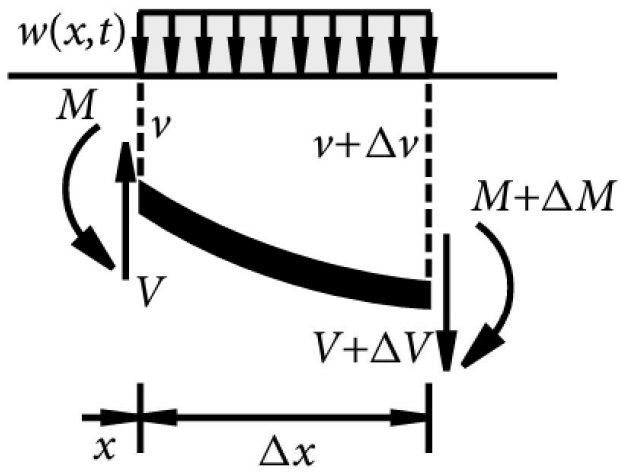
ELEMENT INELASTIC MATERIAL

$$\rho A \frac{\partial^2 v}{\partial t^2} + EI \frac{\partial^4 v}{\partial x^4} = w(x,t).$$

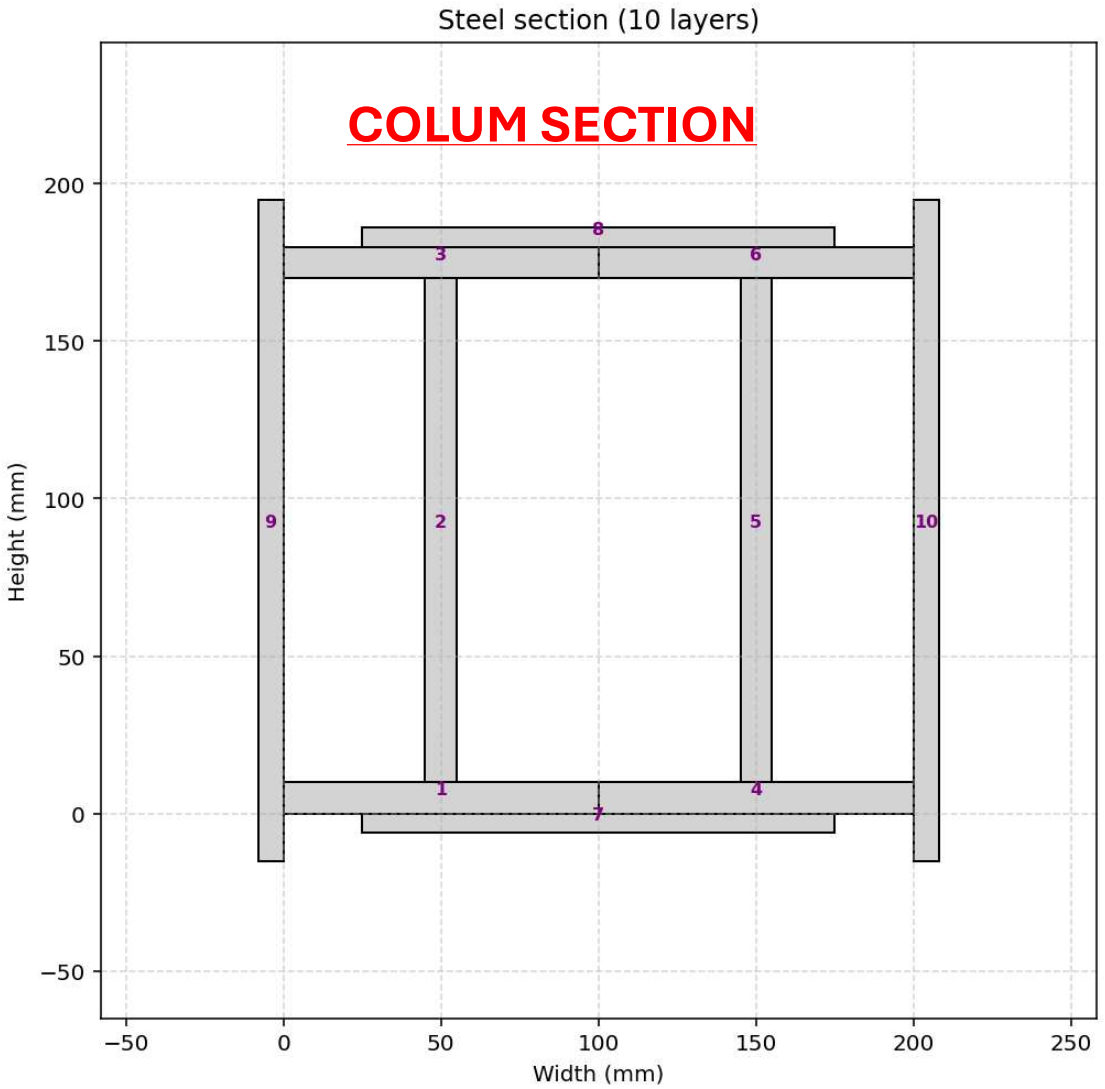


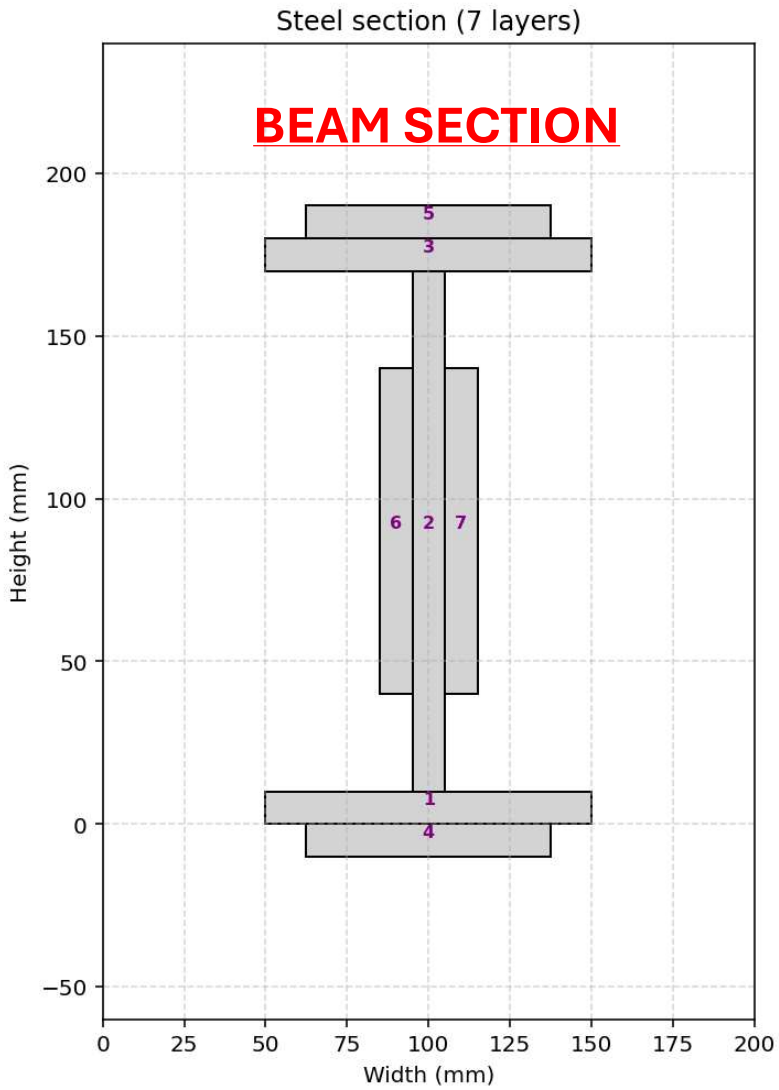
$$W(x,t) = P(t) = W_0 e^{-0.05\bar{\omega}t} sin(\bar{\omega}t)$$

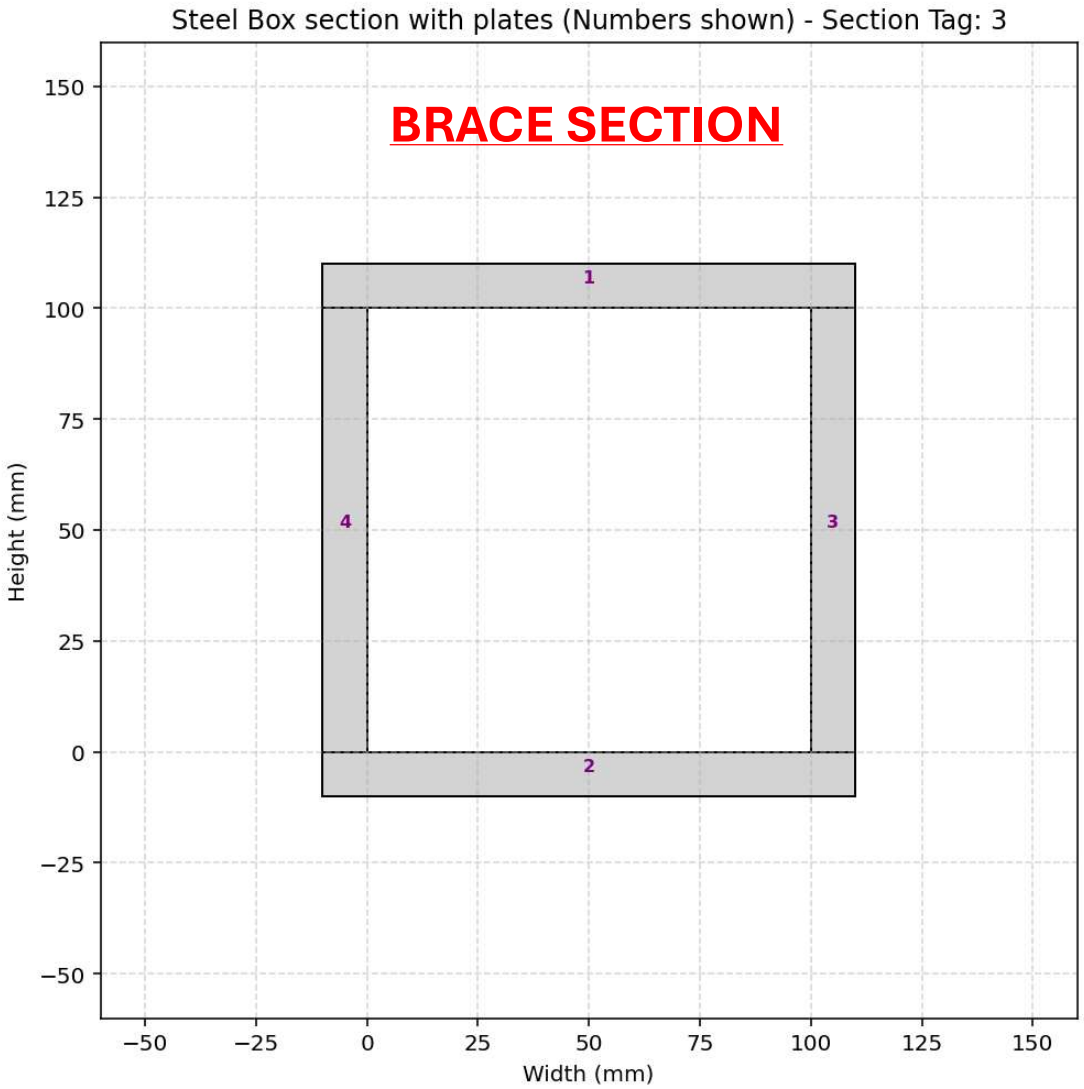




Flexural motion of a uniform elastic beam


Steel section (10 layers)
COLUM SECTION
Height (mm)
Width (mm)
200
150
100
50
0
−50
−50
0
50
100
150
200
250
1
2
3
4
5
6
7
8
9
10


Steel section (7 layers)
BEAM SECTION
Height (mm)
Width (mm)
200
150
100
50
0
−50
0
25
50
75
100
125
150
175
200
5
3
6
2
7
1
4


Steel Box section with plates (Numbers shown) - Section Tag: 3
BRACE SECTION
1
2
3
4
Height (mm)
Width (mm)
150
125
100
75
50
25
0
−25
−50
−50
−25
0
25
50
75
100
125
150

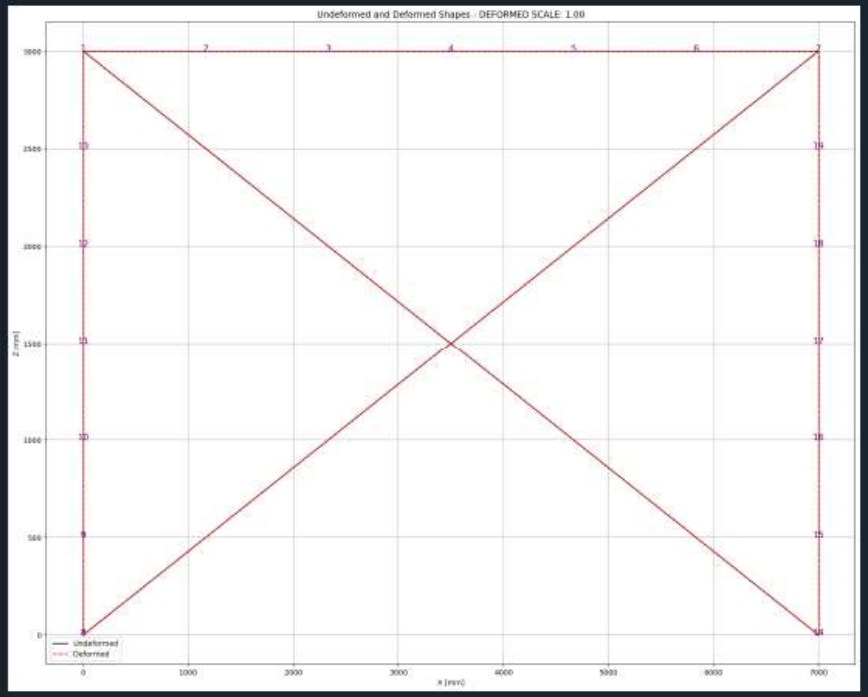
Spyder (Python 3.12)

File Edit Search Source Run Debug Consoles Projects Tools View Help

C:\Users\Dell\Desktop\OPENSEES_FILES\+EIGHT_ANALY...L_BRACED_FRAME_8_ANA\W(x,t)_STEEL_BRACED_FRAME.py

W(x,t)_STEEL_BRACED_FRAME.py | PLATE_DOUBLE_I_STE...EIGHT_FUN_EXTRA.py

```
1   ##########################################################################
2   #                  >> IN THE NAME OF ALLAH, THE MOST GRACIOUS, THE MOST ME
3   #       COMPREHENSIVE NONLINEAR SEISMIC ASSESSMENT OF A STEEL BRACED FRAME
4   #FRAMEWORK FOR STATIC PUSHOVER, CYCLIC DEGRADATION, STATIC TIME-HISTORY AND
5   #--------------------------------------------------------------------------
6   #                                          W(x,t) = W0 sin(wt)
7   #                                   W(x,t) = W0 exp(-0.05wt) sin(wt)
8   #--------------------------------------------------------------------------
9   #         ASSESSMENT OF DUCTILITY DAMAGE INDICES FOR STRUCTURAL ELEMENTS AN
10  #                                   OF ENERGY DISSIPATION CAPACITY IND
11  #--------------------------------------------------------------------------
12  #                    THIS PYTHON SCRIPT WRITTEN BY SALAR DELAVAR GHASHGH
13  #                                 EMAIL: salar.d.ghashghaei@gmail.com
14  ##########################################################################
15  """
16  Nonlinear Seismic Performance Assessment of a steel braced frame:
17  An OpenSeesPy Framework for Material and Geometric Nonlinearity Under Stati
18
19  This OpenSeesPy script performs rigorous nonlinear static and dynamic analy
20   for performance-based earthquake engineering. The 2D model incorporates bo
21   (elastic-perfectly plastic or hysteretic steel with strain hardening)
22   and geometric nonlinearity to capture P-delta effects
23   and large displacements—essential for collapse assessment.
24
25  Eight analysis protocols are implemented:
26  (1) [PERIOD] : Structural Period
27  (2) [STATIC] : Gravity load analysis establishing dead load state
28  (3) [PUSHOVER] : Displacement-controlled pushover generating full capacity
29   and plastic mechanism identification
30  (4) [CYCLIC_DISPLACEMENT] : Symmetric cyclic displacement protocol capturin
31   pinching behavior, and energy dissipation degradation
32  (5) [STATIC_EXTERNAL_TIME-DEPENDENT_LOADING] : Static Analysis of External
33  (6) [DYNAMIC_EXTERNAL_TIME-DEPENDENT_LOADING] : Dynamic Analysis of Externa
34  (7) [FREE-VIBRATION] : Free-vibration with initial conditions extracting da
```



IPython Console | Files | Help | Variable Explorer | Debugger | Plots | History

Save file | Inline | Conda: anaconda3 (Python 3.12.7) | LSP: Python | Line 12, Col 105 | UTF-8 | CRLF | RW | Mem 39%

STATIC ANALYSIS

Spyder (Python 3.12)

File Edit Search Source Run Debug Consoles Projects Tools View Help

C:\Users\Dell\Desktop\OPENSEES_FILES\+EIGHT_ANALY...L_BRACED_FRAME_8_ANA\W(x,t)_STEEL_BRACED_FRAME.py

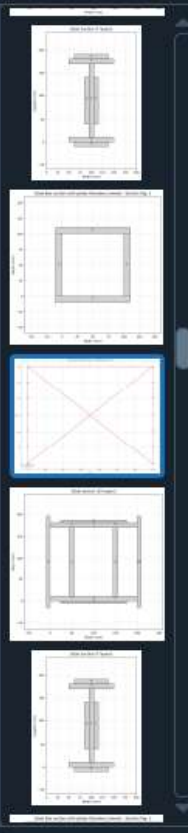
W(x,t)_STEEL_BRACED_FRAME.py | PLATE_DOUBLE_I_STE...EIGHT_FUN_EXTRA.py

```
1172  #%%-----------------------------------------------------
1173  # PERIOD ANALYSIS
1174  #ELE_TYPE = 'elasticBeamColumn'      # MATERIAL LINEARITY
1175  ELE_TYPE = 'nonlinearBeamColumn'     # MATERIAL AND GEOMETRIC NONLINEARITIE
1176  MAT_TYPE = 'INELASTIC'   # 'ELASTIC' OR 'INELASTIC'
1177  ANAL_TYPE = 'PERIOD'
1178  BRACE = True                         # True: WITH BRACE - False: WITHOUT BR
1179
1180  DATA = STEEL_BRACED_FRAME(LENGTH, HEIGHT, MAT_TYPE, ELE_TYPE, ANAL_TYPE, TO
1181  (PERIOD_MIN_X, PERIOD_MAX_X) = DATA
1182  print('Structure First Period:  ', PERIOD_MIN_X)
1183  print('Structure Second Period: ', PERIOD_MAX_X)
1184
1185  S01.PLOT_2D_FRAME(deformed_scale=1.0)
1186  #%%-----------------------------------------------------
1187  # STATIC ANALYSIS
1188  #ELE_TYPE = 'elasticBeamColumn'      # MATERIAL LINEARITY
1189  ELE_TYPE = 'nonlinearBeamColumn'     # MATERIAL AND GEOMETRIC NONLINEARITIE
1190  MAT_TYPE = 'INELASTIC'   # 'ELASTIC' OR 'INELASTIC'
1191  ANAL_TYPE = 'STATIC'
1192  BRACE = True                         # True: WITH BRACE - False: WITHOUT BR
1193
1194  DATA = STEEL_BRACED_FRAME(LENGTH, HEIGHT, MAT_TYPE, ELE_TYPE, ANAL_TYPE, TO
1195  (reactionX, reactionY, disp_mid,
1196   ele_axialforce, ele_shearforce, ele_momentforce,
1197   node_displacementsX, node_displacementsY,
1198   dispX, dispY) = DATA
1199
1200  S01.PLOT_2D_FRAME(deformed_scale=1.0)
1201  #%%-----------------------------------------------------
1202  # PUSHOVER ANALYSIS (STATIC TIME-HISTORY ANALYSIS)
1203  #ELE_TYPE = 'elasticBeamColumn'      # MATERIAL LINEARITY
1204  ELE_TYPE = 'nonlinearBeamColumn'     # MATERIAL AND GEOMETRIC NONLINEARITIE
1205  MAT_TYPE = 'INELASTIC'   # 'ELASTIC' OR 'INELASTIC'
```

...RGY_DISSIPATION_CAPACITY_INDEX\08_STEEL_BRACED_FRAME_8_ANA

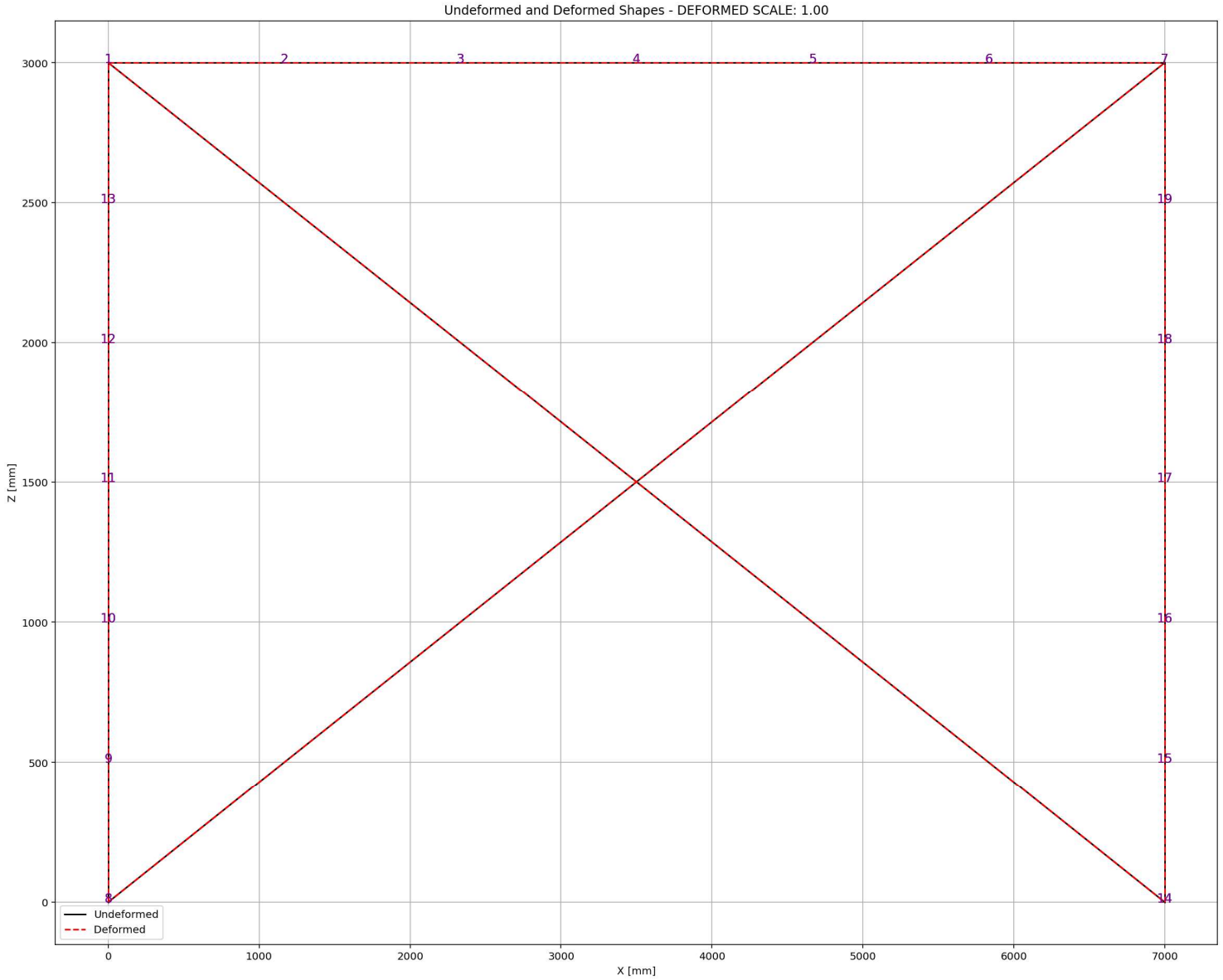
23 %





IPython Console | Files | Help | Variable Explorer | Debugger | Plots | History

Inline Conda: anaconda3 (Python 3.12.7) LSP: Python Line 12, Col 105 UTF-8 CRLF RW Mem 39%


Undeformed and Deformed Shapes - DEFORMED SCALE: 1.00
Z [mm]
X [mm]
0
500
1000
1500
2000
2500
3000
0
1000
2000
3000
4000
5000
6000
7000
1
2
3
4
5
6
7
8
9
10
11
12
13
14
15
16
17
18
19
Undeformed
Deformed

PUSHOVER ANALYSIS

Spyder (Python 3.12)

File Edit Search Source Run Debug Consoles Projects Tools View Help

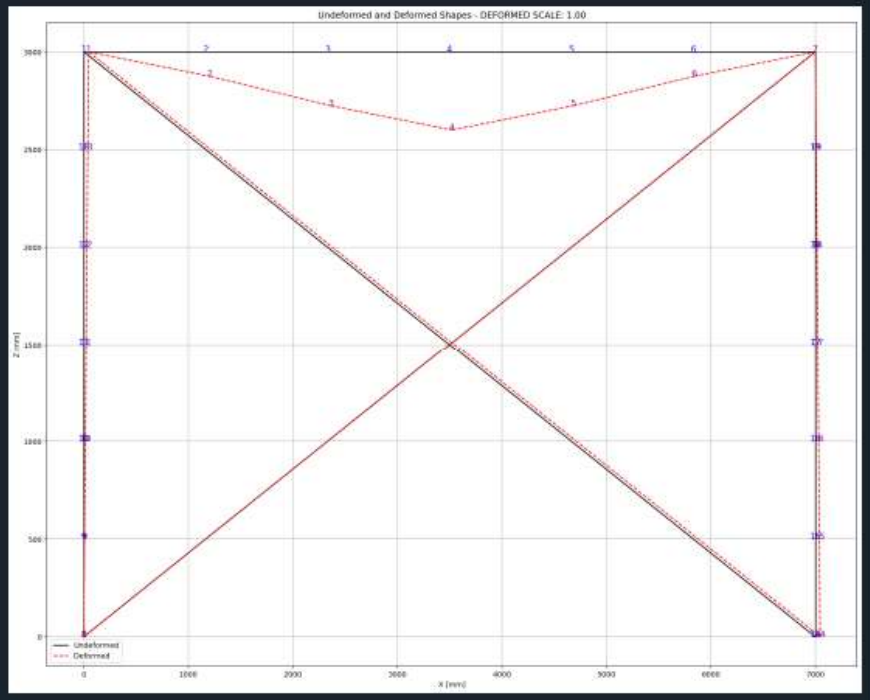
...RGY_DISSIPATION_CAPACITY_INDEX\08_STEEL_BRACED_FRAME_8_ANA

C:\Users\Dell\Desktop\OPENSEES_FILES\+EIGHT_ANALY...L_BRACED_FRAME_8_ANA\W(x,t)_STEEL_BRACED_FRAME.py

W(x,t)_STEEL_BRACED_FRAME.py | PLATE_DOUBLE_I_STE...EIGHT_FUN_EXTRA.py

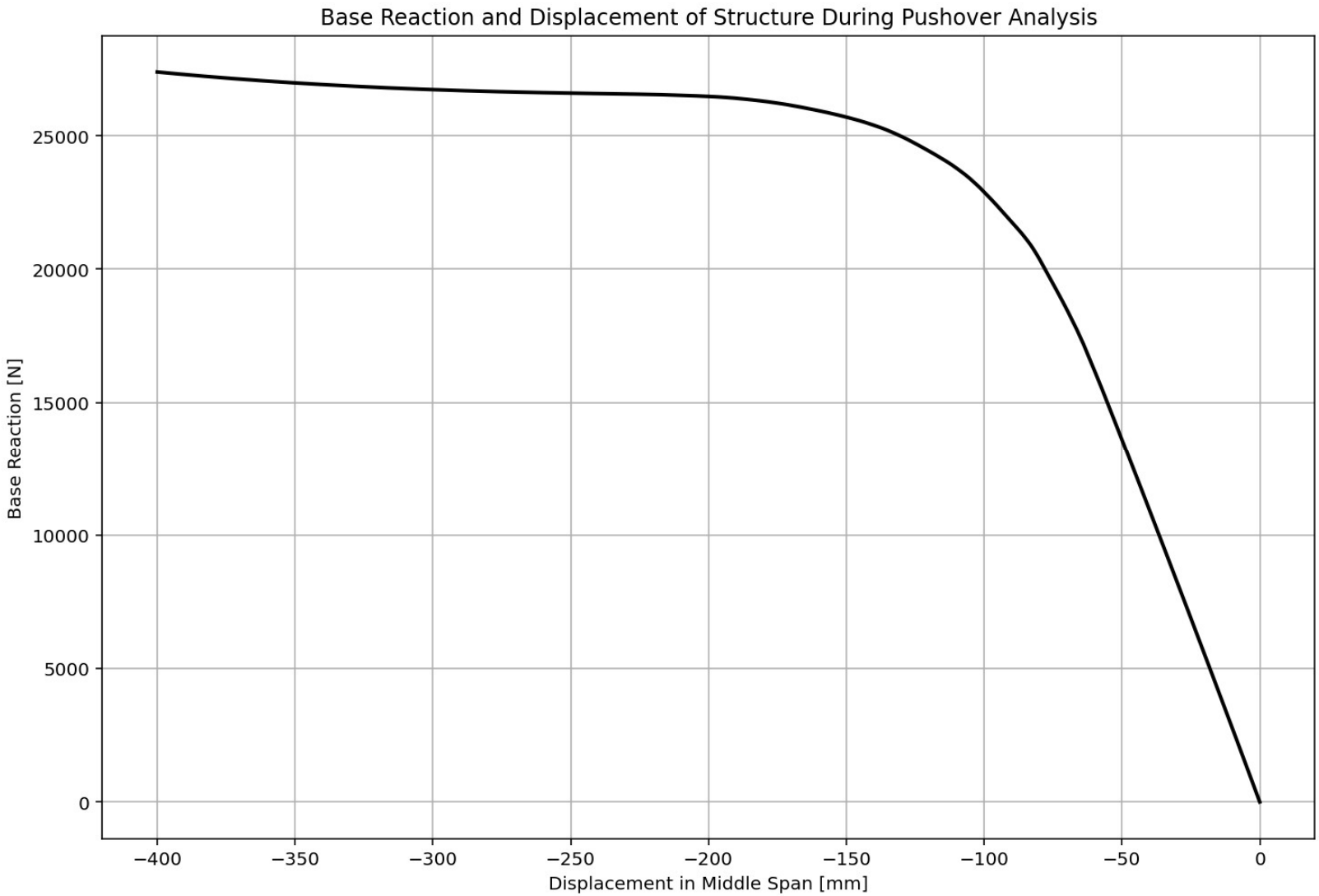
```
1202    # PUSHOVER ANALYSIS (STATIC TIME-HISTORY ANALYSIS)
1203    #ELE_TYPE = 'elasticBeamColumn'      # MATERIAL LINEARITY
1204    ELE_TYPE = 'nonlinearBeamColumn'     # MATERIAL AND GEOMETRIC NONLINEARITIE
1205    MAT_TYPE = 'INELASTIC'   # 'ELASTIC' OR 'INELASTIC'
1206    ANAL_TYPE = 'PUSHOVER'
1207    BRACE = True                         # True: WITH BRACE - False: WITHOUT BR
1208
1209    DATA = STEEL_BRACED_FRAME(LENGTH, HEIGHT, MAT_TYPE, ELE_TYPE, ANAL_TYPE, TO
1210    (reactionX_PUSH, reactionY_PUSH, disp_mid_PUSH,
1211     ele_axialforce_PUSH, ele_shearforce_PUSH, ele_momentforce_PUSH,
1212     node_displacementsX_PUSH, node_displacementsY_PUSH,
1213      dispX_PUSH, dispY_PUSH,
1214     PERIOD_MIN_PUSH, PERIOD_MAX_PUSH,
1215     STRESSb_PUSH, STRAINb_PUSH, STRESSt_PUSH, STRAINt_PUSH, CURVATURE_PUSH) =
1216
1217
1218    XDATA = disp_mid_PUSH
1219    YDATA = reactionY_PUSH
1220    XLABEL = 'Displacement in Middle Span [mm]'
1221    YLABEL = 'Base Reaction [N]'
1222    TITLE = 'Base Reaction and Displacement of Structure During Pushover Analys
1223    COLOR = 'black'
1224    SEMILOGY = False
1225    PLOT(XDATA, YDATA, TITLE, XLABEL, YLABEL, COLOR, SEMILOGY)
1226
1227    DATA = S07.BILNEAR_CURVE(np.abs(disp_mid_PUSH), np.abs(reactionY_PUSH), SLO
1228    (X_PUSH, Y_PUSH, Elastic_ST, Plastic_ST, Tangent_ST, Ductility_Rito, Over_S
1229
1230    # PLOT STRUCTURAL PERIOD DURING THE ANALYSIS
1231    plt.figure(0, figsize=(12, 8))
1232    plt.plot(disp_mid_PUSH, PERIOD_MIN_PUSH, linewidth=3)
1233    plt.plot(disp_mid_PUSH, PERIOD_MAX_PUSH, linewidth=3)
1234    plt.title('Period of Structure During Pushover Analysis')
1235    plt.ylabel('Structural Period [s]')
```

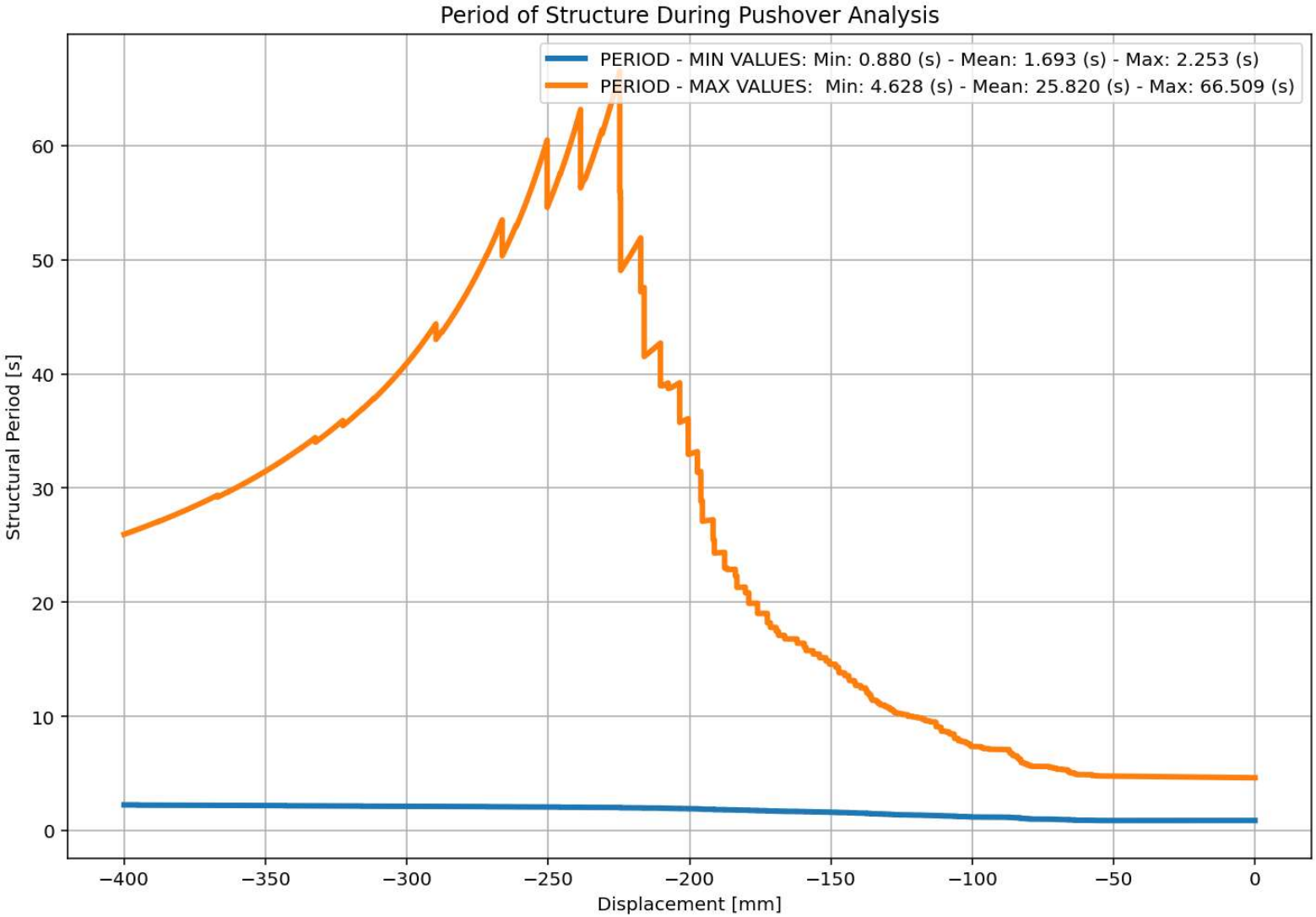
23 %

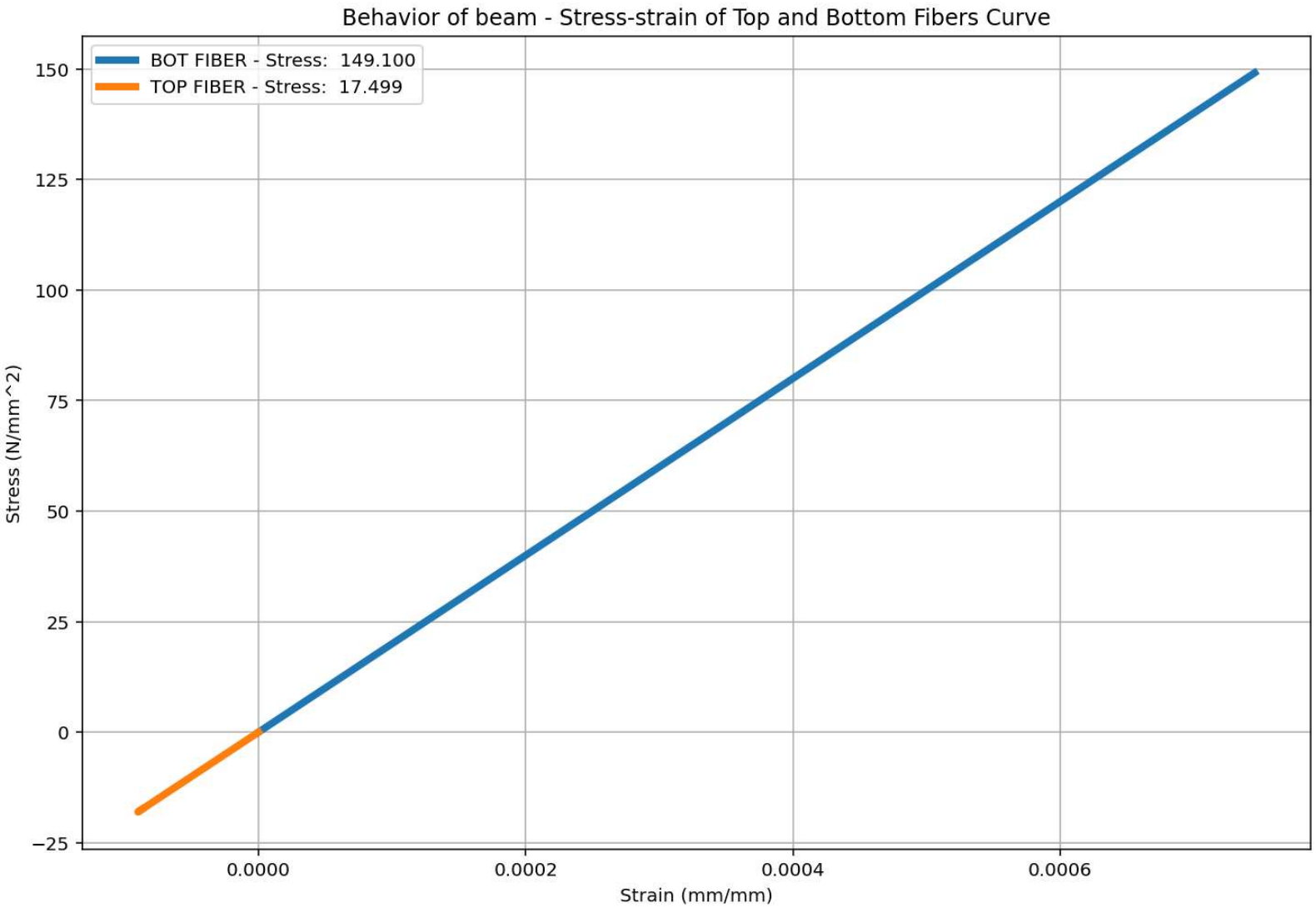


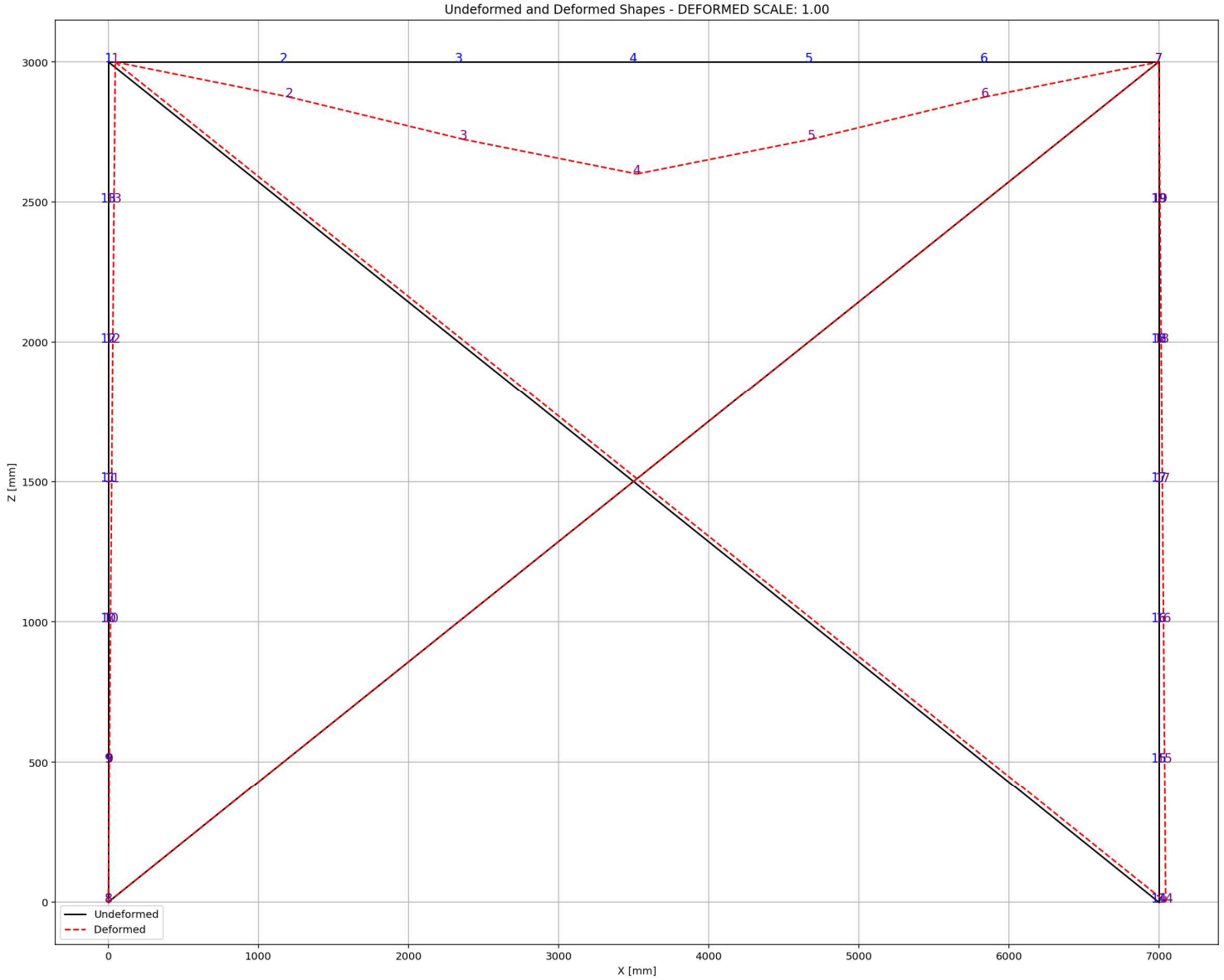
IPython Console | Files | Help | Variable Explorer | Debugger | Plots | History

Inline Conda: anaconda3 (Python 3.12.7) LSP: Python Line 12, Col 105 UTF-8 CRLF RW Mem 40%


Base Reaction and Displacement of Structure During Pushover Analysis
Base Reaction [N]
25000
20000
15000
10000
5000
0
−400
−350
−300
−250
−200
−150
−100
−50
0
Displacement in Middle Span [mm]


Period of Structure During Pushover Analysis
PERIOD - MIN VALUES: Min: 0.880 (s) - Mean: 1.693 (s) - Max: 2.253 (s)
PERIOD - MAX VALUES: Min: 4.628 (s) - Mean: 25.820 (s) - Max: 66.509 (s)
Structural Period [s]
0
10
20
30
40
50
60
−400
−350
−300
−250
−200
−150
−100
−50
0
Displacement [mm]


Behavior of beam - Stress-strain of Top and Bottom Fibers Curve
BOT FIBER - Stress: 149.100
TOP FIBER - Stress: 17.499
150
125
100
75
50
25
0
−25
Stress (N/mm^2)
0.0000
0.0002
0.0004
0.0006
Strain (mm/mm)


Undeformed and Deformed Shapes - DEFORMED SCALE: 1.00
Z [mm]
X [mm]
Undeformed
Deformed
0
500
1000
1500
2000
2500
3000
0
1000
2000
3000
4000
5000
6000
7000

CYCLIC DISPLACEMENT ANALYSIS

Spyder (Python 3.12)

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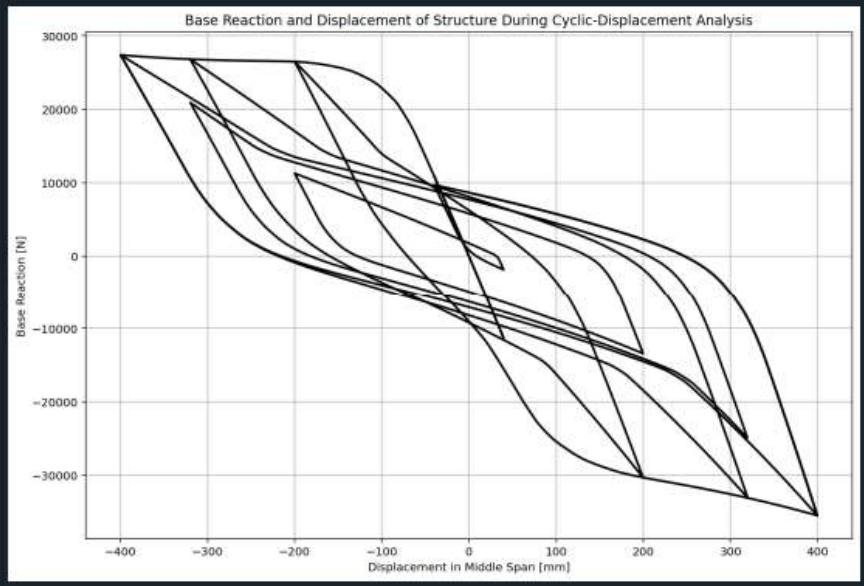
C:\Users\Dell\Desktop\OPENSEES_FILES\+EIGHT_ANALY...L_BRACED_FRAME_8_ANA\W(x,t)_STEEL_BRACED_FRAME.py

W(x,t)_STEEL_BRACED_FRAME.py | PLATE_DOUBLE_I_STE...EIGHT_FUN_EXTRA.py

```
1257  # CYCLIC DISPLACEMENT ANALYSIS (STATIC TIME-HISTORY ANALYSIS)
1258  #ELE_TYPE = 'elasticBeamColumn'      # MATERIAL LINEARITY
1259  ELE_TYPE = 'nonlinearBeamColumn'     # MATERIAL AND GEOMETRIC NONLINEARITIE
1260  MAT_TYPE = 'INELASTIC'   # 'ELASTIC' OR 'INELASTIC'
1261  ANAL_TYPE = 'CYCLIC_DISPLACEMENT'
1262  BRACE = True                         # True: WITH BRACE - False: WITHOUT BR
1263
1264  DATA = STEEL_BRACED_FRAME(LENGTH, HEIGHT, MAT_TYPE, ELE_TYPE, ANAL_TYPE, TO
1265  (reactionX_CP, reactionY_CP, disp_mid_CP,
1266   ele_axialforce_CP, ele_shearforce_CP, ele_momentforce_CP,
1267   node_displacementsX_CP, node_displacementsY_CP,
1268    dispX_CP, dispY_CP,
1269   PERIOD_MIN_CP, PERIOD_MAX_CP,
1270   STRESSb_CP, STRAINb_CP, STRESSt_CP, STRAINt_CP, CURVATURE_CP) = DATA
1271
1272
1273  XDATA = disp_mid_CP
1274  YDATA = reactionY_CP
1275  XLABEL = 'Displacement in Middle Span [mm]'
1276  YLABEL = 'Base Reaction [N]'
1277  TITLE = 'Base Reaction and Displacement of Structure During Cyclic-Displace
1278  COLOR = 'black'
1279  SEMILOGY = False
1280  PLOT(XDATA, YDATA, TITLE, XLABEL, YLABEL, COLOR, SEMILOGY)
1281
1282  # PLOT STRUCTURAL PERIOD DURING THE ANALYSIS
1283  plt.figure(0, figsize=(12, 8))
1284  plt.plot(disp_mid_CP, PERIOD_MIN_CP, linewidth=3)
1285  plt.plot(disp_mid_CP, PERIOD_MAX_CP, linewidth=3)
1286  plt.title('Period of Structure During Cyclic Displacement Analysis')
1287  plt.ylabel('Structural Period [s]')
1288  plt.xlabel('Displacement [mm]')
1289  #plt.semilogy()
1290  plt.grid()
```

...RGY_DISSIPATION_CAPACITY_INDEX\08_STEEL_BRACED_FRAME_8_ANA

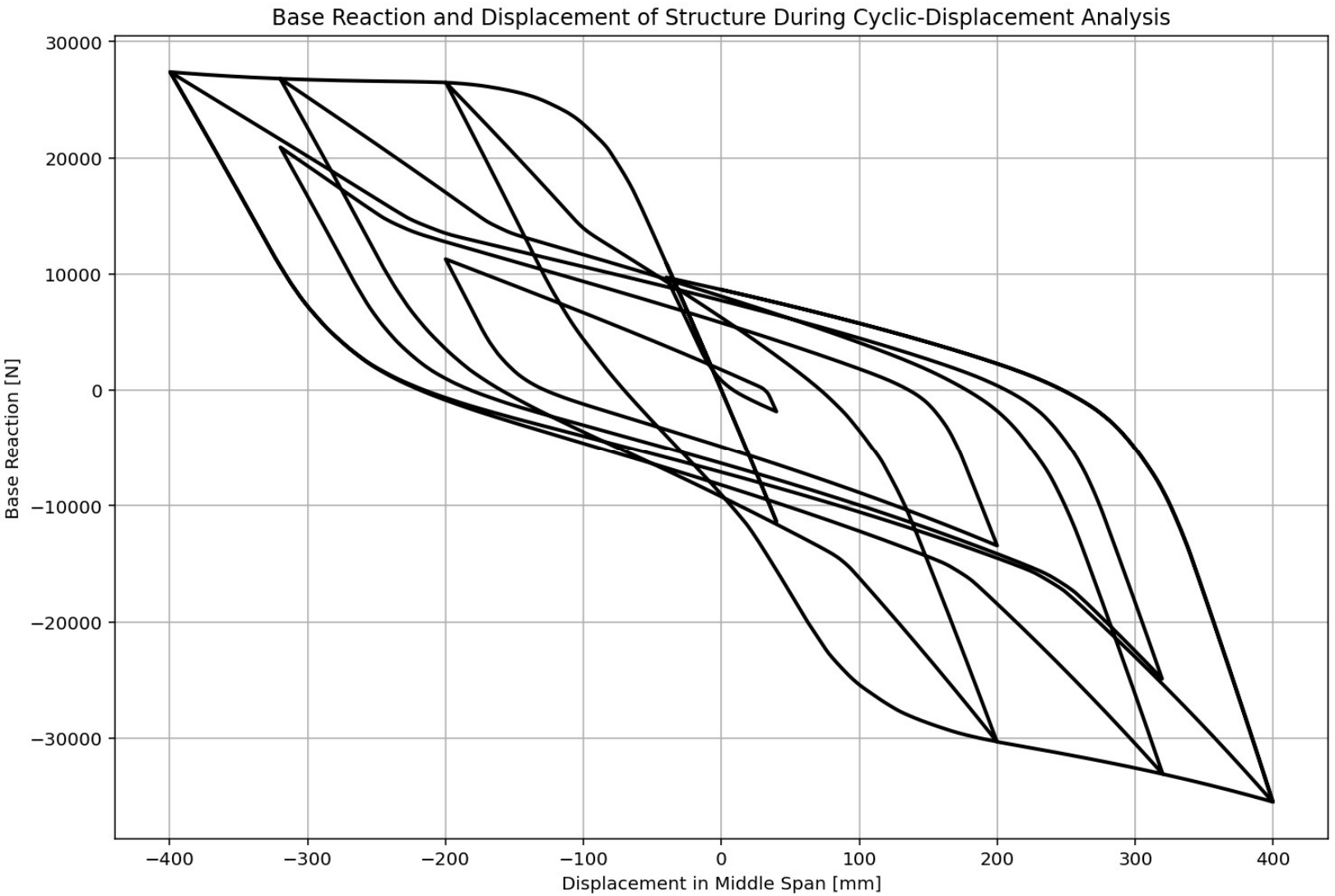
37 %

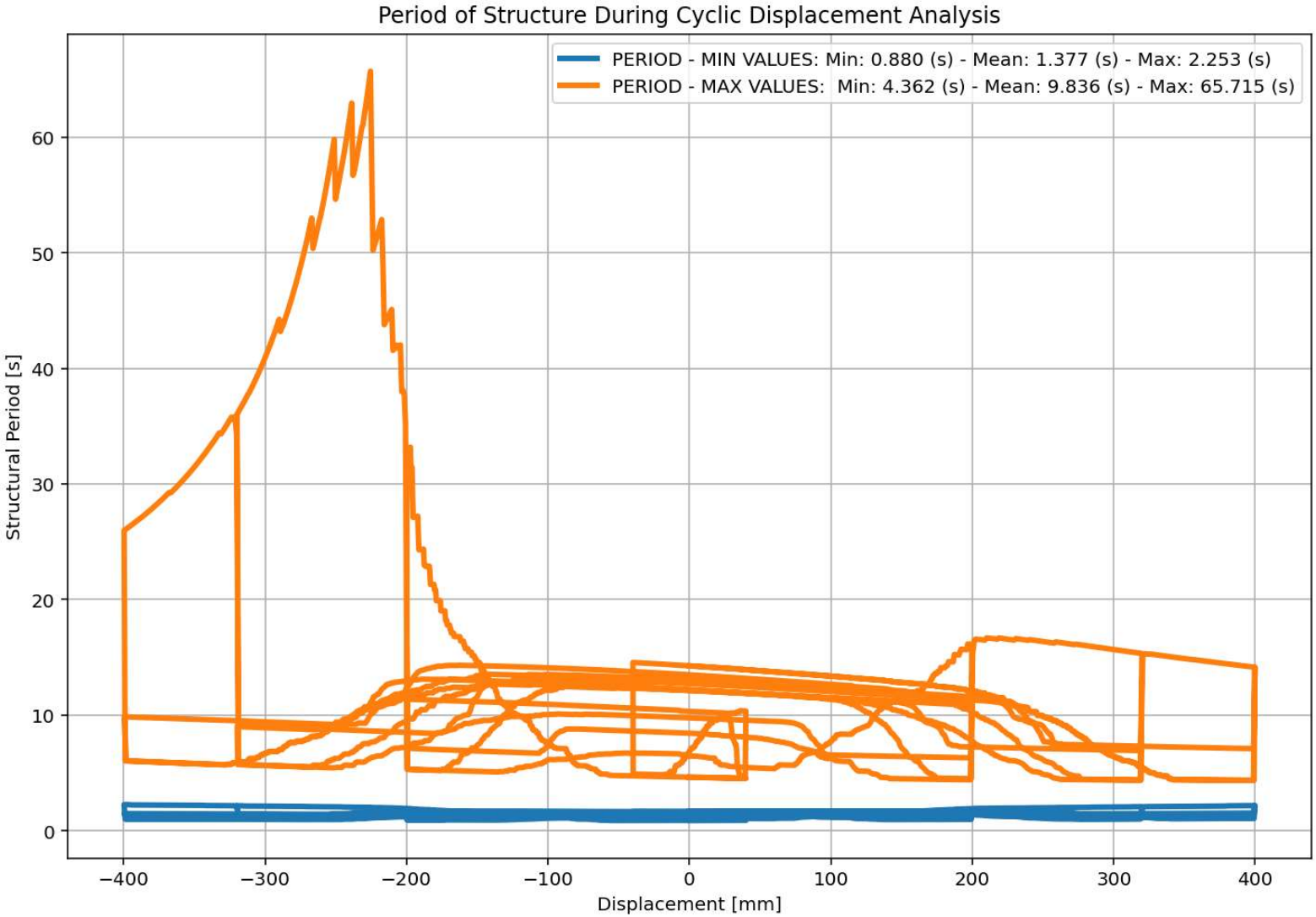


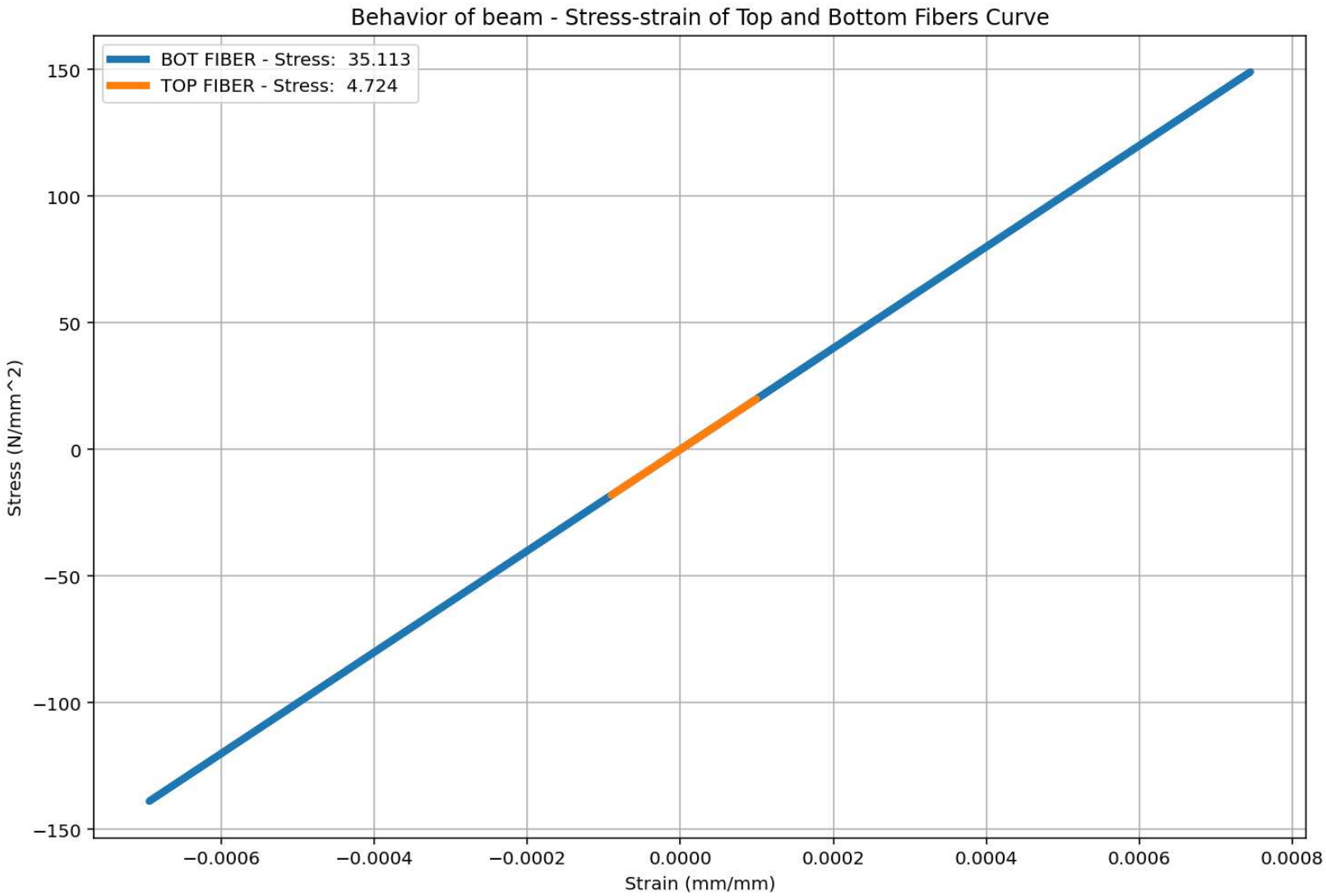


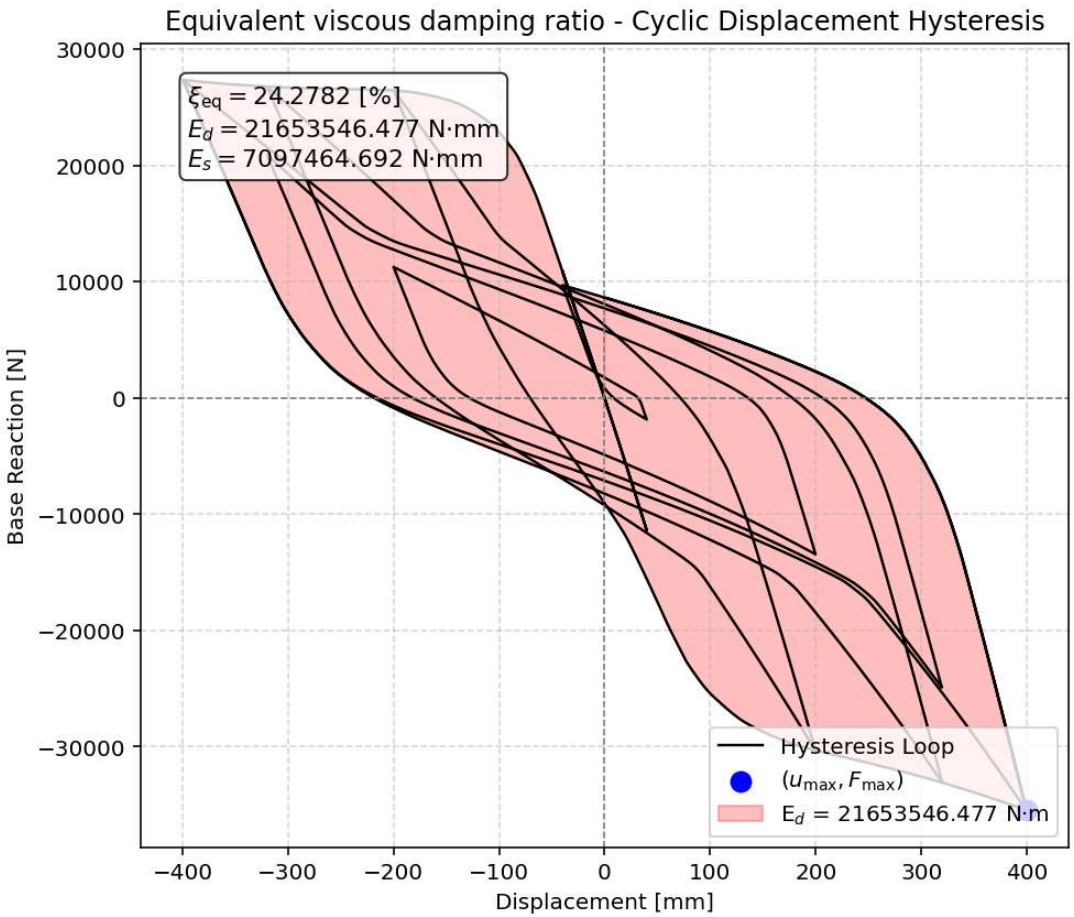
IPython Console | Files | Help | Variable Explorer | Debugger | Plots | History

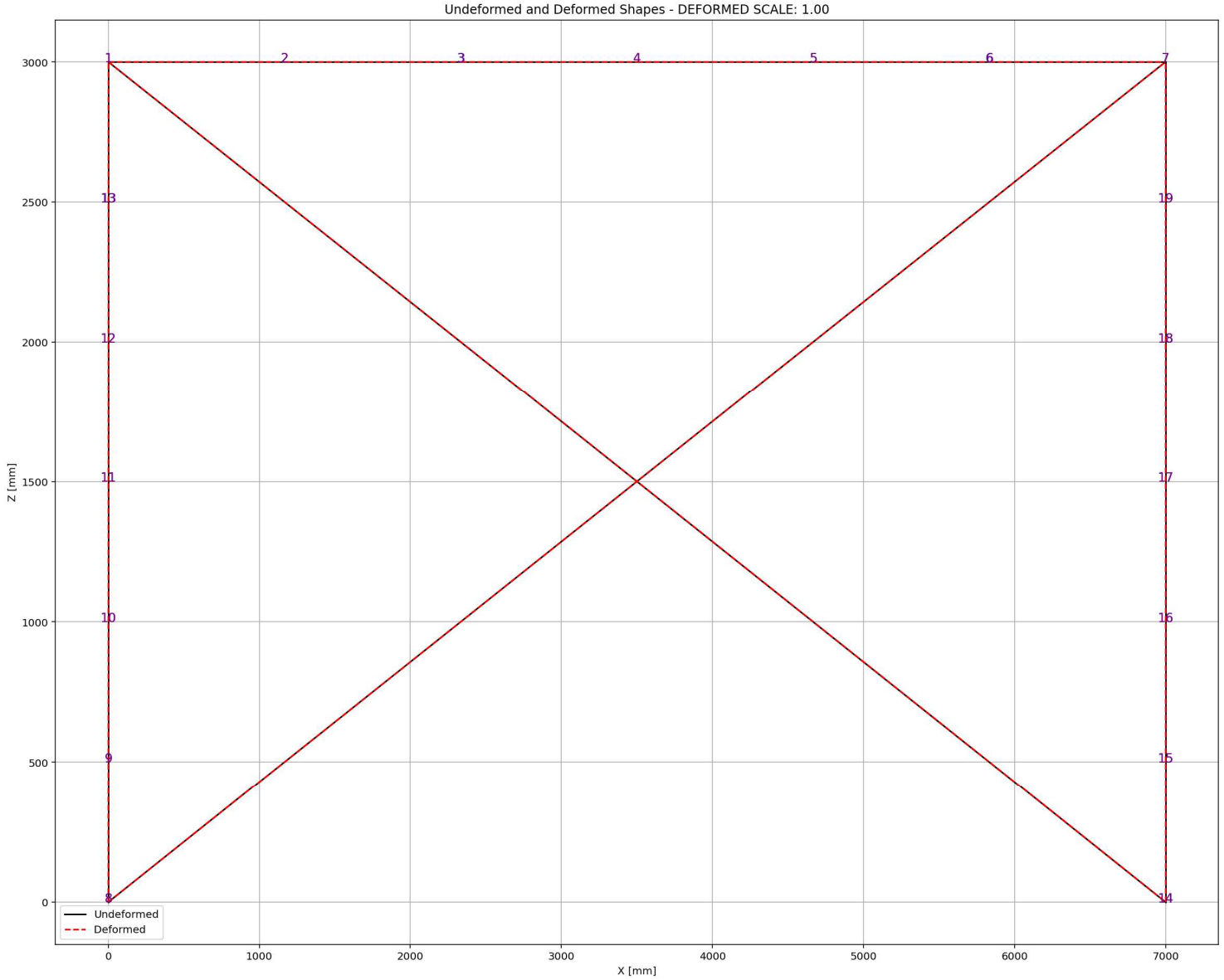
Inline Conda: anaconda3 (Python 3.12.7) LSP: Python Line 12, Col 105 UTF-8 CRLF RW Mem 41%


Base Reaction and Displacement of Structure During Cyclic-Displacement Analysis
Base Reaction [N]
30000
20000
10000
0
−10000
−20000
−30000
−400
−300
−200
−100
0
100
200
300
400
Displacement in Middle Span [mm]


Period of Structure During Cyclic Displacement Analysis
PERIOD - MIN VALUES: Min: 0.880 (s) - Mean: 1.377 (s) - Max: 2.253 (s)
PERIOD - MAX VALUES: Min: 4.362 (s) - Mean: 9.836 (s) - Max: 65.715 (s)
Structural Period [s]
0
10
20
30
40
50
60
−400
−300
−200
−100
0
100
200
300
400
Displacement [mm]


Behavior of beam - Stress-strain of Top and Bottom Fibers Curve
BOT FIBER - Stress: 35.113
TOP FIBER - Stress: 4.724
150
100
50
0
−50
−100
−150
Stress (N/mm^2)
−0.0006
−0.0004
−0.0002
0.0000
0.0002
0.0004
0.0006
0.0008
Strain (mm/mm)


Equivalent viscous damping ratio - Cyclic Displacement Hysteresis
ξ_{eq} = 24.2782 [%]
E_d = 21653546.477 N·mm
E_s = 7097464.692 N·mm
Base Reaction [N]
Displacement [mm]
30000
20000
10000
0
−10000
−20000
−30000
−400
−300
−200
−100
0
100
200
300
400
Hysteresis Loop
(u_{max}, F_{max})
E_d = 21653546.477 N·m


Undeformed and Deformed Shapes - DEFORMED SCALE: 1.00
Z [mm]
X [mm]
0
500
1000
1500
2000
2500
3000
0
1000
2000
3000
4000
5000
6000
7000
1
2
3
4
5
6
7
8
9
10
11
12
13
14
15
16
17
18
19
Undeformed
Deformed

STATIC TIME-HISTORY WITH EXTERNAL TIME-DEPENDENT LOADING ANALYSIS

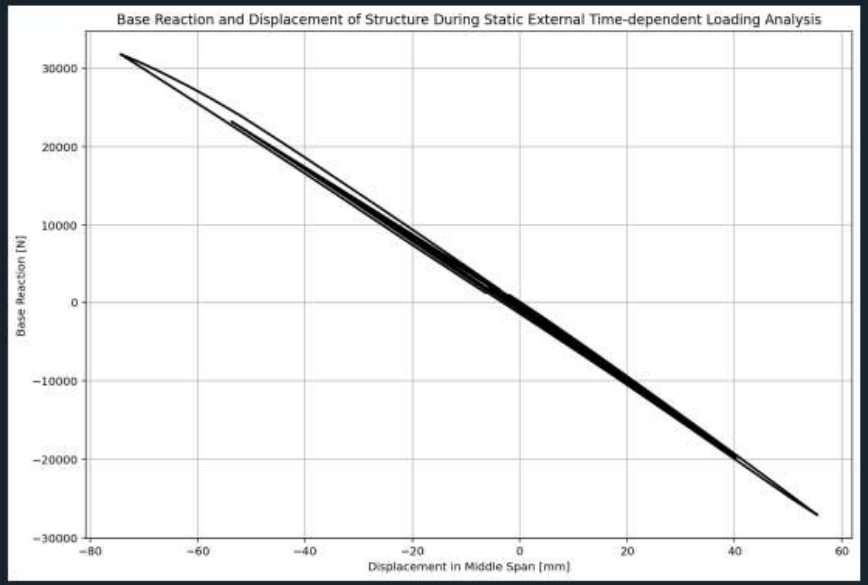
Spyder (Python 3.12)

File Edit Search Source Run Debug Consoles Projects Tools View Help

C:\Users\Dell\Desktop\OPENSEES_FILES\+EIGHT_ANALY...L_BRACED_FRAME_8_ANA\W(x,t)_STEEL_BRACED_FRAME.py

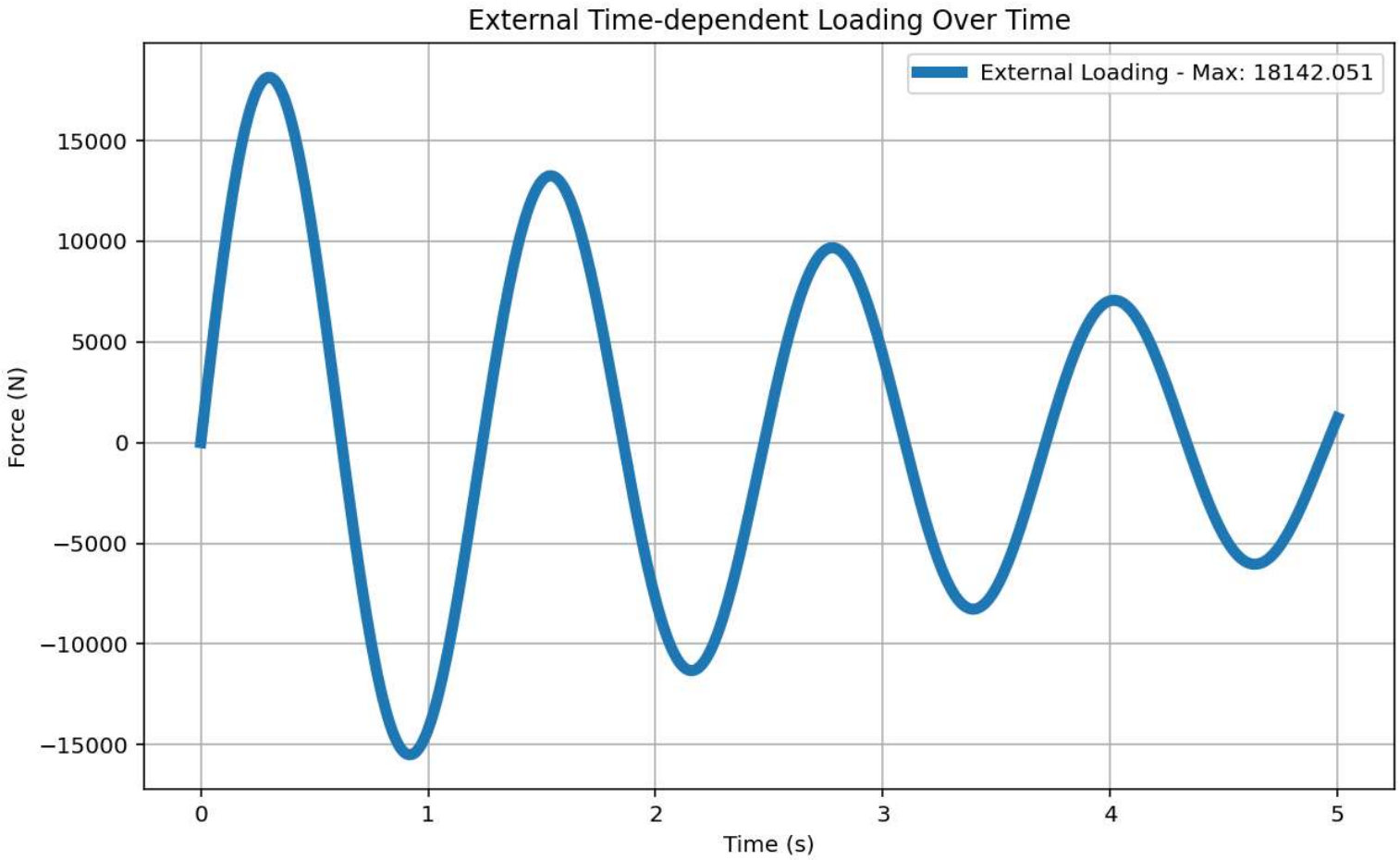
W(x,t)_STEEL_BRACED_FRAME.py | PLATE_DOUBLE_I_STE...EIGHT_FUN_EXTRA.py

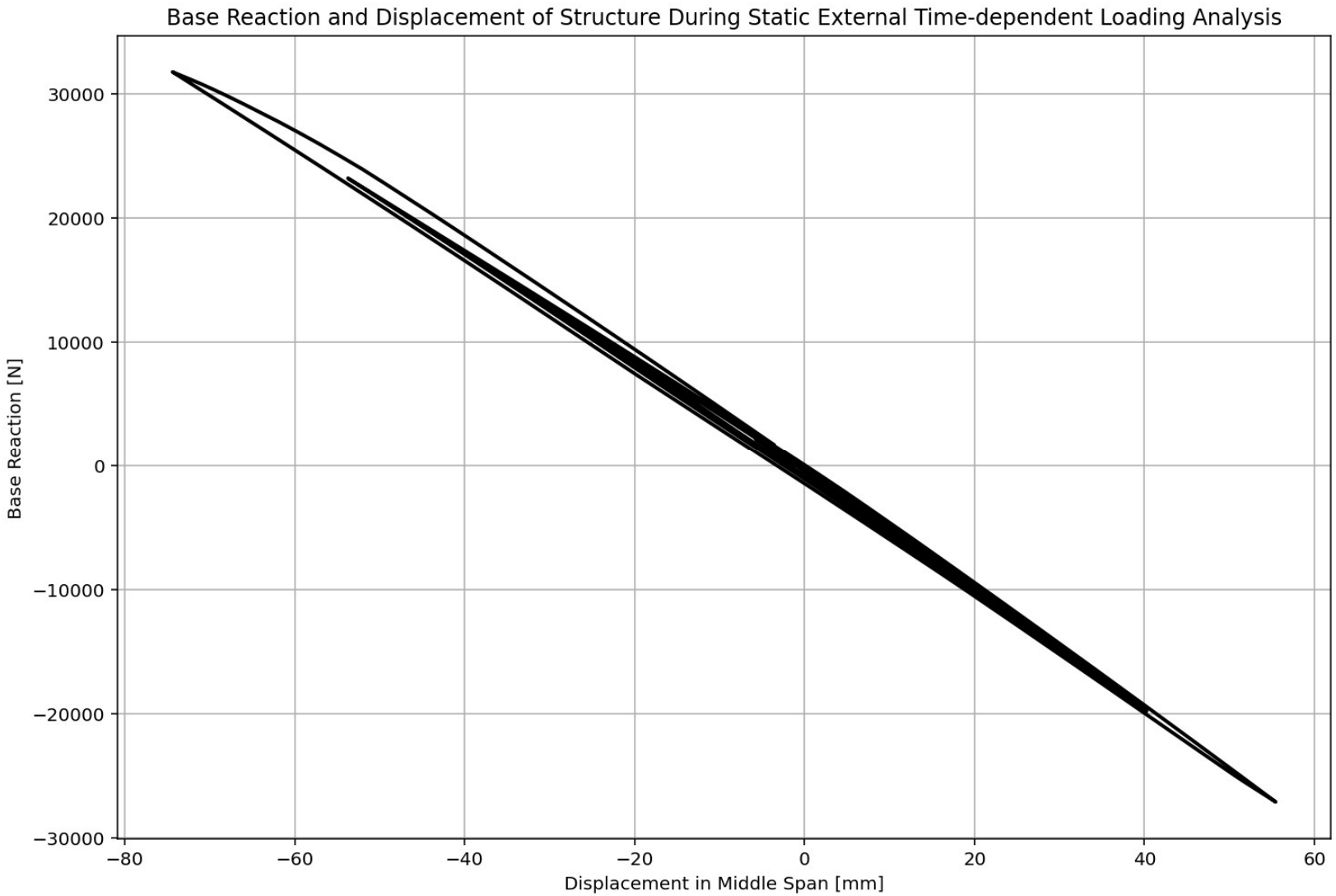
```
1309  # EXTERNAL TIME-DEPENDENT LOADING ANALYSIS (STATIC TIME-HISTORY ANALYSIS)
1310  #ELE_TYPE = 'elasticBeamColumn'      # MATERIAL LINEARITY
1311  ELE_TYPE = 'nonlinearBeamColumn'     # MATERIAL AND GEOMETRIC NONLINEARITIE
1312  MAT_TYPE = 'INELASTIC'   # 'ELASTIC' OR 'INELASTIC'
1313  ANAL_TYPE = 'STATIC_EXTERNAL_TIME-DEPENDENT_LOADING'
1314  BRACE = True                         # True: WITH BRACE - False: WITHOUT BR
1315
1316  DATA = STEEL_BRACED_FRAME(LENGTH, HEIGHT, MAT_TYPE, ELE_TYPE, ANAL_TYPE, TO
1317
1318  (reactionX_ETDLS, reactionY_ETDLS, disp_mid_ETDLS,
1319   ele_axialforce_ETDLS, ele_shearforce_ETDLS, ele_momentforce_ETDLS,
1320   node_displacementsX_ETDLS, node_displacementsY_ETDLS,
1321    dispX_ETDLS, dispY_ETDLS,
1322   PERIOD_MIN_ETDLS, PERIOD_MAX_ETDLS,
1323   STRESSb_ETDLS, STRAINb_ETDLS, STRESSt_ETDLS, STRAINt_ETDLS, CURVATURE_ETDL
1324
1325
1326  XDATA = disp_mid_ETDLS
1327  YDATA = reactionY_ETDLS
1328  XLABEL = 'Displacement in Middle Span [mm]'
1329  YLABEL = 'Base Reaction [N]'
1330  TITLE = 'Base Reaction and Displacement of Structure During Static External
1331  COLOR = 'black'
1332  SEMILOGY = False
1333  PLOT(XDATA, YDATA, TITLE, XLABEL, YLABEL, COLOR, SEMILOGY)
1334
1335  # PLOT STRUCTURAL PERIOD DURING THE ANALYSIS
1336  plt.figure(0, figsize=(12, 8))
1337  plt.plot(disp_mid_ETDLS, PERIOD_MIN_ETDLS, linewidth=3)
1338  plt.plot(disp_mid_ETDLS, PERIOD_MAX_ETDLS, linewidth=3)
1339  plt.title('Period of Structure During Static External Time-dependent Loadin
1340  plt.ylabel('Structural Period [s]')
1341  plt.xlabel('Displacement [mm]')
1342  #plt.semilogy()
```

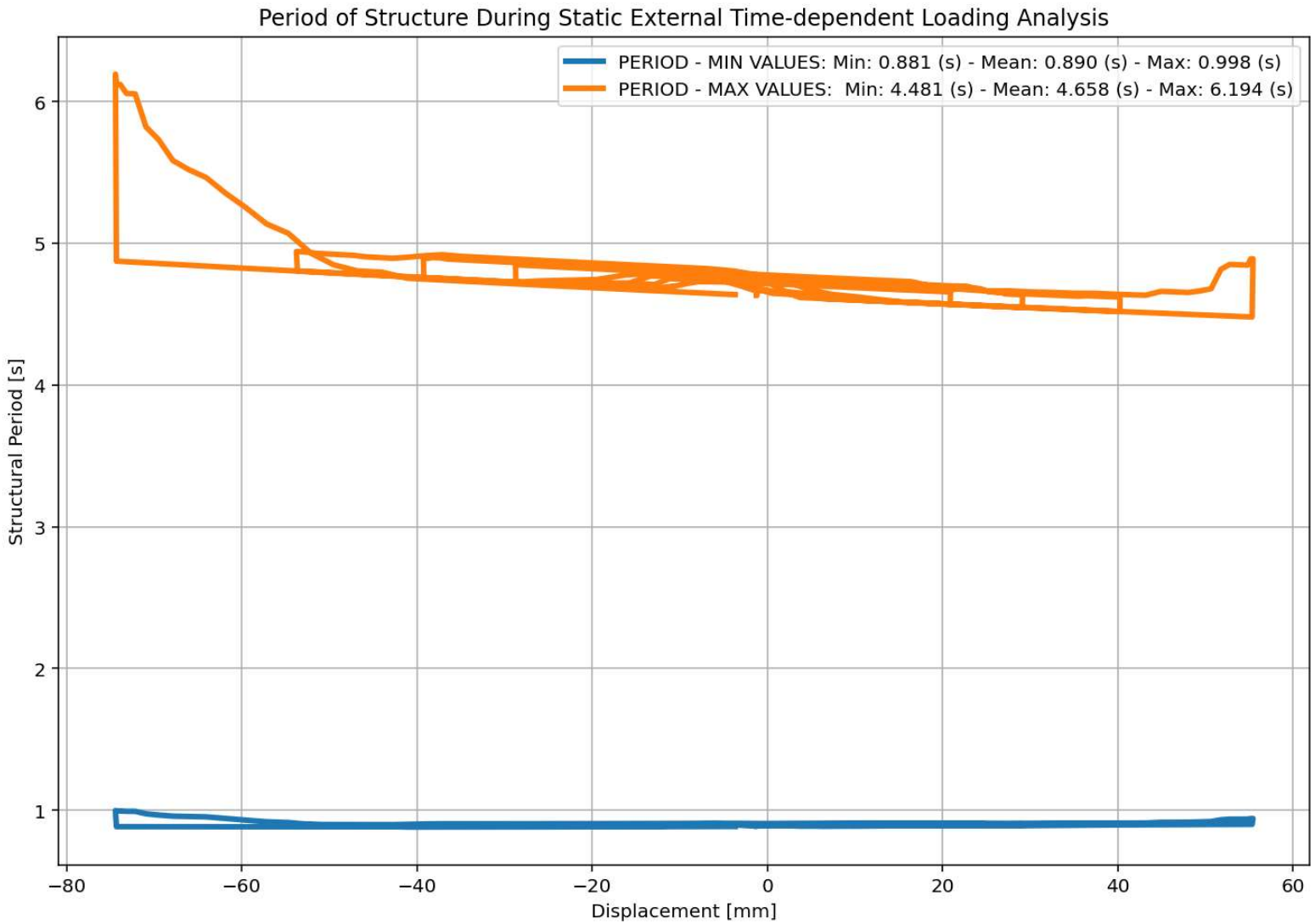


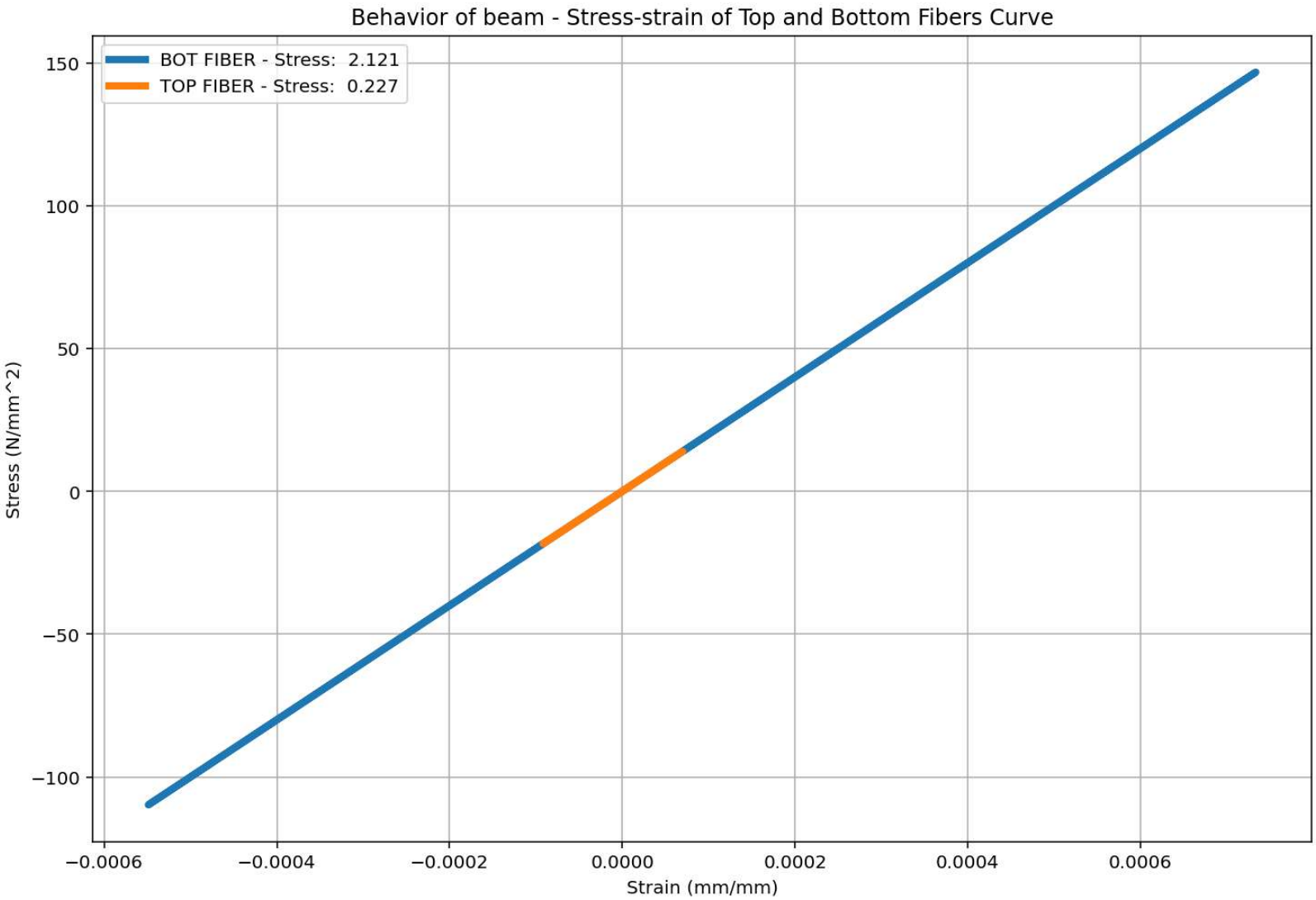
IPython Console | Files | Help | Variable Explorer | Debugger | Plots | History

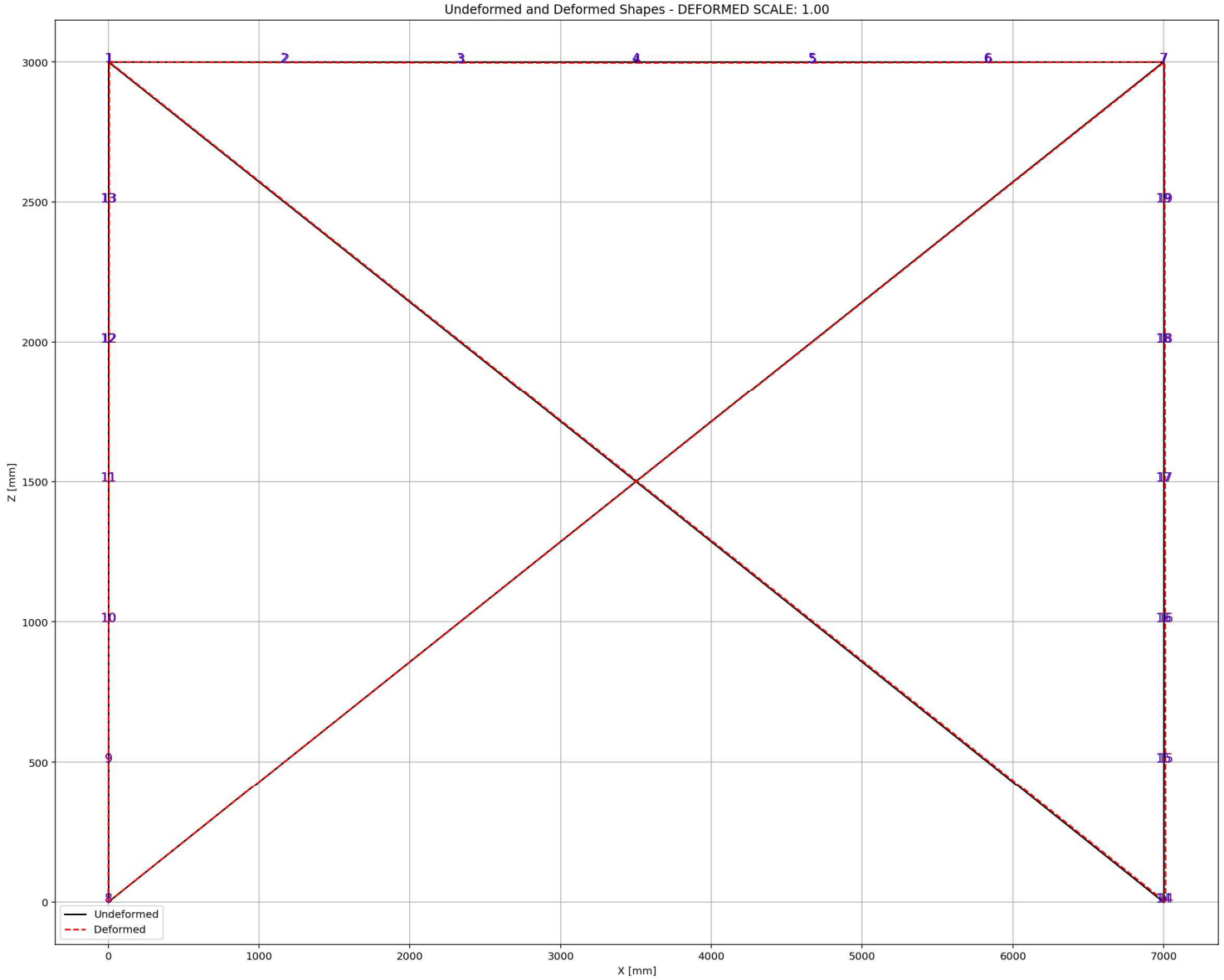
Inline Conda: anaconda3 (Python 3.12.7) LSP: Python Line 12, Col 105 UTF-8 CRLF RW Mem 40%


External Time-dependent Loading Over Time
External Loading - Max: 18142.051
Force (N)
15000
10000
5000
0
−5000
−10000
−15000
0
1
2
3
4
5
Time (s)


Base Reaction and Displacement of Structure During Static External Time-dependent Loading Analysis
Base Reaction [N]
30000
20000
10000
0
−10000
−20000
−30000
−80
−60
−40
−20
0
20
40
60
Displacement in Middle Span [mm]


Period of Structure During Static External Time-dependent Loading Analysis
PERIOD - MIN VALUES: Min: 0.881 (s) - Mean: 0.890 (s) - Max: 0.998 (s)
PERIOD - MAX VALUES: Min: 4.481 (s) - Mean: 4.658 (s) - Max: 6.194 (s)
Structural Period [s]
Displacement [mm]
1
2
3
4
5
6
−80
−60
−40
−20
0
20
40
60


Behavior of beam - Stress-strain of Top and Bottom Fibers Curve
BOT FIBER - Stress: 2.121
TOP FIBER - Stress: 0.227
150
100
50
0
−50
−100
Stress (N/mm^2)
−0.0006
−0.0004
−0.0002
0.0000
0.0002
0.0004
0.0006
Strain (mm/mm)


Undeformed and Deformed Shapes - DEFORMED SCALE: 1.00
Z [mm]
X [mm]
Undeformed
Deformed
0
500
1000
1500
2000
2500
3000
0
1000
2000
3000
4000
5000
6000
7000
1
2
3
4
5
6
7
8
9
10
11
12
13
14
15
16
17
18
19

DYNAMIC TIME-HISTORY WITH EXTERNAL TIME-DEPENDENT LOADING ANALYSIS

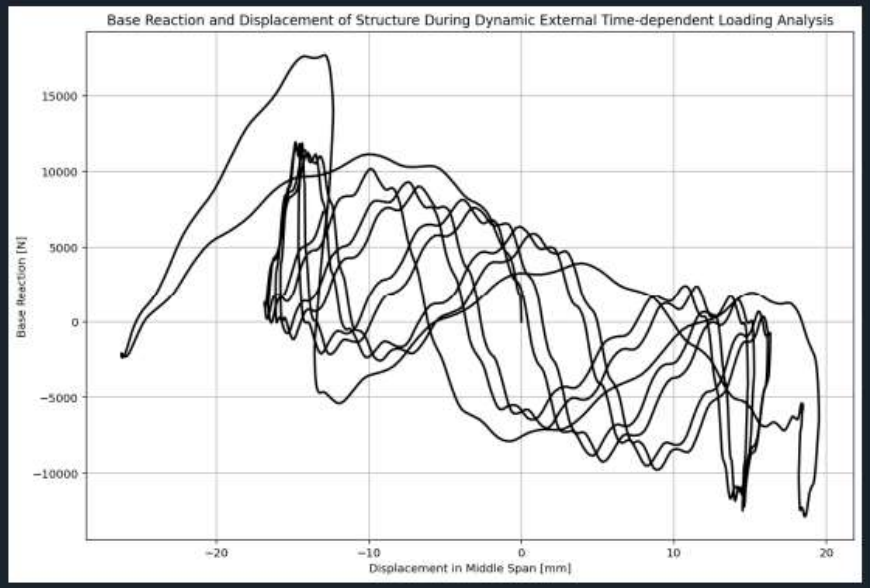
Spyder (Python 3.12)

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C:\Users\Dell\Desktop\OPENSEES_FILES\+EIGHT_ANALY...L_BRACED_FRAME_8_ANA\W(x,t)_STEEL_BRACED_FRAME.py

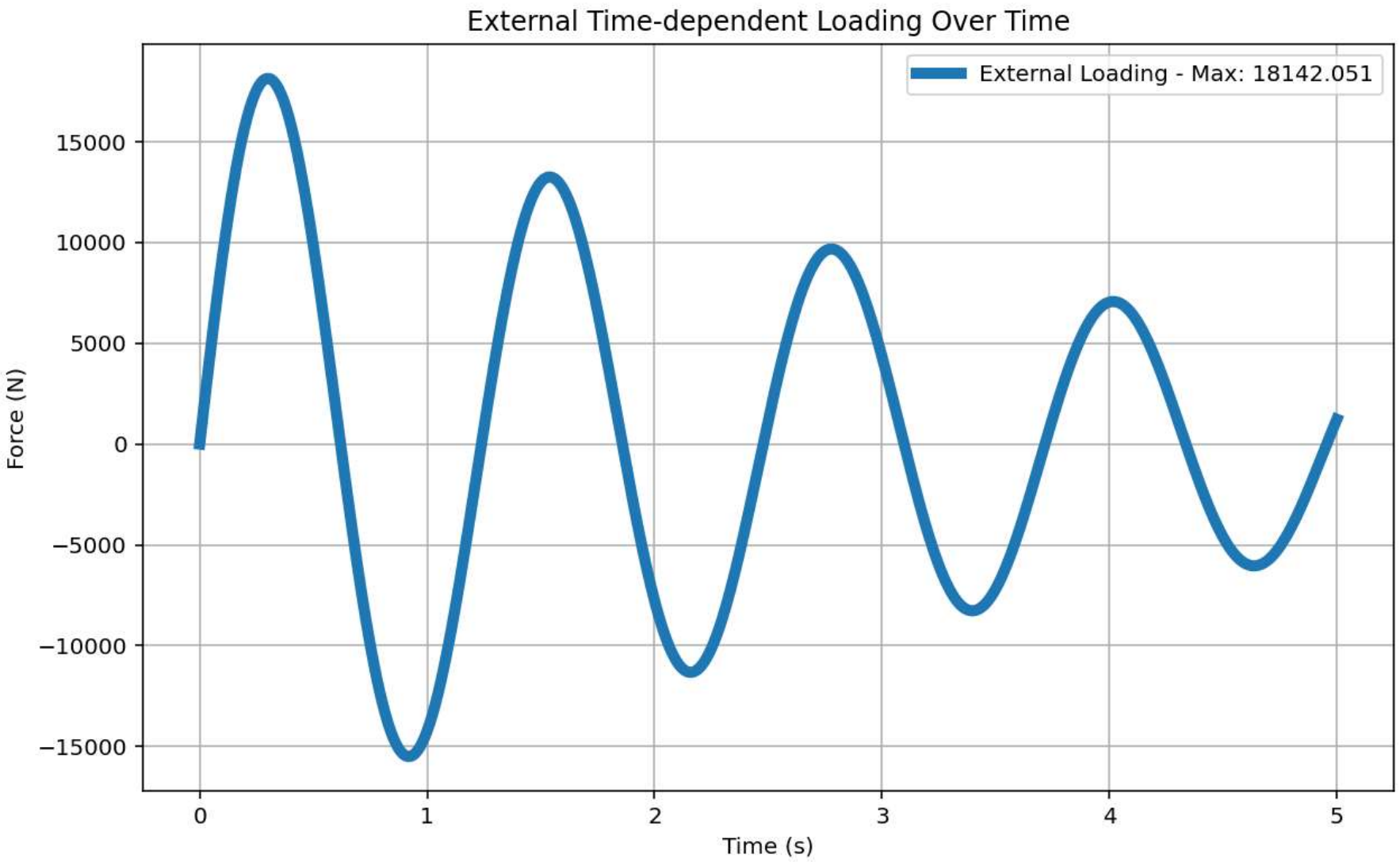
W(x,t)_STEEL_BRACED_FRAME.py | PLATE_DOUBLE_I_STE...EIGHT_FUN_EXTRA.py

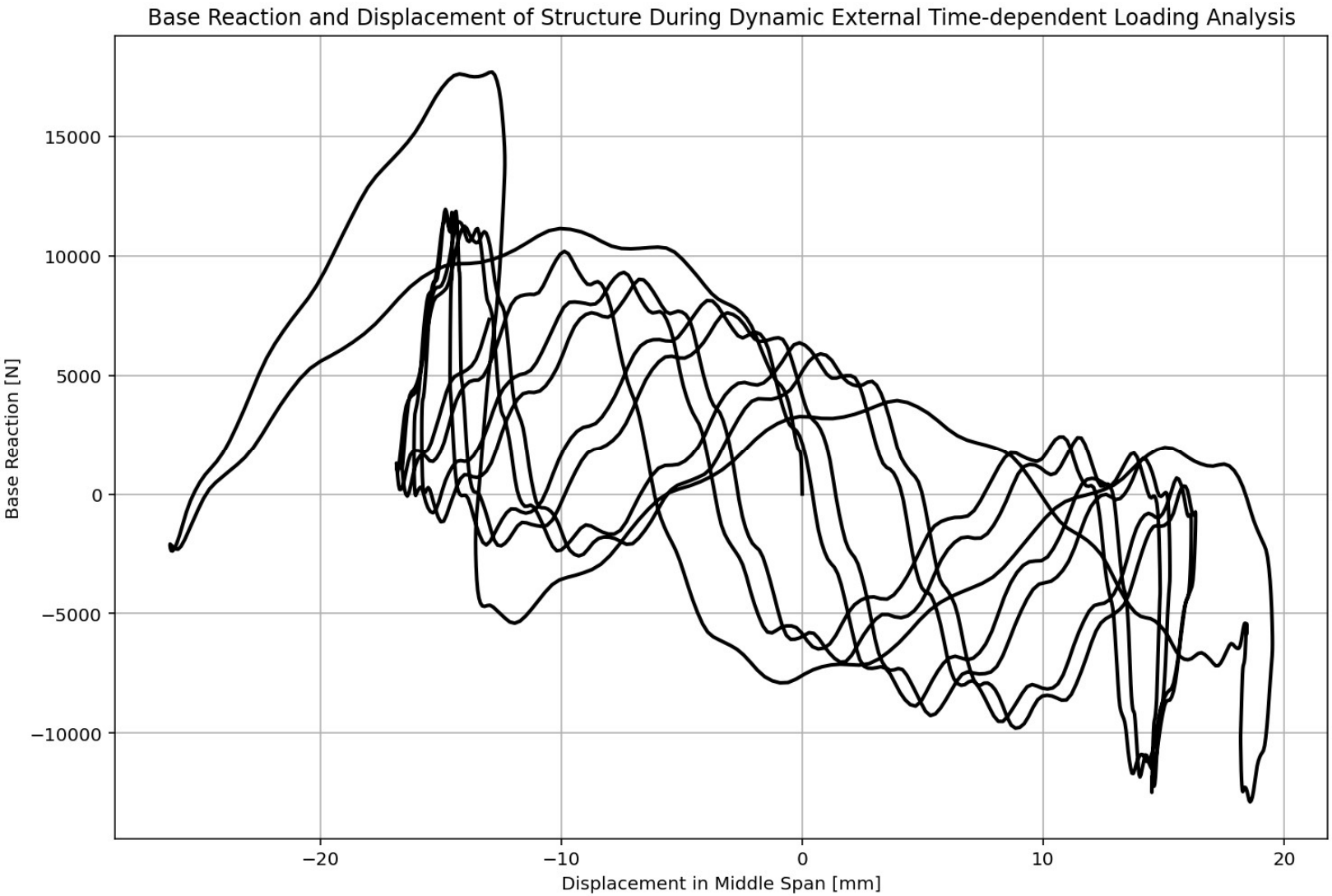
```
1362    # EXTERNAL TIME-DEPENDENT LOADING ANALYSIS (DYNAMIC TIME-HISTORY ANALYSIS)
1363    #ELE_TYPE = 'elasticBeamColumn'      # MATERIAL LINEARITY
1364    ELE_TYPE = 'nonlinearBeamColumn'     # MATERIAL AND GEOMETRIC NONLINEARITIE
1365    MAT_TYPE = 'INELASTIC'   # 'ELASTIC' OR 'INELASTIC'
1366    ANAL_TYPE = 'DYNAMIC_EXTERNAL_TIME-DEPENDENT_LOADING'
1367    BRACE = True                         # True: WITH BRACE - False: WITHOUT BR
1368
1369    DATA = STEEL_BRACED_FRAME(LENGTH, HEIGHT, MAT_TYPE, ELE_TYPE, ANAL_TYPE, TO
1370
1371    (time_ETDLD, reactionX_ETDLD, reactionY_ETDLD, disp_mid_ETDLD,
1372    ele_axialforce_ETDLD, ele_shearforce_ETDLD, ele_momentforce_ETDLD,
1373    node_displacementsX_ETDLD, node_displacementsY_ETDLD,
1374    dispX_ETDLD, dispY_ETDLD,
1375    veloX_ETDLD, veloY_ETDLD,
1376    accX_ETDLD, accY_ETDLD,
1377    stiffness_ETDLD, PERIOD_ETDLD, damping_ratio_ETDLD,
1378    PERIOD_MIN_ETDLD, PERIOD_MAX_ETDLD,
1379    STRESSb_ETDLD, STRAINb_ETDLD, STRESSt_ETDLD, STRAINt_ETDLD, CURVATURE_ETDLD
1380
1381
1382    XDATA = disp_mid_ETDLD
1383    YDATA = reactionY_ETDLD
1384    XLABEL = 'Displacement in Middle Span [mm]'
1385    YLABEL = 'Base Reaction [N]'
1386    TITLE = 'Base Reaction and Displacement of Structure During Dynamic Externa
1387    COLOR = 'black'
1388    SEMILOGY = False
1389    PLOT(XDATA, YDATA, TITLE, XLABEL, YLABEL, COLOR, SEMILOGY)
1390
1391    PLOT_TIME_HISTORY(time_ETDLD, reactionY_ETDLD, disp_mid_ETDLD,
1392                      dispX_ETDLD, dispY_ETDLD,
1393                      veloX_ETDLD, veloY_ETDLD,
1394                      accX_ETDLD, accY_ETDLD)
1395
```

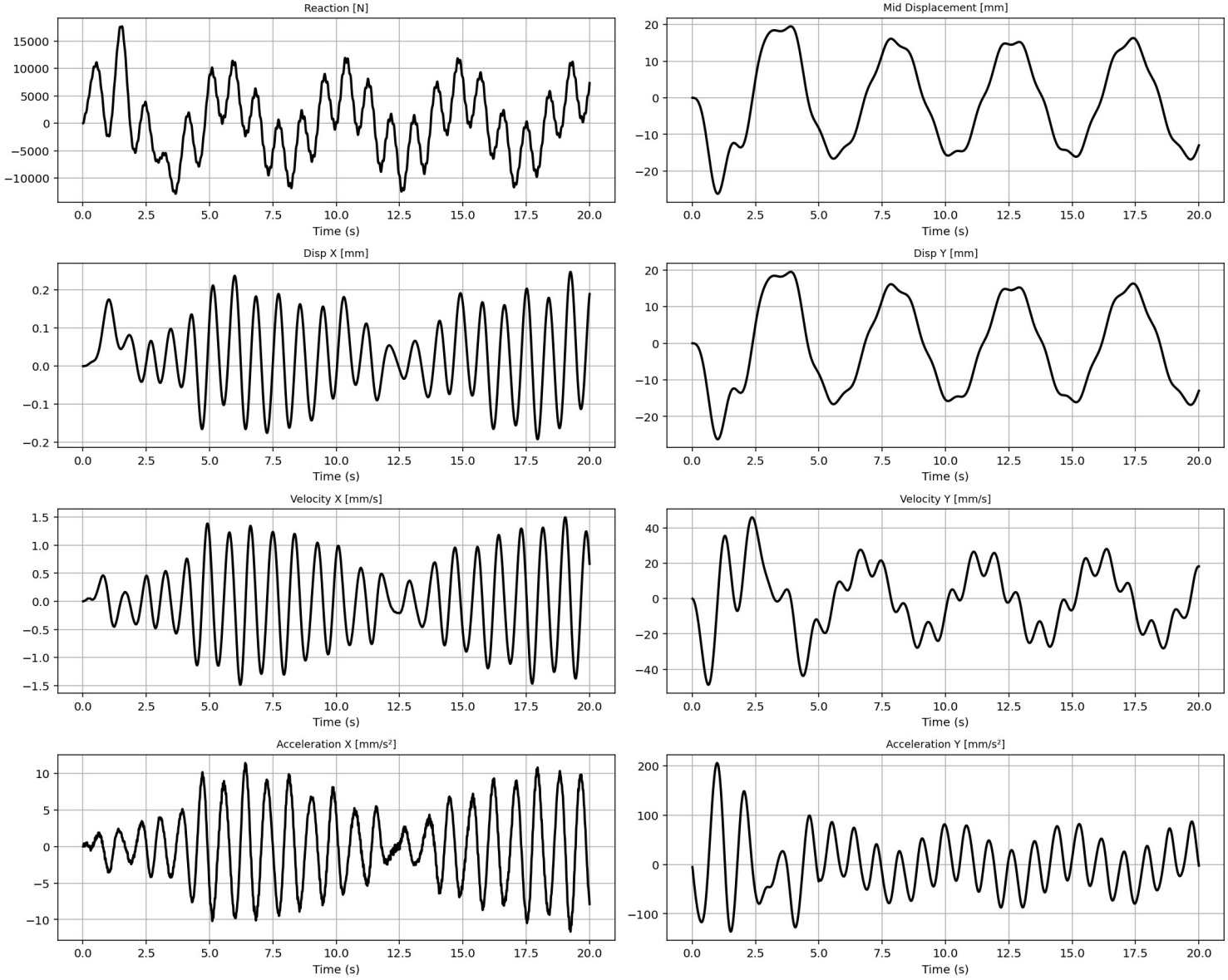


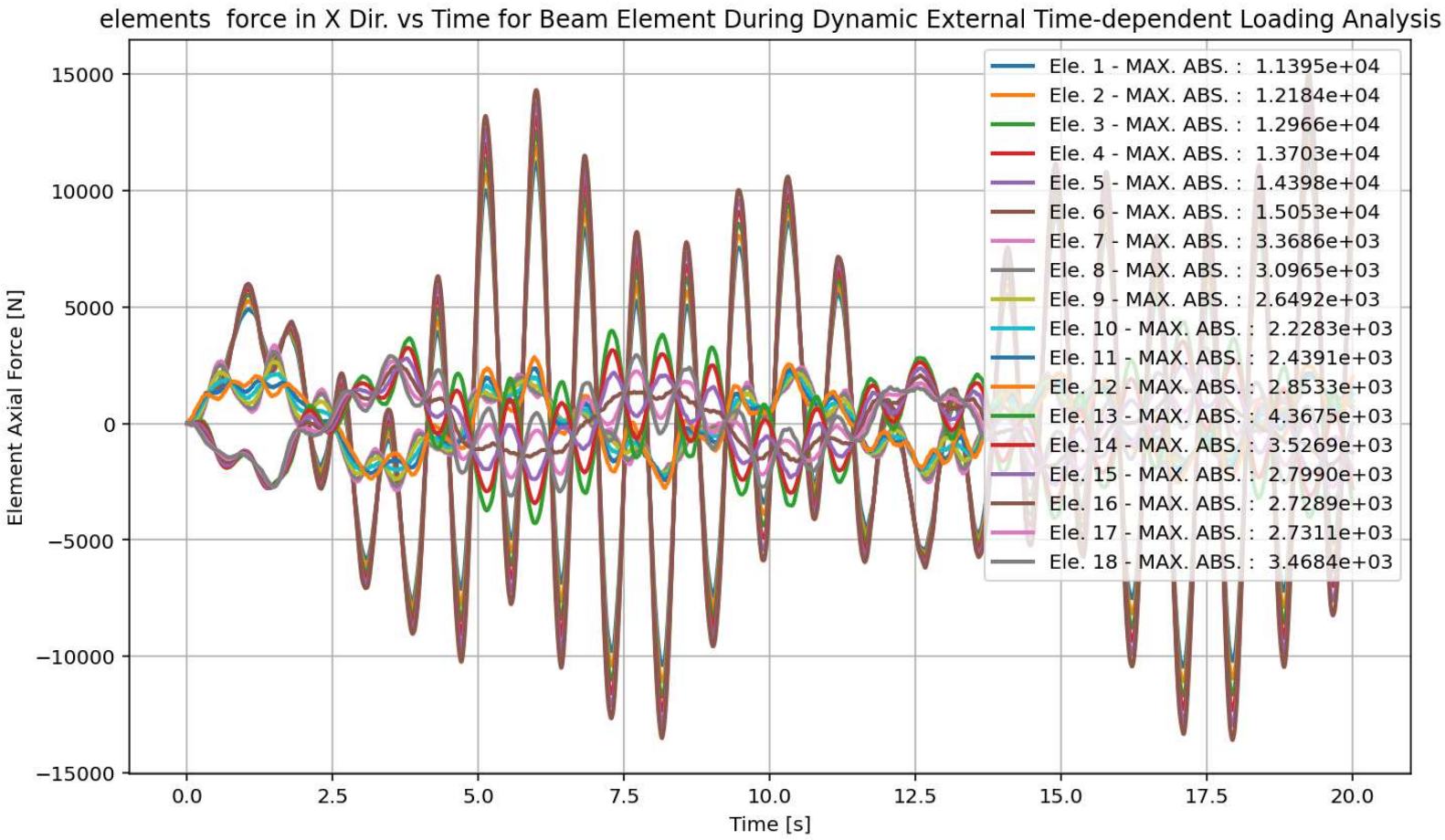
IPython Console | Files | Help | Variable Explorer | Debugger | Plots | History

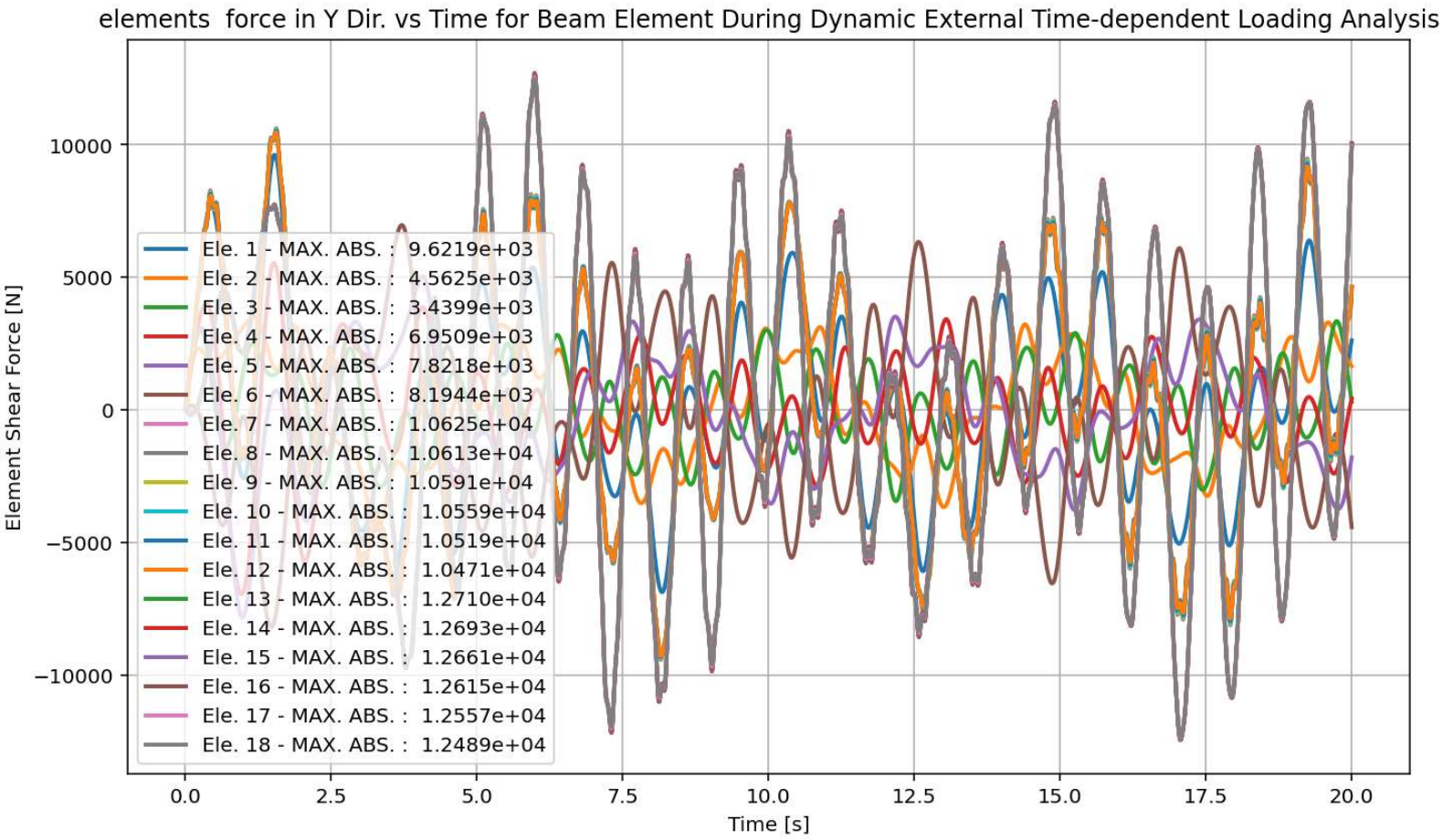
Inline Conda: anaconda3 (Python 3.12.7) LSP: Python Line 12, Col 105 UTF-8 CRLF RW Mem 40%

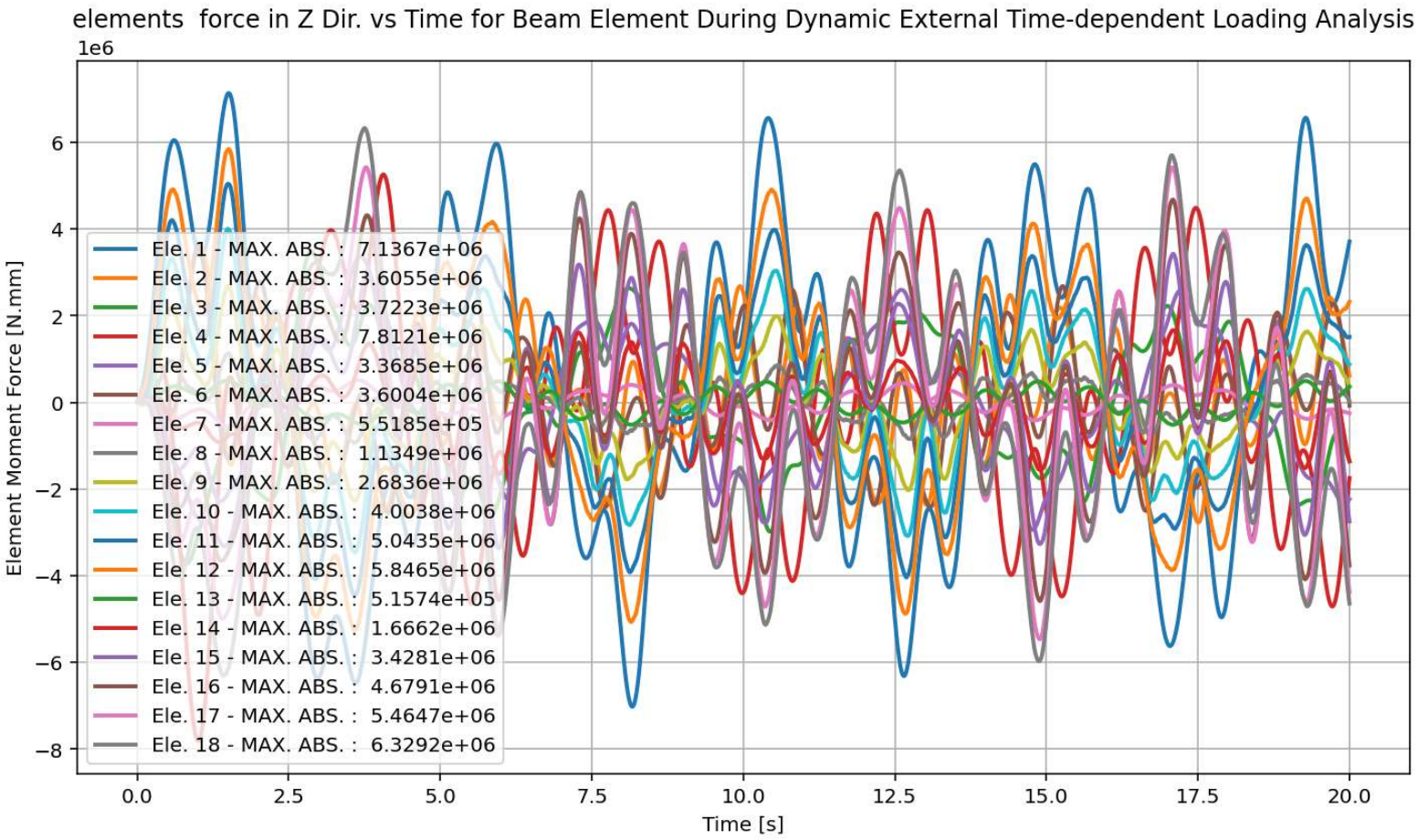

External Time-dependent Loading Over Time
External Loading - Max: 18142.051
Force (N)
15000
10000
5000
0
−5000
−10000
−15000
0
1
2
3
4
5
Time (s)

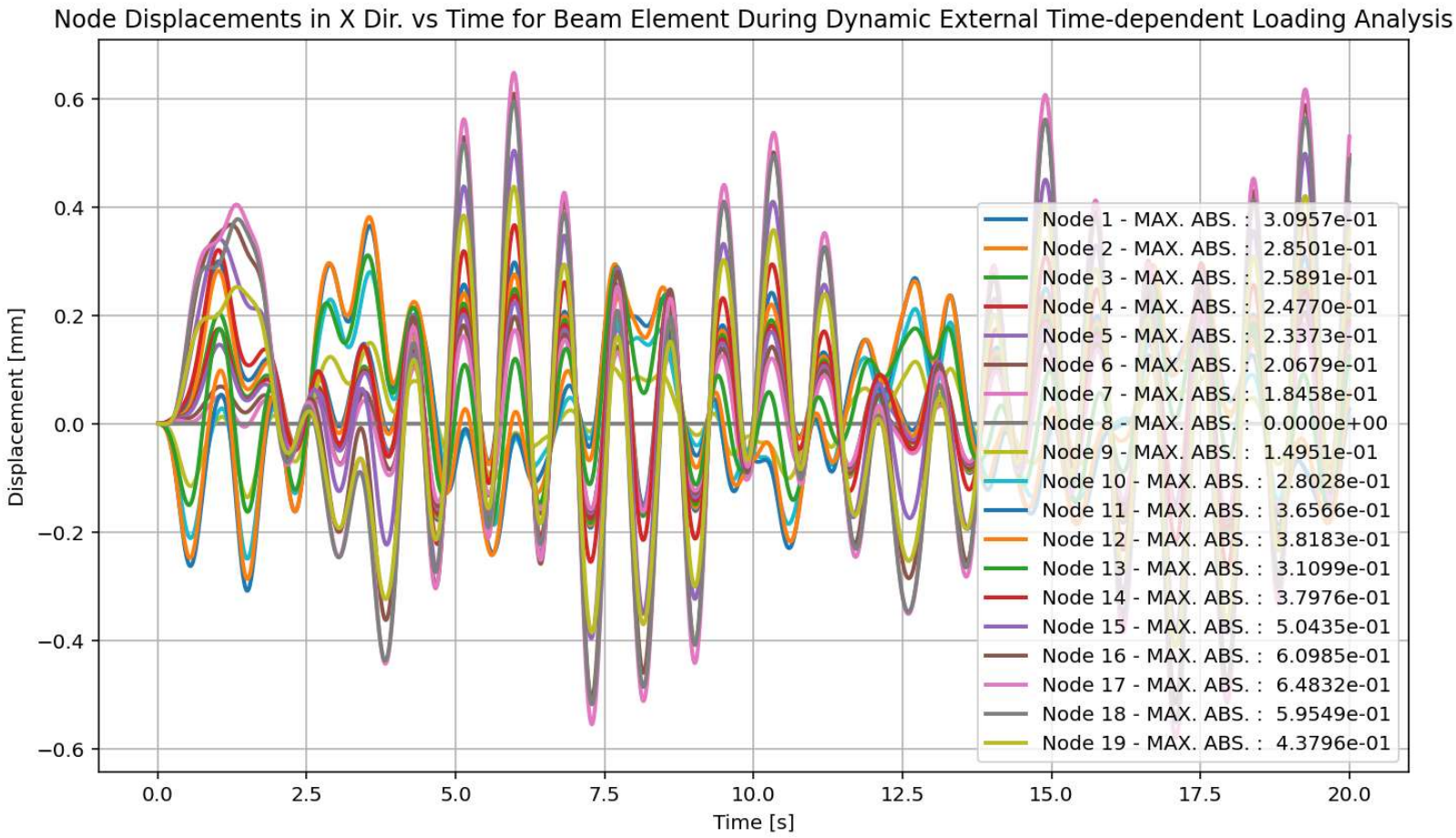

Base Reaction and Displacement of Structure During Dynamic External Time-dependent Loading Analysis
Base Reaction [N]
15000
10000
5000
0
−5000
−10000
−20
−10
0
10
20
Displacement in Middle Span [mm]

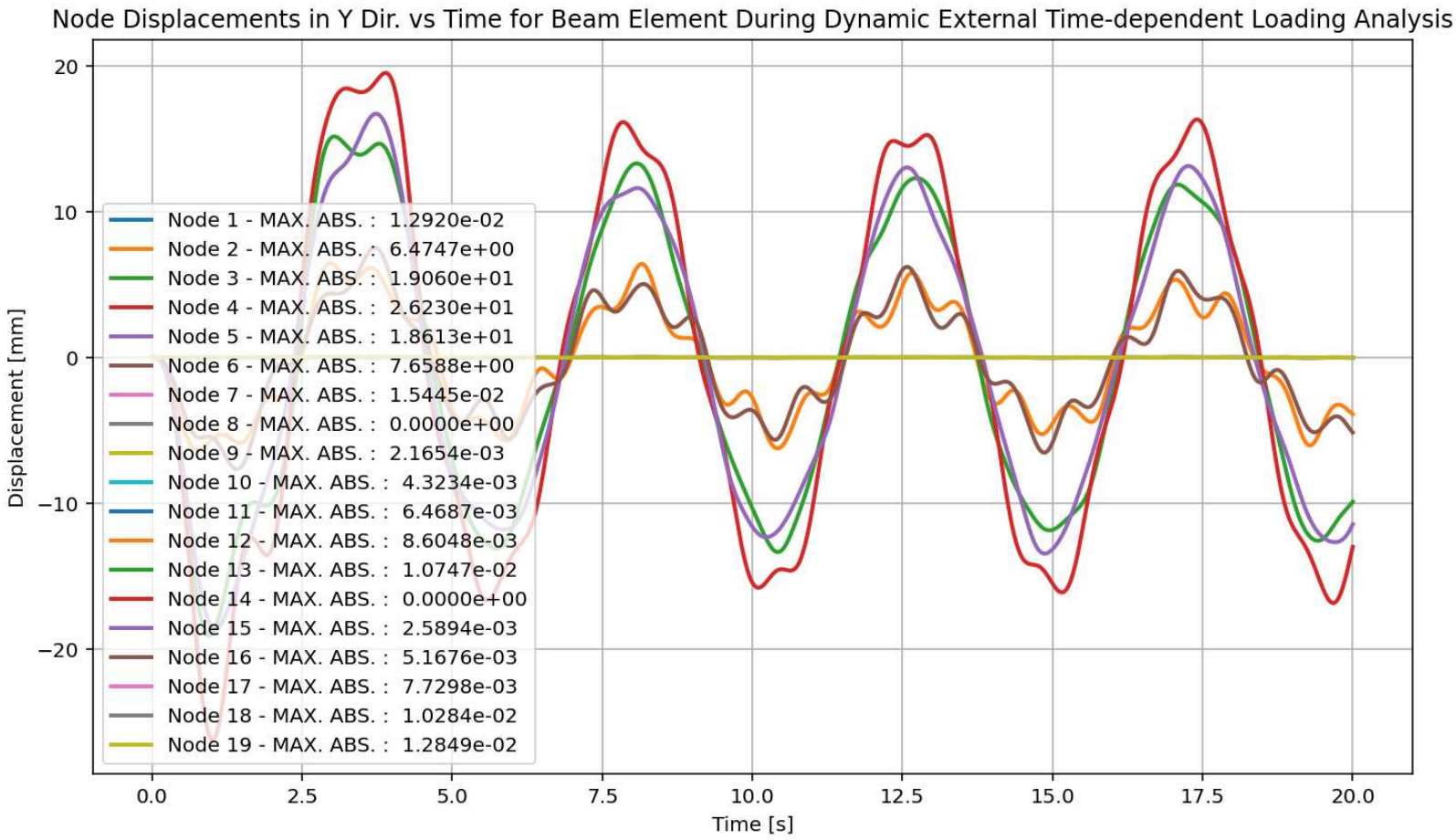

Reaction [N]
Time (s)
Mid Displacement [mm]
Time (s)
Disp X [mm]
Time (s)
Disp Y [mm]
Time (s)
Velocity X [mm/s]
Time (s)
Velocity Y [mm/s]
Time (s)
Acceleration X [mm/s²]
Time (s)
Acceleration Y [mm/s²]
Time (s)

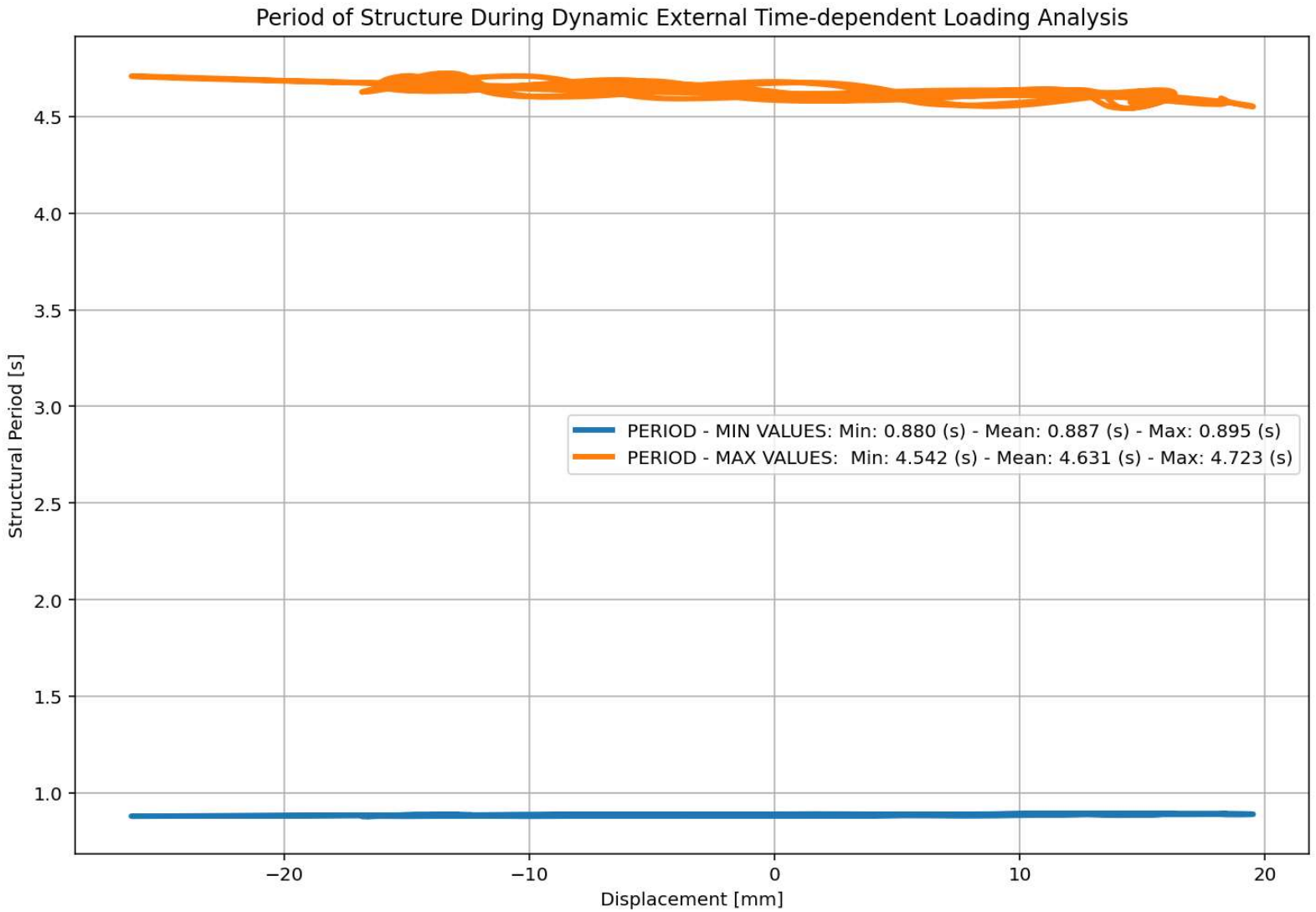

elements force in X Dir. vs Time for Beam Element During Dynamic External Time-dependent Loading Analysis
Element Axial Force [N]
Time [s]
15000
10000
5000
0
−5000
−10000
−15000
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Ele. 1 - MAX. ABS. : 1.1395e+04
Ele. 2 - MAX. ABS. : 1.2184e+04
Ele. 3 - MAX. ABS. : 1.2966e+04
Ele. 4 - MAX. ABS. : 1.3703e+04
Ele. 5 - MAX. ABS. : 1.4398e+04
Ele. 6 - MAX. ABS. : 1.5053e+04
Ele. 7 - MAX. ABS. : 3.3686e+03
Ele. 8 - MAX. ABS. : 3.0965e+03
Ele. 9 - MAX. ABS. : 2.6492e+03
Ele. 10 - MAX. ABS. : 2.2283e+03
Ele. 11 - MAX. ABS. : 2.4391e+03
Ele. 12 - MAX. ABS. : 2.8533e+03
Ele. 13 - MAX. ABS. : 4.3675e+03
Ele. 14 - MAX. ABS. : 3.5269e+03
Ele. 15 - MAX. ABS. : 2.7990e+03
Ele. 16 - MAX. ABS. : 2.7289e+03
Ele. 17 - MAX. ABS. : 2.7311e+03
Ele. 18 - MAX. ABS. : 3.4684e+03

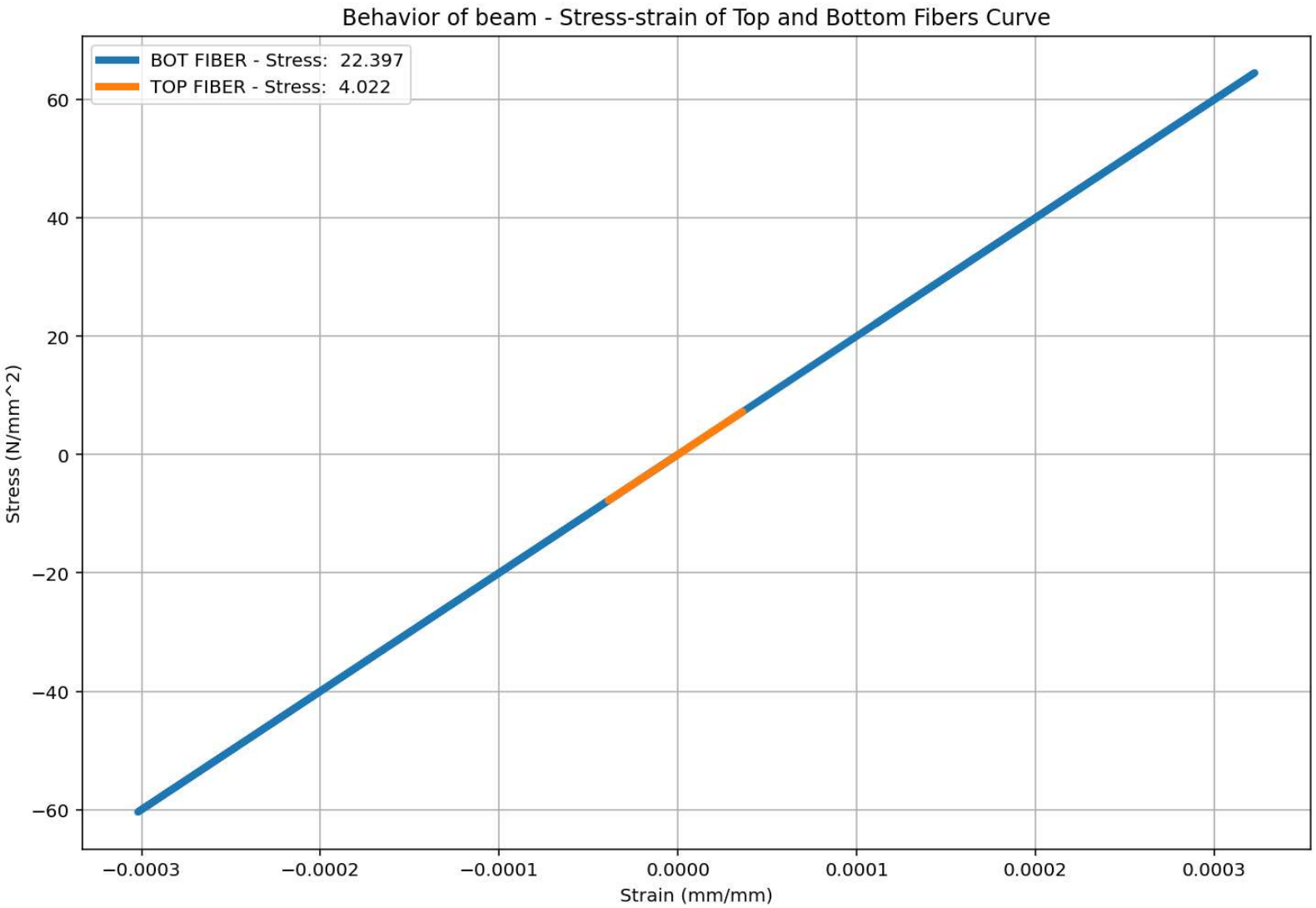

elements force in Y Dir. vs Time for Beam Element During Dynamic External Time-dependent Loading Analysis
Element Shear Force [N]
Time [s]
10000
5000
0
−5000
−10000
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Ele. 1 - MAX. ABS. : 9.6219e+03
Ele. 2 - MAX. ABS. : 4.5625e+03
Ele. 3 - MAX. ABS. : 3.4399e+03
Ele. 4 - MAX. ABS. : 6.9509e+03
Ele. 5 - MAX. ABS. : 7.8218e+03
Ele. 6 - MAX. ABS. : 8.1944e+03
Ele. 7 - MAX. ABS. : 1.0625e+04
Ele. 8 - MAX. ABS. : 1.0613e+04
Ele. 9 - MAX. ABS. : 1.0591e+04
Ele. 10 - MAX. ABS. : 1.0559e+04
Ele. 11 - MAX. ABS. : 1.0519e+04
Ele. 12 - MAX. ABS. : 1.0471e+04
Ele. 13 - MAX. ABS. : 1.2710e+04
Ele. 14 - MAX. ABS. : 1.2693e+04
Ele. 15 - MAX. ABS. : 1.2661e+04
Ele. 16 - MAX. ABS. : 1.2615e+04
Ele. 17 - MAX. ABS. : 1.2557e+04
Ele. 18 - MAX. ABS. : 1.2489e+04

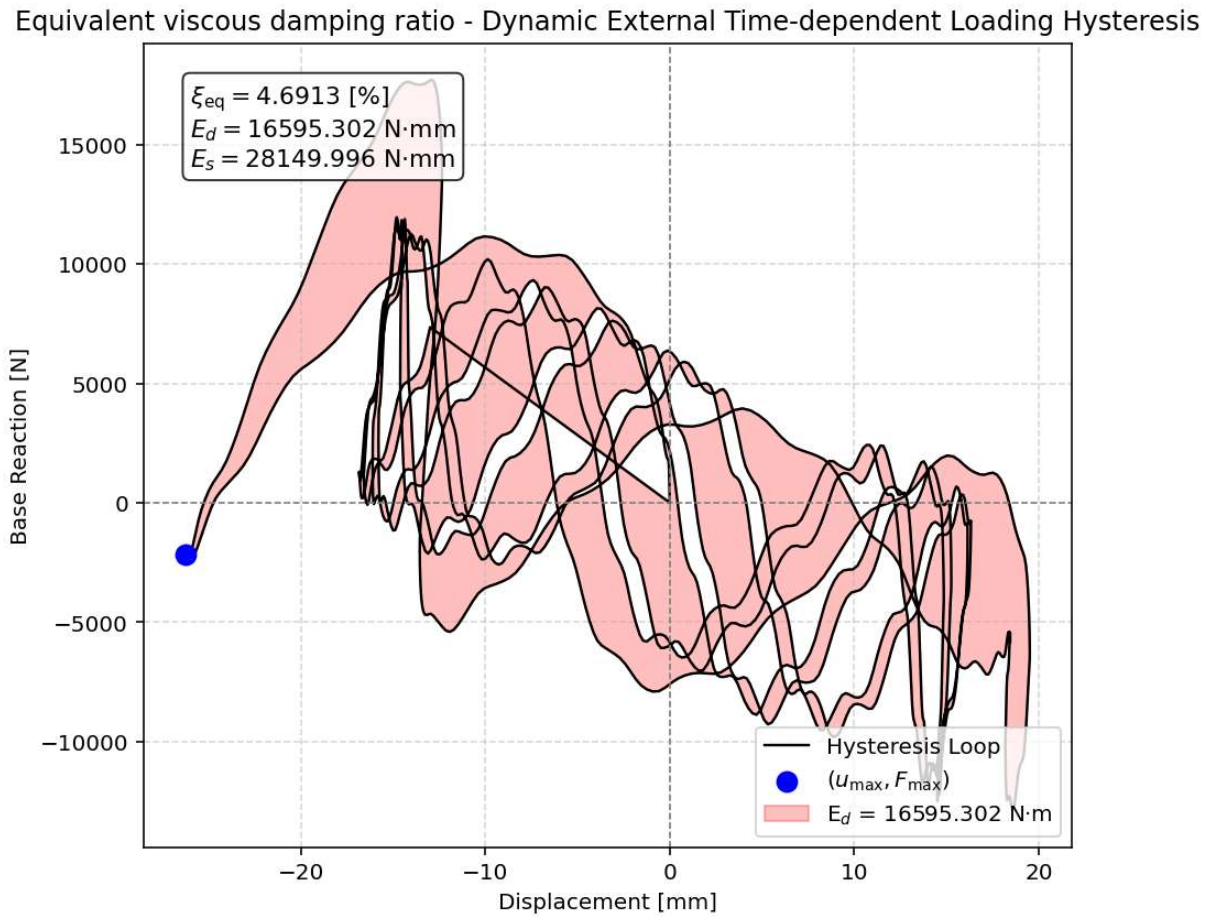

elements force in Z Dir. vs Time for Beam Element During Dynamic External Time-dependent Loading Analysis
1e6
Element Moment Force [N.mm]
Time [s]
Ele. 1 - MAX. ABS. : 7.1367e+06
Ele. 2 - MAX. ABS. : 3.6055e+06
Ele. 3 - MAX. ABS. : 3.7223e+06
Ele. 4 - MAX. ABS. : 7.8121e+06
Ele. 5 - MAX. ABS. : 3.3685e+06
Ele. 6 - MAX. ABS. : 3.6004e+06
Ele. 7 - MAX. ABS. : 5.5185e+05
Ele. 8 - MAX. ABS. : 1.1349e+06
Ele. 9 - MAX. ABS. : 2.6836e+06
Ele. 10 - MAX. ABS. : 4.0038e+06
Ele. 11 - MAX. ABS. : 5.0435e+06
Ele. 12 - MAX. ABS. : 5.8465e+06
Ele. 13 - MAX. ABS. : 5.1574e+05
Ele. 14 - MAX. ABS. : 1.6662e+06
Ele. 15 - MAX. ABS. : 3.4281e+06
Ele. 16 - MAX. ABS. : 4.6791e+06
Ele. 17 - MAX. ABS. : 5.4647e+06
Ele. 18 - MAX. ABS. : 6.3292e+06

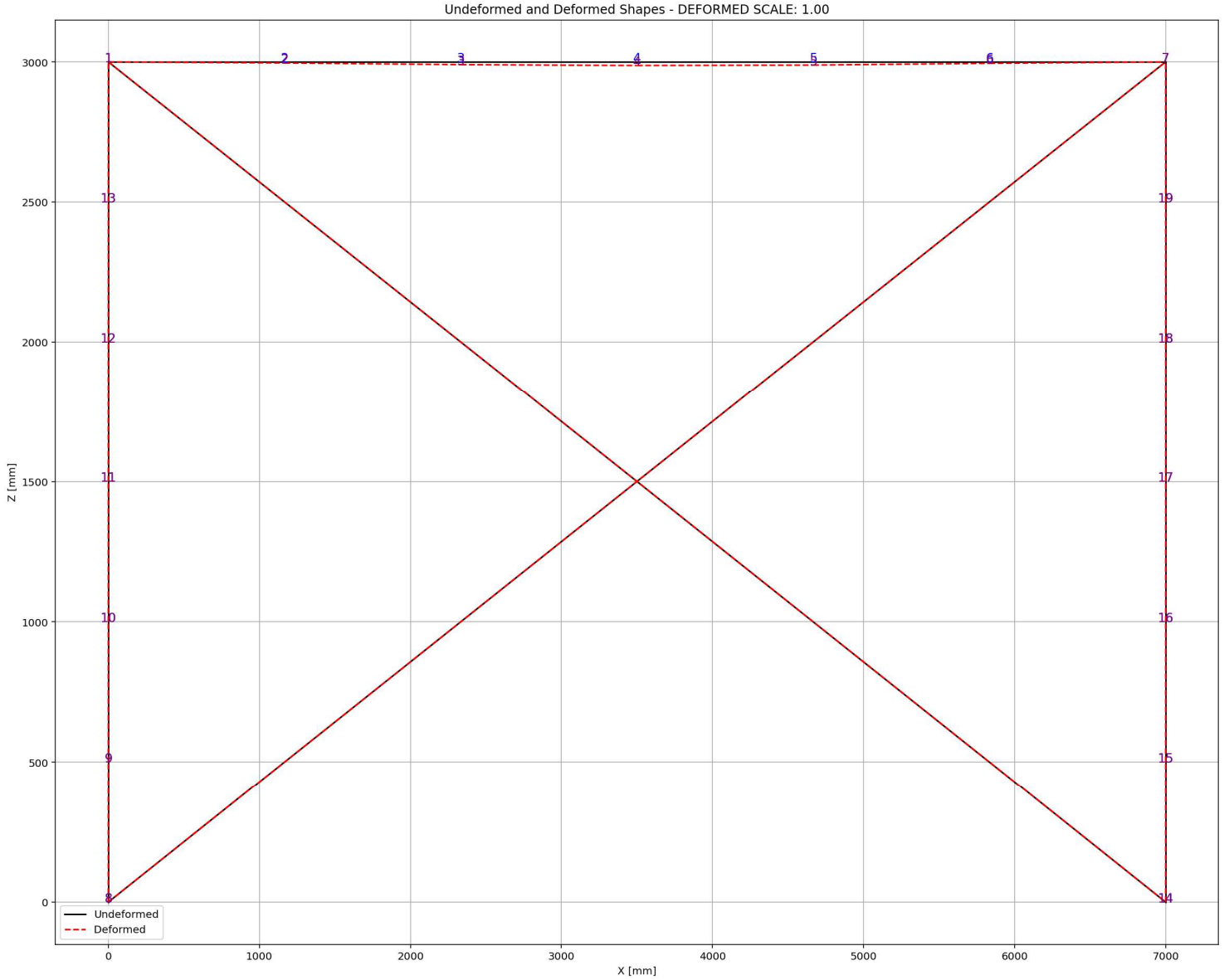

Node Displacements in X Dir. vs Time for Beam Element During Dynamic External Time-dependent Loading Analysis
Displacement [mm]
Time [s]
0.6
0.4
0.2
0.0
−0.2
−0.4
−0.6
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Node 1 - MAX. ABS. : 3.0957e-01
Node 2 - MAX. ABS. : 2.8501e-01
Node 3 - MAX. ABS. : 2.5891e-01
Node 4 - MAX. ABS. : 2.4770e-01
Node 5 - MAX. ABS. : 2.3373e-01
Node 6 - MAX. ABS. : 2.0679e-01
Node 7 - MAX. ABS. : 1.8458e-01
Node 8 - MAX. ABS. : 0.0000e+00
Node 9 - MAX. ABS. : 1.4951e-01
Node 10 - MAX. ABS. : 2.8028e-01
Node 11 - MAX. ABS. : 3.6566e-01
Node 12 - MAX. ABS. : 3.8183e-01
Node 13 - MAX. ABS. : 3.1099e-01
Node 14 - MAX. ABS. : 3.7976e-01
Node 15 - MAX. ABS. : 5.0435e-01
Node 16 - MAX. ABS. : 6.0985e-01
Node 17 - MAX. ABS. : 6.4832e-01
Node 18 - MAX. ABS. : 5.9549e-01
Node 19 - MAX. ABS. : 4.3796e-01


Node Displacements in Y Dir. vs Time for Beam Element During Dynamic External Time-dependent Loading Analysis
Displacement [mm]
Time [s]
20
10
0
−10
−20
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Node 1 - MAX. ABS. : 1.2920e-02
Node 2 - MAX. ABS. : 6.4747e+00
Node 3 - MAX. ABS. : 1.9060e+01
Node 4 - MAX. ABS. : 2.6230e+01
Node 5 - MAX. ABS. : 1.8613e+01
Node 6 - MAX. ABS. : 7.6588e+00
Node 7 - MAX. ABS. : 1.5445e-02
Node 8 - MAX. ABS. : 0.0000e+00
Node 9 - MAX. ABS. : 2.1654e-03
Node 10 - MAX. ABS. : 4.3234e-03
Node 11 - MAX. ABS. : 6.4687e-03
Node 12 - MAX. ABS. : 8.6048e-03
Node 13 - MAX. ABS. : 1.0747e-02
Node 14 - MAX. ABS. : 0.0000e+00
Node 15 - MAX. ABS. : 2.5894e-03
Node 16 - MAX. ABS. : 5.1676e-03
Node 17 - MAX. ABS. : 7.7298e-03
Node 18 - MAX. ABS. : 1.0284e-02
Node 19 - MAX. ABS. : 1.2849e-02


Period of Structure During Dynamic External Time-dependent Loading Analysis
Structural Period [s]
Displacement [mm]
PERIOD - MIN VALUES: Min: 0.880 (s) - Mean: 0.887 (s) - Max: 0.895 (s)
PERIOD - MAX VALUES: Min: 4.542 (s) - Mean: 4.631 (s) - Max: 4.723 (s)
4.5
4.0
3.5
3.0
2.5
2.0
1.5
1.0
−20
−10
0
10
20


Behavior of beam - Stress-strain of Top and Bottom Fibers Curve
BOT FIBER - Stress: 22.397
TOP FIBER - Stress: 4.022
Stress (N/mm^2)
60
40
20
0
−20
−40
−60
−0.0003
−0.0002
−0.0001
0.0000
0.0001
0.0002
0.0003
Strain (mm/mm)


Equivalent viscous damping ratio - Dynamic External Time-dependent Loading Hysteresis
$\xi_{eq} = 4.6913$ [%]
$E_d = 16595.302$ N·mm
$E_s = 28149.996$ N·mm
Base Reaction [N]
Displacement [mm]
15000
10000
5000
0
−5000
−10000
−20
−10
0
10
20
Hysteresis Loop
(u_{max}, F_{max})
E_d = 16595.302 N·m


Undeformed and Deformed Shapes - DEFORMED SCALE: 1.00
Z [mm]
X [mm]
Undeformed
Deformed
0
500
1000
1500
2000
2500
3000
0
1000
2000
3000
4000
5000
6000
7000
1
2
3
4
5
6
7
8
9
10
11
12
13
14
15
16
17
18
19

FREE-VIBRATION ANALYSIS

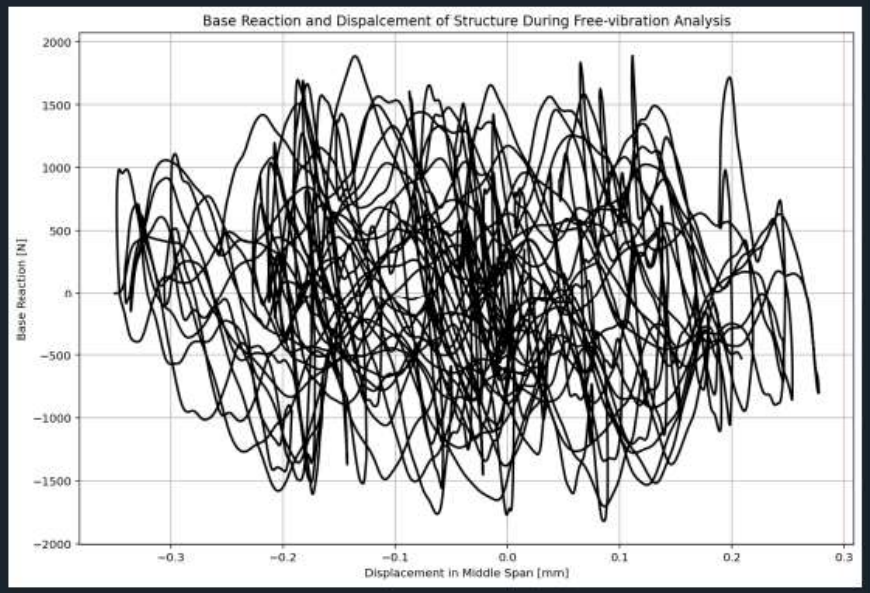
Spyder (Python 3.12)

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C:\Users\Dell\Desktop\OPENSEES_FILES\+EIGHT_ANALY...L_BRACED_FRAME_8_ANA\W(x,t)_STEEL_BRACED_FRAME.py

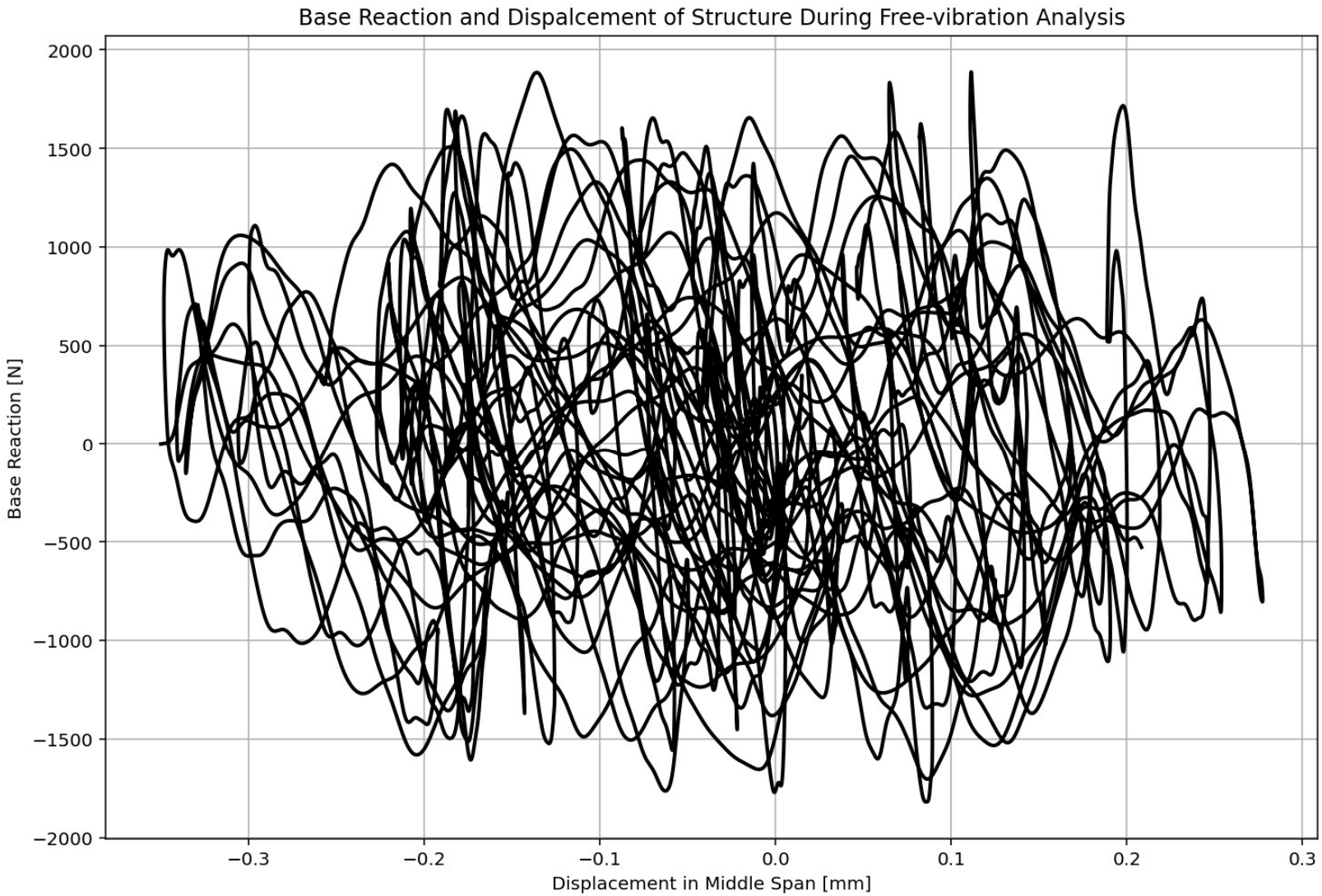
W(x,t)_STEEL_BRACED_FRAME.py | PLATE_DOUBLE_I_STE...EIGHT_FUN_EXTRA.py

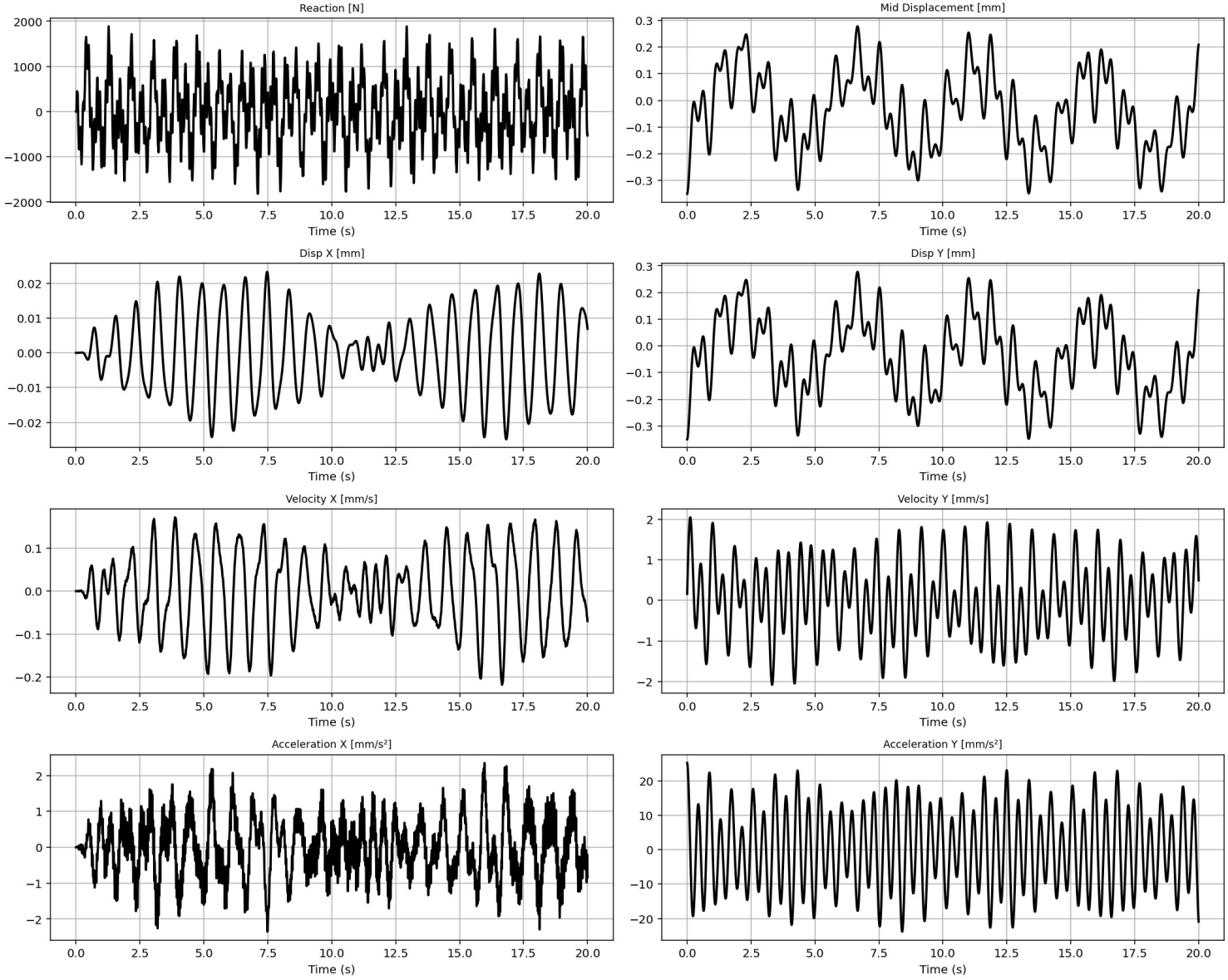
```
1441  # FREE-VIBRATION ANALYSIS (DYNAMIC TIME-HISTORY ANALYSIS)
1442  #ELE_TYPE = 'elasticBeamColumn'      # MATERIAL LINEARITY
1443  ELE_TYPE = 'nonlinearBeamColumn'     # MATERIAL AND GEOMETRIC NONLINEARITIE
1444  MAT_TYPE = 'INELASTIC'               # 'ELASTIC' OR 'INELASTIC'
1445  ANAL_TYPE = 'FREE-VIBRATION'
1446  BRACE = True                         # True: WITH BRACE - False: WITHOUT BR
1447
1448  DATA = STEEL_BRACED_FRAME(LENGTH, HEIGHT, MAT_TYPE, ELE_TYPE, ANAL_TYPE, TO
1449
1450  (time_FV, reactioXn_FV, reactionY_FV, disp_mid_FV,
1451  ele_axialforce_FV, ele_shearforce_FV, ele_momentforce_FV,
1452  node_displacementsX_FV, node_displacementsY_FV,
1453  dispX_FV, dispY_FV,
1454  veloX_FV, veloY_FV,
1455  accX_FV, accY_FV,
1456  stiffness_FV, PERIOD_FV, damping_ratio_FV,
1457  PERIOD_MIN_FV, PERIOD_MAX_FV,
1458  STRESSb_FV, STRAINb_FV, STRESSt_FV, STRAINt_FV, CURVATURE_FV) = DATA
1459
1460  XDATA = disp_mid_FV
1461  YDATA = reactionY_FV
1462  XLABEL = 'Displacement in Middle Span [mm]'
1463  YLABEL = 'Base Reaction [N]'
1464  TITLE = 'Base Reaction and Dispalcement of Structure During Free-vibration
1465  COLOR = 'black'
1466  SEMILOGY = False
1467  PLOT(XDATA, YDATA, TITLE, XLABEL, YLABEL, COLOR, SEMILOGY)
1468
1469  PLOT_TIME_HISTORY(time_FV, reactionY_FV, disp_mid_FV,
1470                    dispX_FV, dispY_FV,
1471                    veloX_FV, veloY_FV,
1472                    accX_FV, accY_FV)
1473
1474  # PLOT ELEMENTS AXIAL FORCE
```

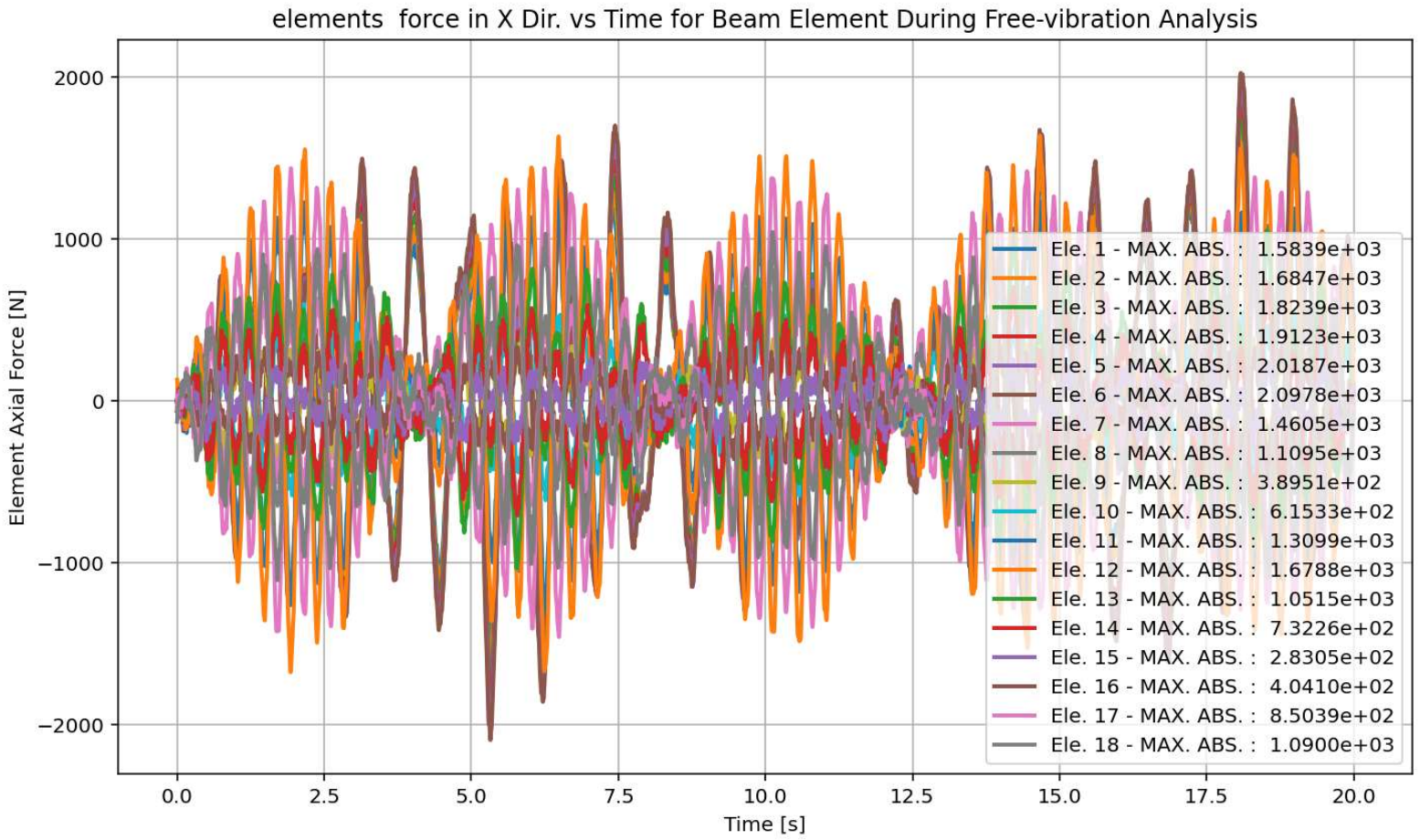


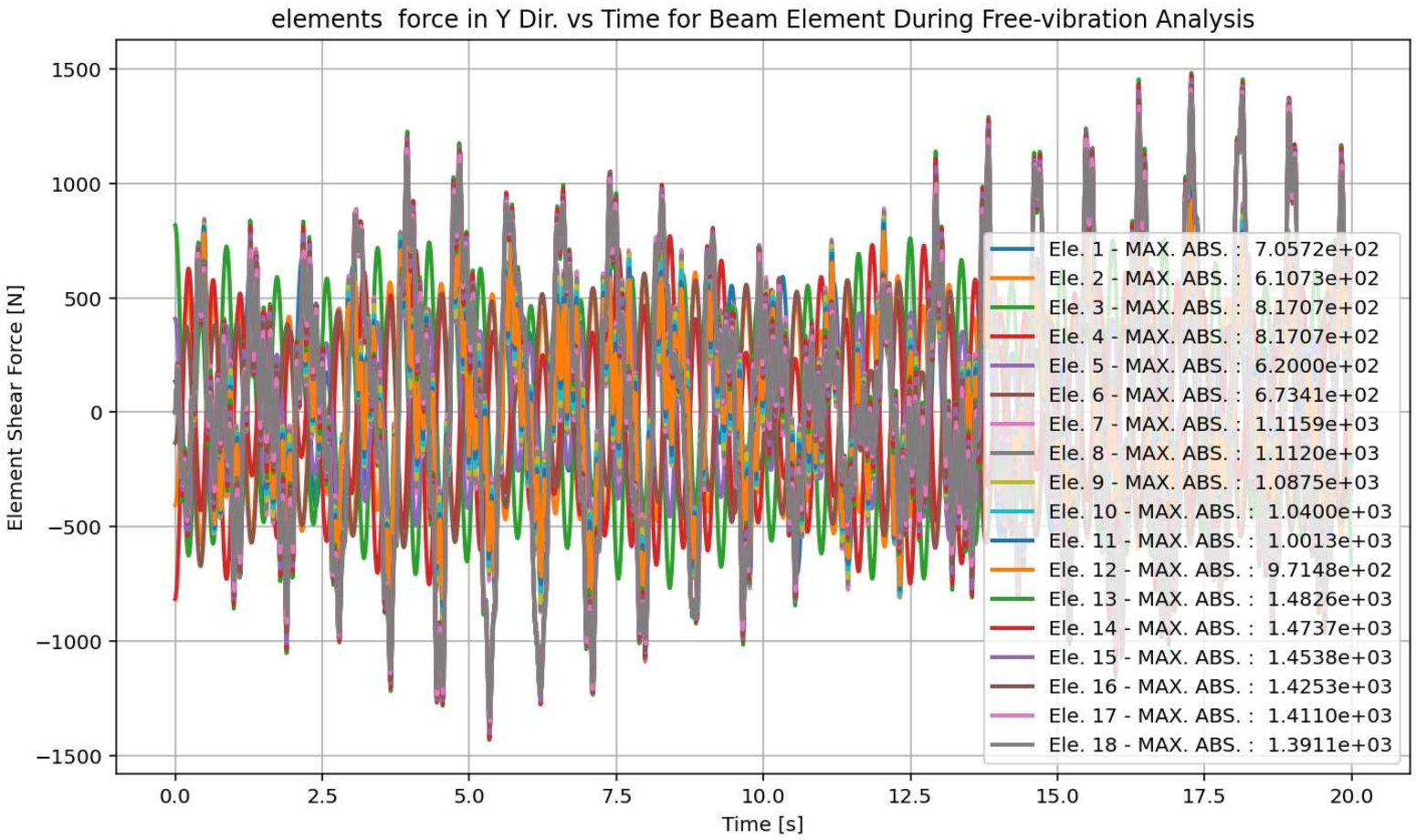
IPython Console | Files | Help | Variable Explorer | Debugger | Plots | History

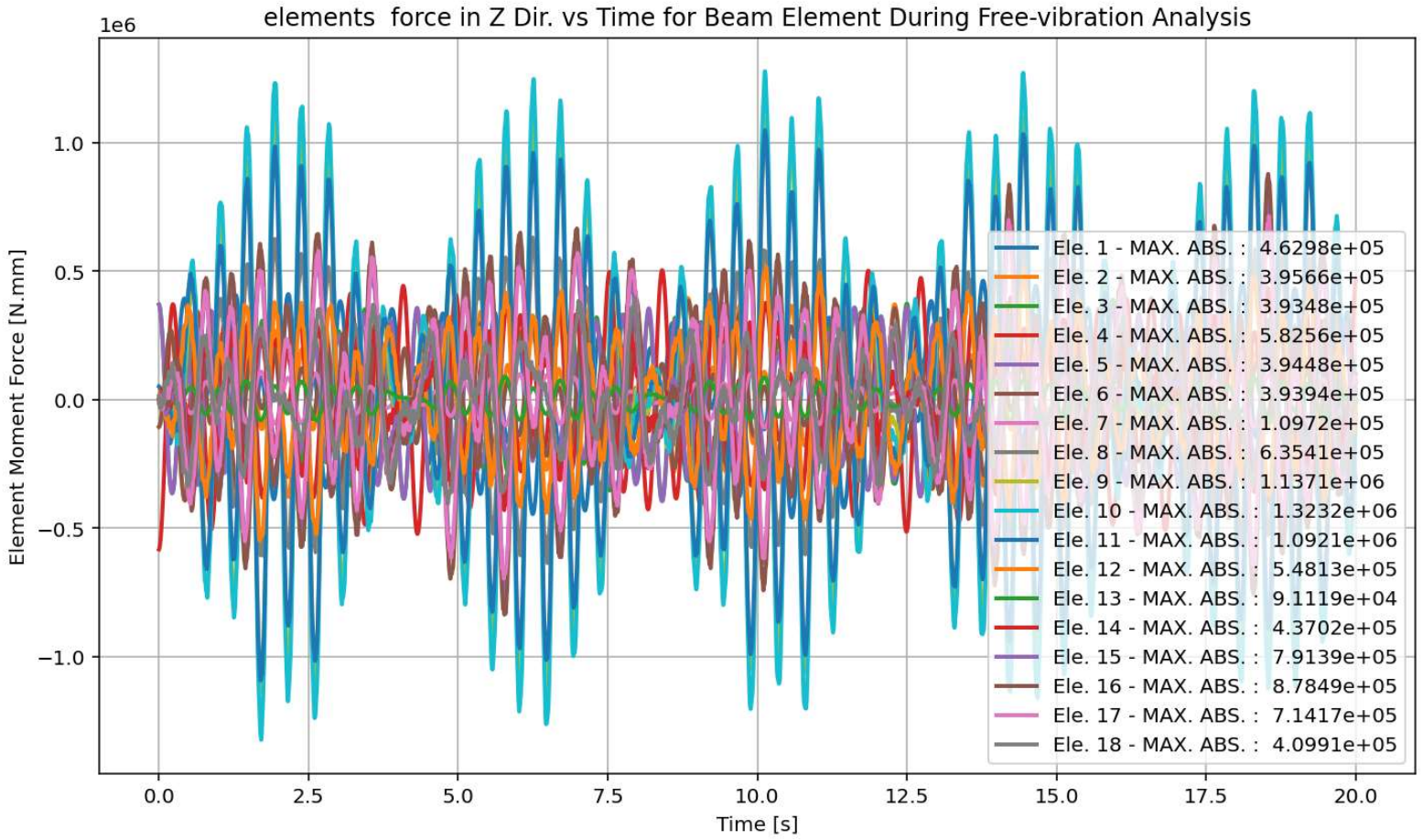
Inline Conda: anaconda3 (Python 3.12.7) LSP: Python Line 12, Col 105 UTF-8 CRLF RW Mem 42%

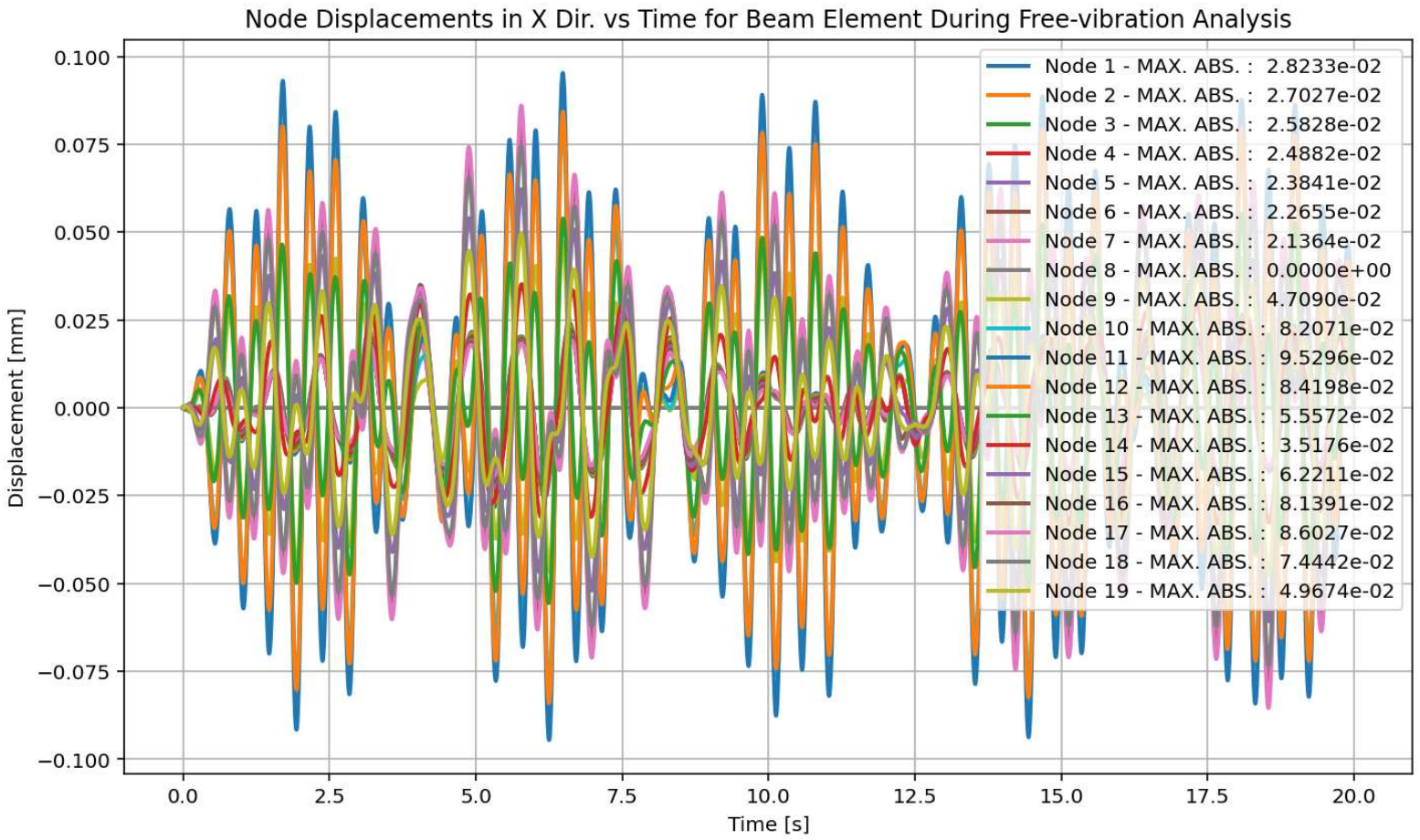

Base Reaction and Dispalcement of Structure During Free-vibration Analysis
Base Reaction [N]
2000
1500
1000
500
0
−500
−1000
−1500
−2000
−0.3
−0.2
−0.1
0.0
0.1
0.2
0.3
Displacement in Middle Span [mm]

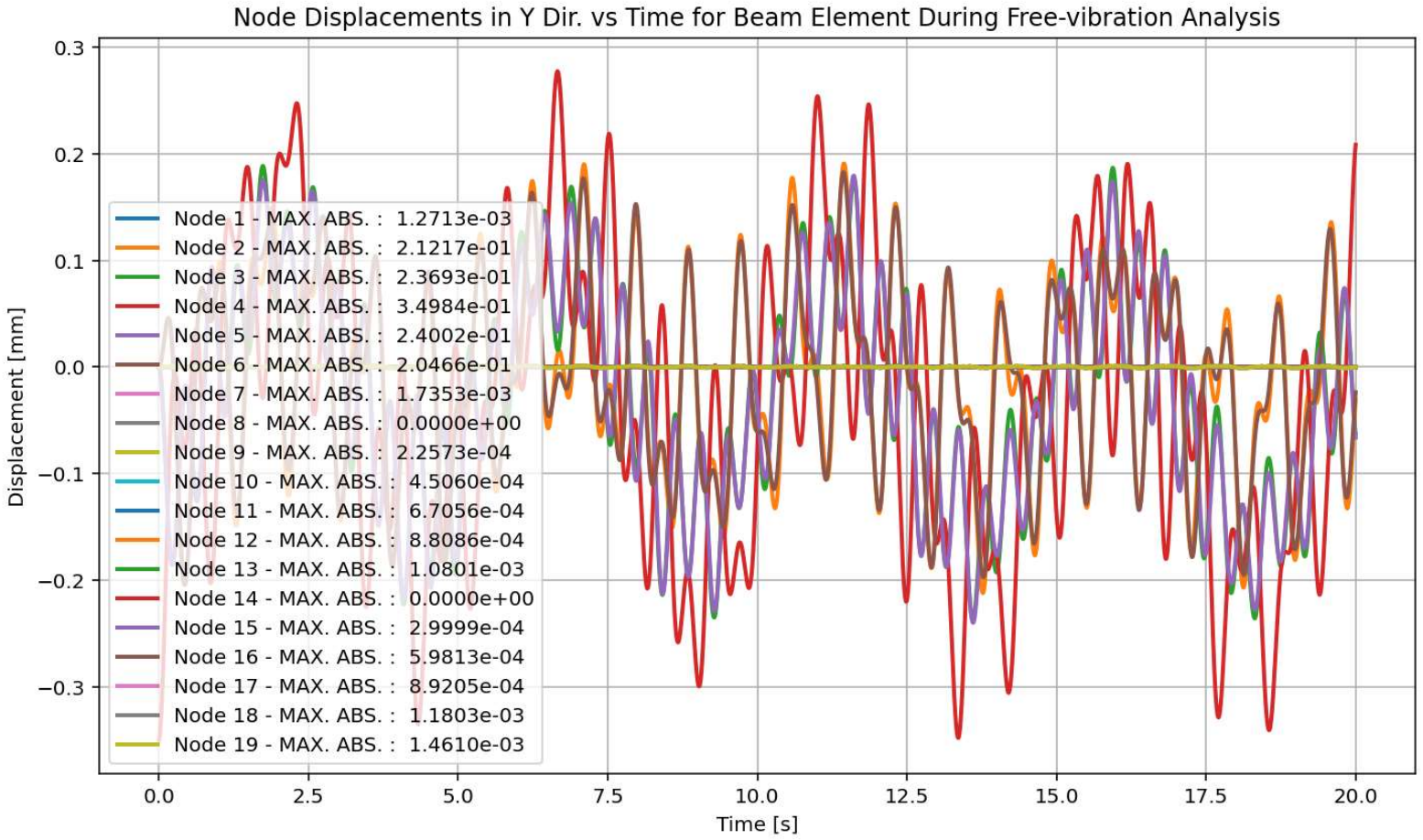

Reaction [N]
Mid Displacement [mm]
Disp X [mm]
Disp Y [mm]
Velocity X [mm/s]
Velocity Y [mm/s]
Acceleration X [mm/s²]
Acceleration Y [mm/s²]
Time (s)

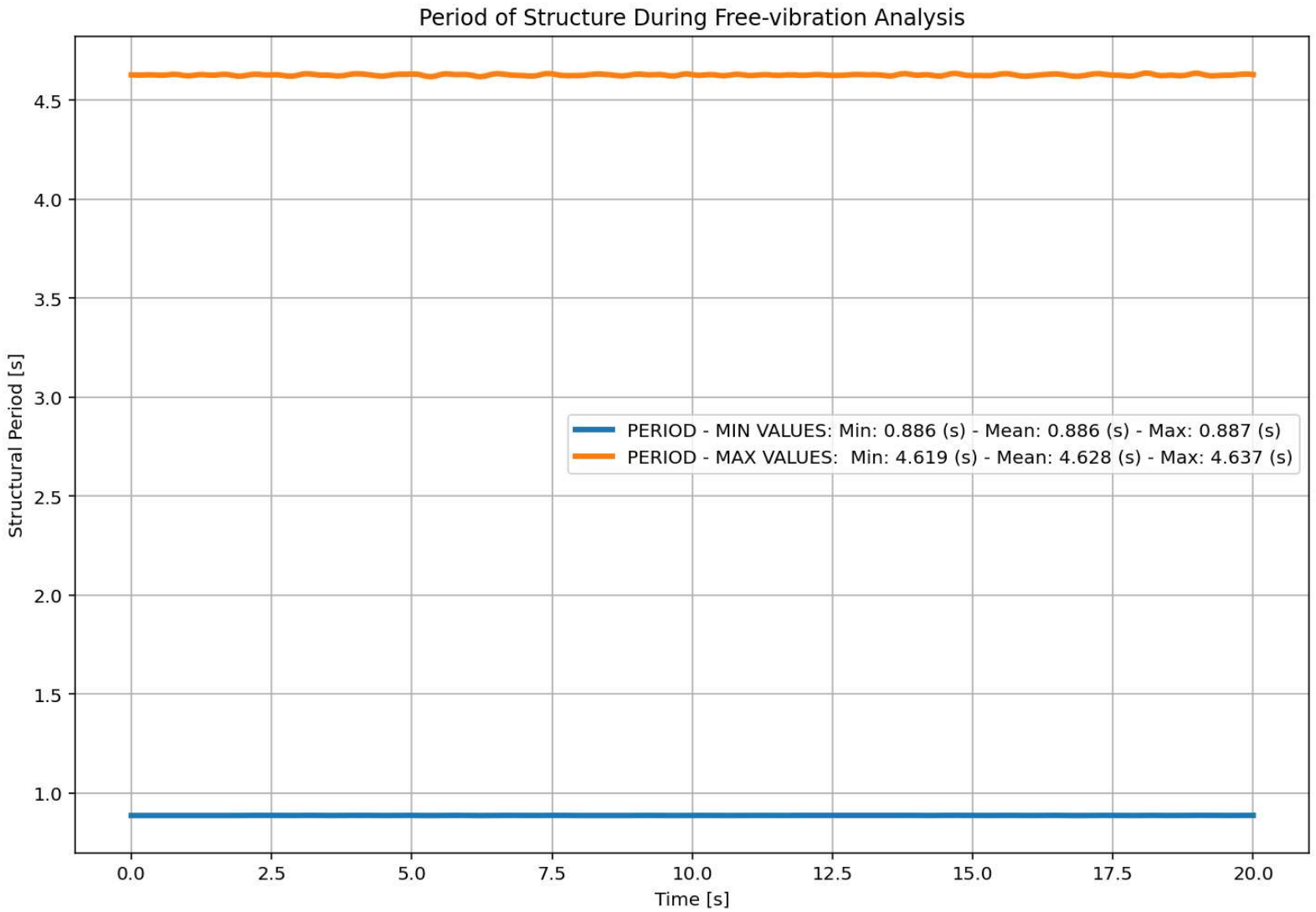

elements force in X Dir. vs Time for Beam Element During Free-vibration Analysis
Element Axial Force [N]
Time [s]
2000
1000
0
−1000
−2000
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Ele. 1 - MAX. ABS. : 1.5839e+03
Ele. 2 - MAX. ABS. : 1.6847e+03
Ele. 3 - MAX. ABS. : 1.8239e+03
Ele. 4 - MAX. ABS. : 1.9123e+03
Ele. 5 - MAX. ABS. : 2.0187e+03
Ele. 6 - MAX. ABS. : 2.0978e+03
Ele. 7 - MAX. ABS. : 1.4605e+03
Ele. 8 - MAX. ABS. : 1.1095e+03
Ele. 9 - MAX. ABS. : 3.8951e+02
Ele. 10 - MAX. ABS. : 6.1533e+02
Ele. 11 - MAX. ABS. : 1.3099e+03
Ele. 12 - MAX. ABS. : 1.6788e+03
Ele. 13 - MAX. ABS. : 1.0515e+03
Ele. 14 - MAX. ABS. : 7.3226e+02
Ele. 15 - MAX. ABS. : 2.8305e+02
Ele. 16 - MAX. ABS. : 4.0410e+02
Ele. 17 - MAX. ABS. : 8.5039e+02
Ele. 18 - MAX. ABS. : 1.0900e+03

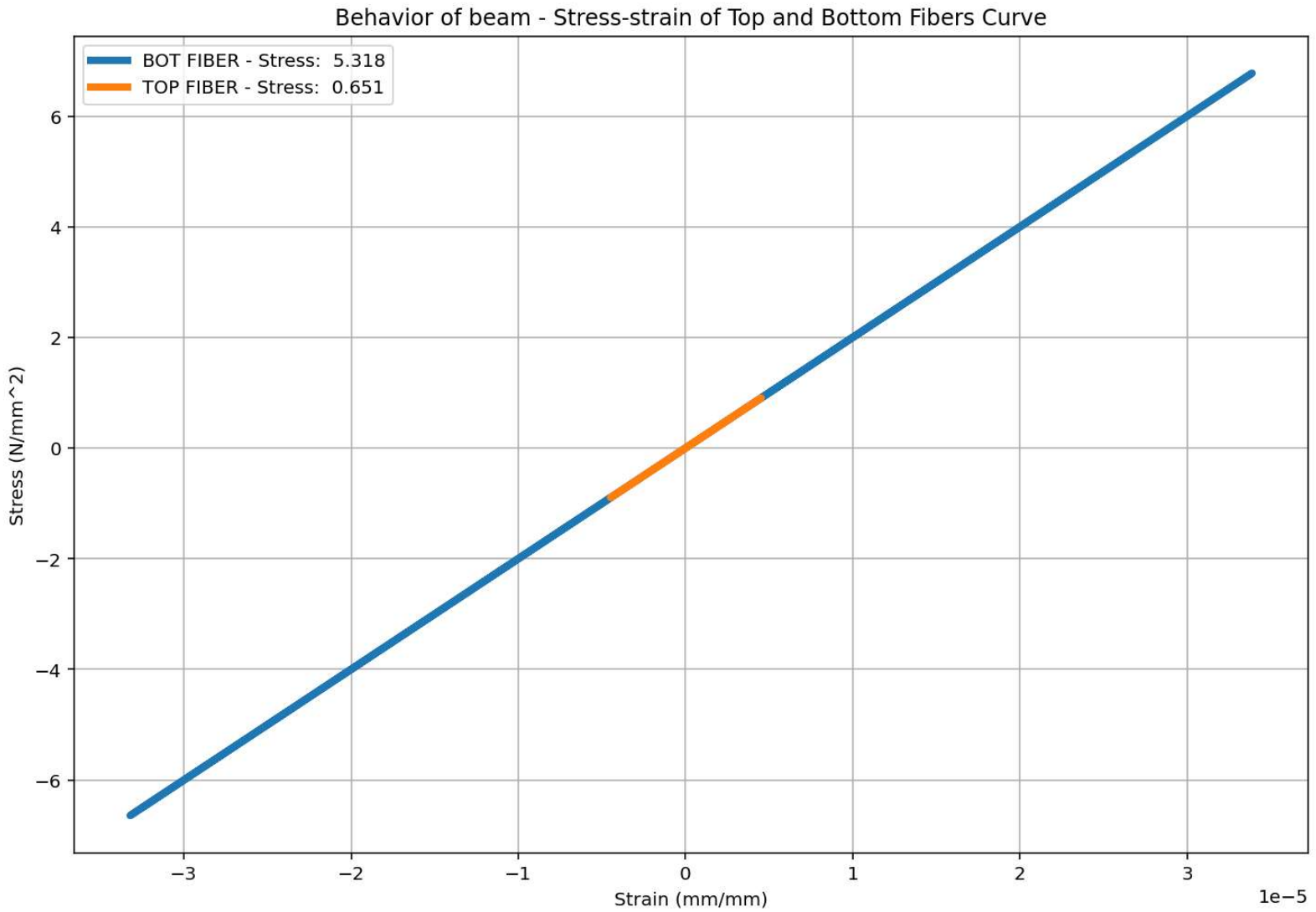

elements force in Y Dir. vs Time for Beam Element During Free-vibration Analysis
Element Shear Force [N]
Time [s]
1500
1000
500
0
−500
−1000
−1500
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Ele. 1 - MAX. ABS. : 7.0572e+02
Ele. 2 - MAX. ABS. : 6.1073e+02
Ele. 3 - MAX. ABS. : 8.1707e+02
Ele. 4 - MAX. ABS. : 8.1707e+02
Ele. 5 - MAX. ABS. : 6.2000e+02
Ele. 6 - MAX. ABS. : 6.7341e+02
Ele. 7 - MAX. ABS. : 1.1159e+03
Ele. 8 - MAX. ABS. : 1.1120e+03
Ele. 9 - MAX. ABS. : 1.0875e+03
Ele. 10 - MAX. ABS. : 1.0400e+03
Ele. 11 - MAX. ABS. : 1.0013e+03
Ele. 12 - MAX. ABS. : 9.7148e+02
Ele. 13 - MAX. ABS. : 1.4826e+03
Ele. 14 - MAX. ABS. : 1.4737e+03
Ele. 15 - MAX. ABS. : 1.4538e+03
Ele. 16 - MAX. ABS. : 1.4253e+03
Ele. 17 - MAX. ABS. : 1.4110e+03
Ele. 18 - MAX. ABS. : 1.3911e+03

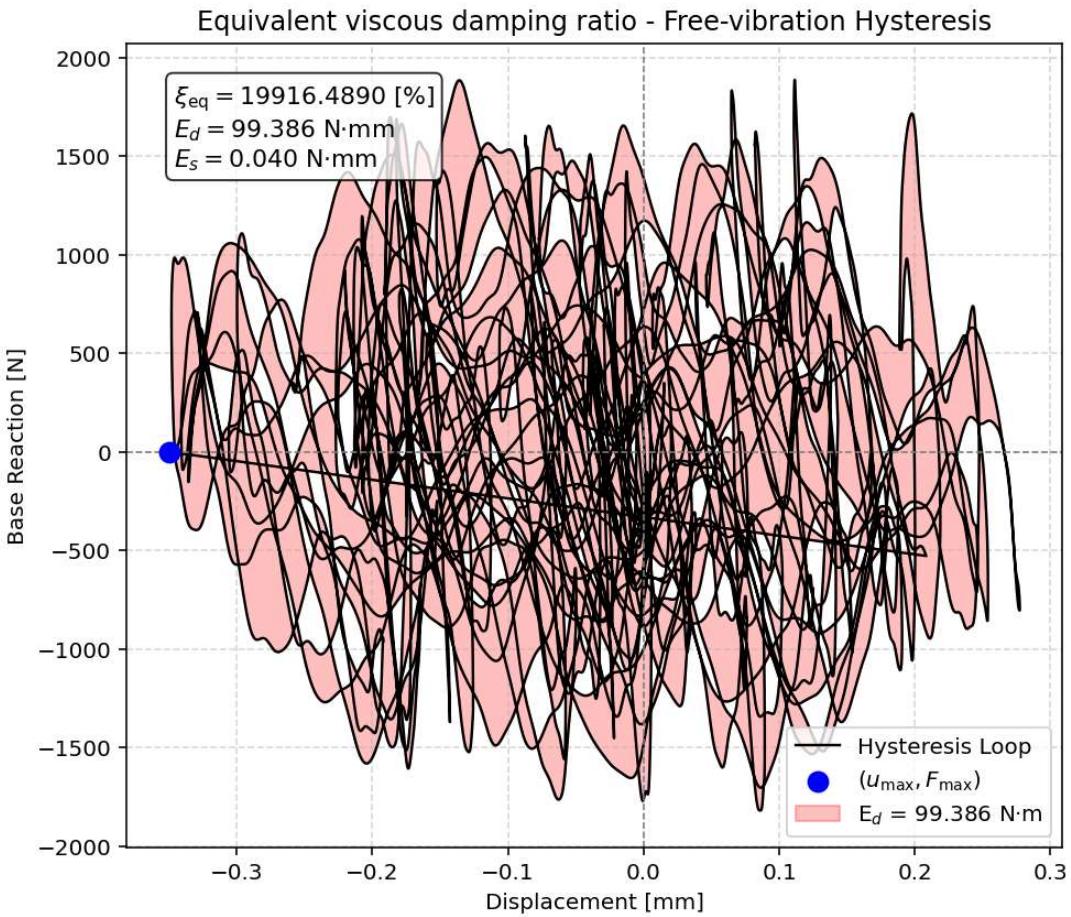

elements force in Z Dir. vs Time for Beam Element During Free-vibration Analysis
1e6
1.0
0.5
0.0
−0.5
−1.0
Element Moment Force [N.mm]
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Time [s]
Ele. 1 - MAX. ABS. : 4.6298e+05
Ele. 2 - MAX. ABS. : 3.9566e+05
Ele. 3 - MAX. ABS. : 3.9348e+05
Ele. 4 - MAX. ABS. : 5.8256e+05
Ele. 5 - MAX. ABS. : 3.9448e+05
Ele. 6 - MAX. ABS. : 3.9394e+05
Ele. 7 - MAX. ABS. : 1.0972e+05
Ele. 8 - MAX. ABS. : 6.3541e+05
Ele. 9 - MAX. ABS. : 1.1371e+06
Ele. 10 - MAX. ABS. : 1.3232e+06
Ele. 11 - MAX. ABS. : 1.0921e+06
Ele. 12 - MAX. ABS. : 5.4813e+05
Ele. 13 - MAX. ABS. : 9.1119e+04
Ele. 14 - MAX. ABS. : 4.3702e+05
Ele. 15 - MAX. ABS. : 7.9139e+05
Ele. 16 - MAX. ABS. : 8.7849e+05
Ele. 17 - MAX. ABS. : 7.1417e+05
Ele. 18 - MAX. ABS. : 4.0991e+05

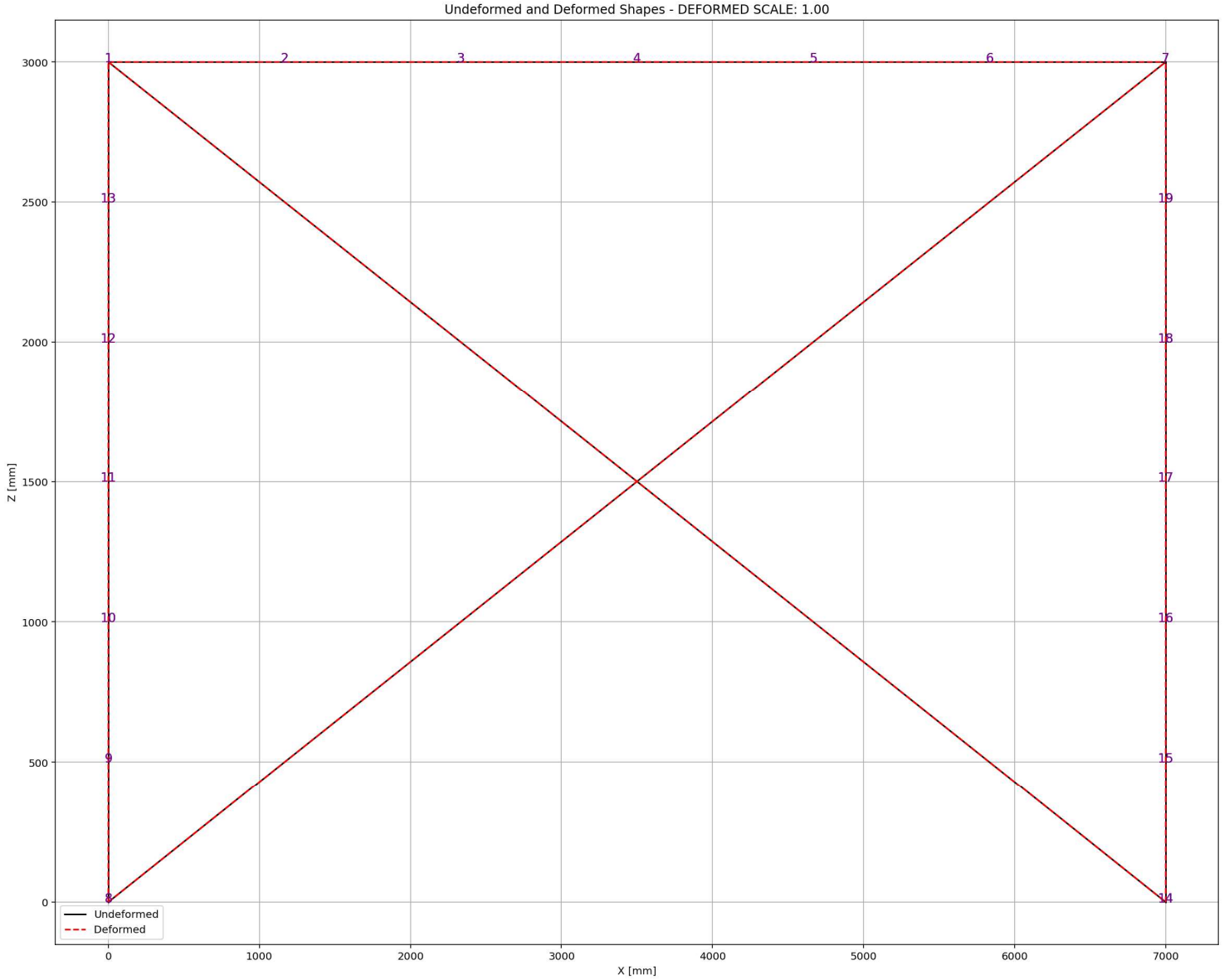

Node Displacements in X Dir. vs Time for Beam Element During Free-vibration Analysis
Displacement [mm]
Time [s]
0.100
0.075
0.050
0.025
0.000
−0.025
−0.050
−0.075
−0.100
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Node 1 - MAX. ABS. : 2.8233e-02
Node 2 - MAX. ABS. : 2.7027e-02
Node 3 - MAX. ABS. : 2.5828e-02
Node 4 - MAX. ABS. : 2.4882e-02
Node 5 - MAX. ABS. : 2.3841e-02
Node 6 - MAX. ABS. : 2.2655e-02
Node 7 - MAX. ABS. : 2.1364e-02
Node 8 - MAX. ABS. : 0.0000e+00
Node 9 - MAX. ABS. : 4.7090e-02
Node 10 - MAX. ABS. : 8.2071e-02
Node 11 - MAX. ABS. : 9.5296e-02
Node 12 - MAX. ABS. : 8.4198e-02
Node 13 - MAX. ABS. : 5.5572e-02
Node 14 - MAX. ABS. : 3.5176e-02
Node 15 - MAX. ABS. : 6.2211e-02
Node 16 - MAX. ABS. : 8.1391e-02
Node 17 - MAX. ABS. : 8.6027e-02
Node 18 - MAX. ABS. : 7.4442e-02
Node 19 - MAX. ABS. : 4.9674e-02


Node Displacements in Y Dir. vs Time for Beam Element During Free-vibration Analysis
Displacement [mm]
Time [s]
0.3
0.2
0.1
0.0
−0.1
−0.2
−0.3
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Node 1 - MAX. ABS. : 1.2713e-03
Node 2 - MAX. ABS. : 2.1217e-01
Node 3 - MAX. ABS. : 2.3693e-01
Node 4 - MAX. ABS. : 3.4984e-01
Node 5 - MAX. ABS. : 2.4002e-01
Node 6 - MAX. ABS. : 2.0466e-01
Node 7 - MAX. ABS. : 1.7353e-03
Node 8 - MAX. ABS. : 0.0000e+00
Node 9 - MAX. ABS. : 2.2573e-04
Node 10 - MAX. ABS. : 4.5060e-04
Node 11 - MAX. ABS. : 6.7056e-04
Node 12 - MAX. ABS. : 8.8086e-04
Node 13 - MAX. ABS. : 1.0801e-03
Node 14 - MAX. ABS. : 0.0000e+00
Node 15 - MAX. ABS. : 2.9999e-04
Node 16 - MAX. ABS. : 5.9813e-04
Node 17 - MAX. ABS. : 8.9205e-04
Node 18 - MAX. ABS. : 1.1803e-03
Node 19 - MAX. ABS. : 1.4610e-03


Period of Structure During Free-vibration Analysis
Structural Period [s]
Time [s]
PERIOD - MIN VALUES: Min: 0.886 (s) - Mean: 0.886 (s) - Max: 0.887 (s)
PERIOD - MAX VALUES: Min: 4.619 (s) - Mean: 4.628 (s) - Max: 4.637 (s)
4.5
4.0
3.5
3.0
2.5
2.0
1.5
1.0
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0


Behavior of beam - Stress-strain of Top and Bottom Fibers Curve
BOT FIBER - Stress: 5.318
TOP FIBER - Stress: 0.651
Stress (N/mm^2)
6
4
2
0
−2
−4
−6
−3
−2
−1
0
1
2
3
Strain (mm/mm)
1e−5


Equivalent viscous damping ratio - Free-vibration Hysteresis
$\xi_{eq} = 19916.4890$ [%]
$E_d = 99.386$ N·mm
$E_s = 0.040$ N·mm
Base Reaction [N]
Displacement [mm]
2000
1500
1000
500
0
−500
−1000
−1500
−2000
−0.3
−0.2
−0.1
0.0
0.1
0.2
0.3
Hysteresis Loop
(u_{max}, F_{max})
E_d = 99.386 N·m


Undeformed and Deformed Shapes - DEFORMED SCALE: 1.00
Z [mm]
X [mm]
0
500
1000
1500
2000
2500
3000
0
1000
2000
3000
4000
5000
6000
7000
1
2
3
4
5
6
7
8
9
10
11
12
13
14
15
16
17
18
19
Undeformed
Deformed

SEISMIC ANALYSIS

Spyder (Python 3.12)

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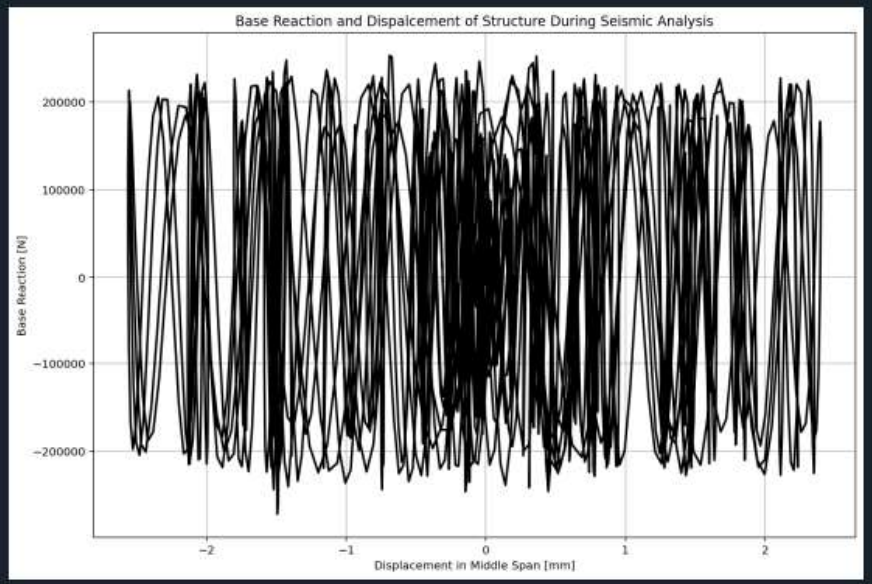
C:\Users\Dell\Desktop\OPENSEES_FILES\+EIGHT_ANALY...L_BRACED_FRAME_8_ANA\W(x,t)_STEEL_BRACED_FRAME.py

W(x,t)_STEEL_BRACED_FRAME.py | PLATE_DOUBLE_I_STE...EIGHT_FUN_EXTRA.py

```
1518  # SEISMIC ANALYSIS (DYNAMIC TIME-HISTORY ANALYSIS)
1519  #ELE_TYPE = 'elasticBeamColumn'      # MATERIAL LINEARITY
1520  ELE_TYPE = 'nonlinearBeamColumn'     # MATERIAL AND GEOMETRIC NONLINEARITIE
1521  MAT_TYPE = 'INELASTIC'   # 'ELASTIC' OR 'INELASTIC'
1522  ANAL_TYPE = 'SEISMIC'
1523  BRACE = True                         # True: WITH BRACE - False: WITHOUT BR
1524
1525  DATA = STEEL_BRACED_FRAME(LENGTH, HEIGHT, MAT_TYPE, ELE_TYPE, ANAL_TYPE, TO
1526
1527  (time_SEI, reactionX_SEI, reactionY_SEI, disp_mid_SEI,
1528   ele_axialforce_SEI, ele_shearforce_SEI, ele_momentforce_SEI,
1529   node_displacementsX_SEI, node_displacementsY_SEI,
1530   dispX_SEI, dispY_SEI,
1531   veloX_SEI, veloY_SEI,
1532   accX_SEI, accY_SEI,
1533   stiffness_SEI, PERIOD_SEI, damping_ratio_SEI,
1534   PERIOD_MIN_SEI, PERIOD_MAX_SEI,
1535   STRESSb_SEI, STRAINb_SEI, STRESSt_SEI, STRAINt_SEI, CURVATURE_SEI) = DATA
1536
1537
1538  XDATA = disp_mid_SEI
1539  YDATA = reactionY_SEI
1540  XLABEL = 'Displacement in Middle Span [mm]'
1541  YLABEL = 'Base Reaction [N]'
1542  TITLE = 'Base Reaction and Dispalcement of Structure During Seismic Analysi
1543  COLOR = 'black'
1544  SEMILOGY = False
1545  PLOT(XDATA, YDATA, TITLE, XLABEL, YLABEL, COLOR, SEMILOGY)
1546
1547  PLOT_TIME_HISTORY(time_SEI, reactionY_SEI, disp_mid_SEI,
1548                    dispX_SEI, dispY_SEI,
1549                    veloX_SEI, veloY_SEI,
1550                    accX_SEI, accY_SEI)
1551
```

...RGY_DISSIPATION_CAPACITY_INDEX\08_STEEL_BRACED_FRAME_8_ANA

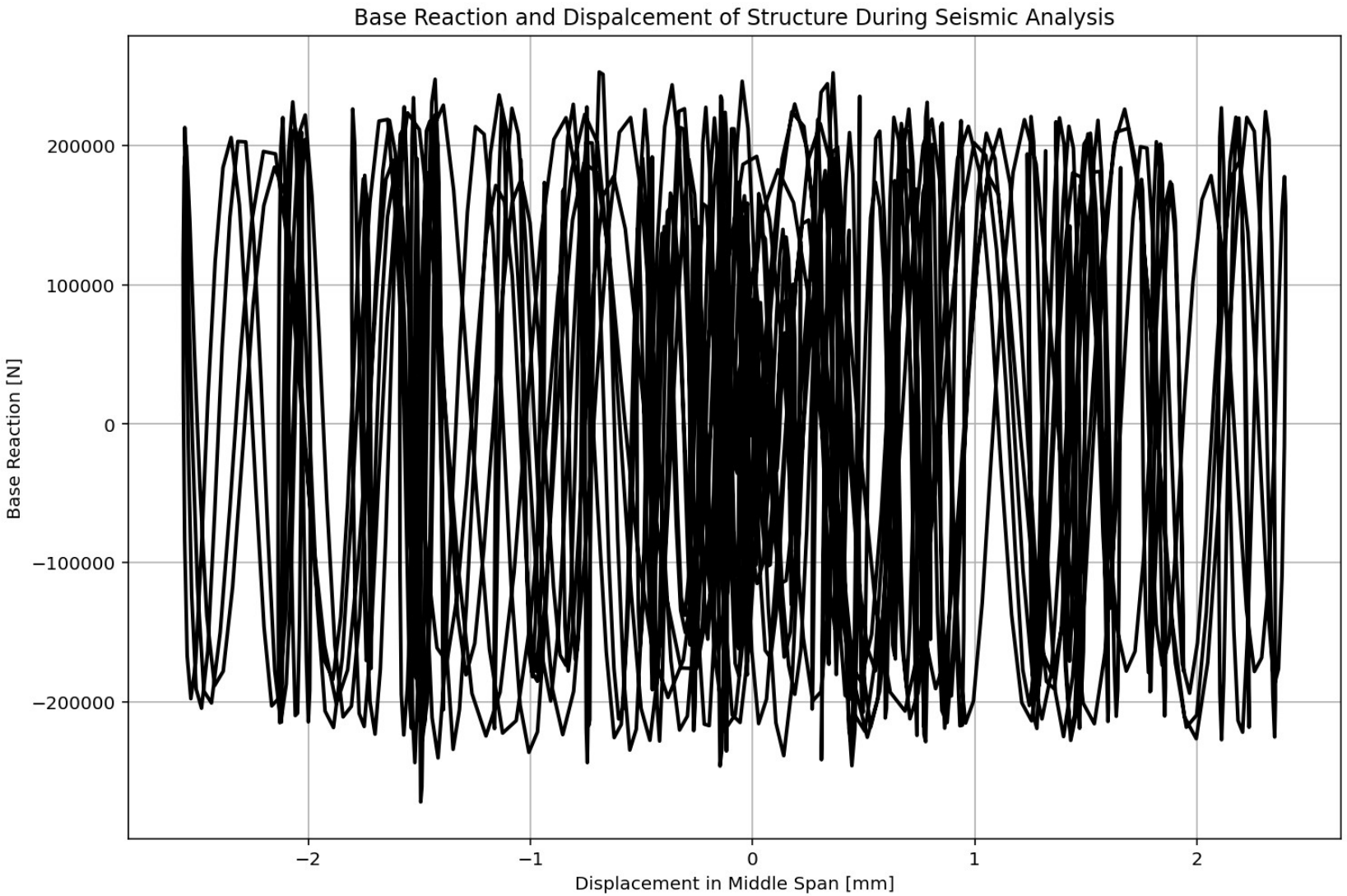
37 %

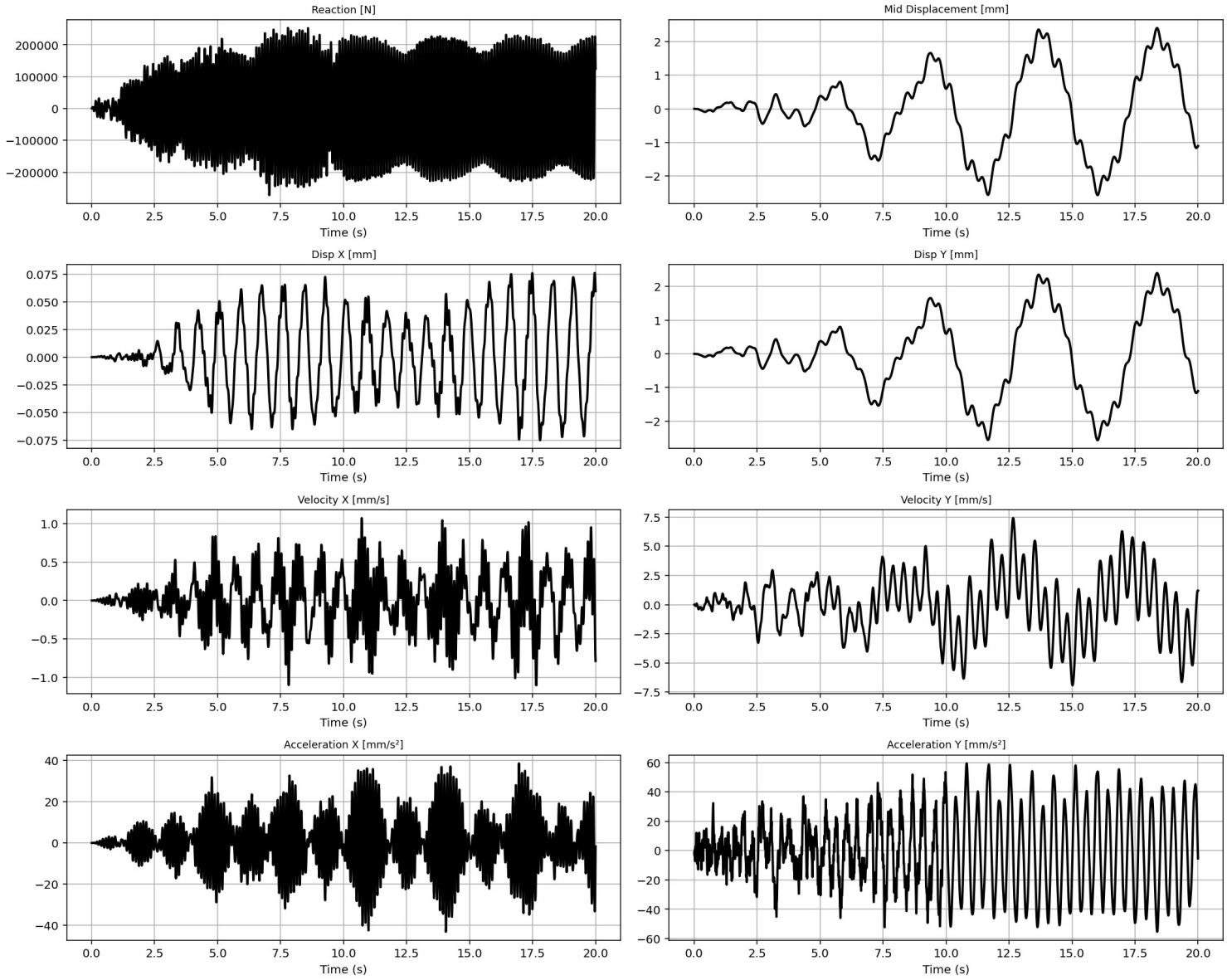


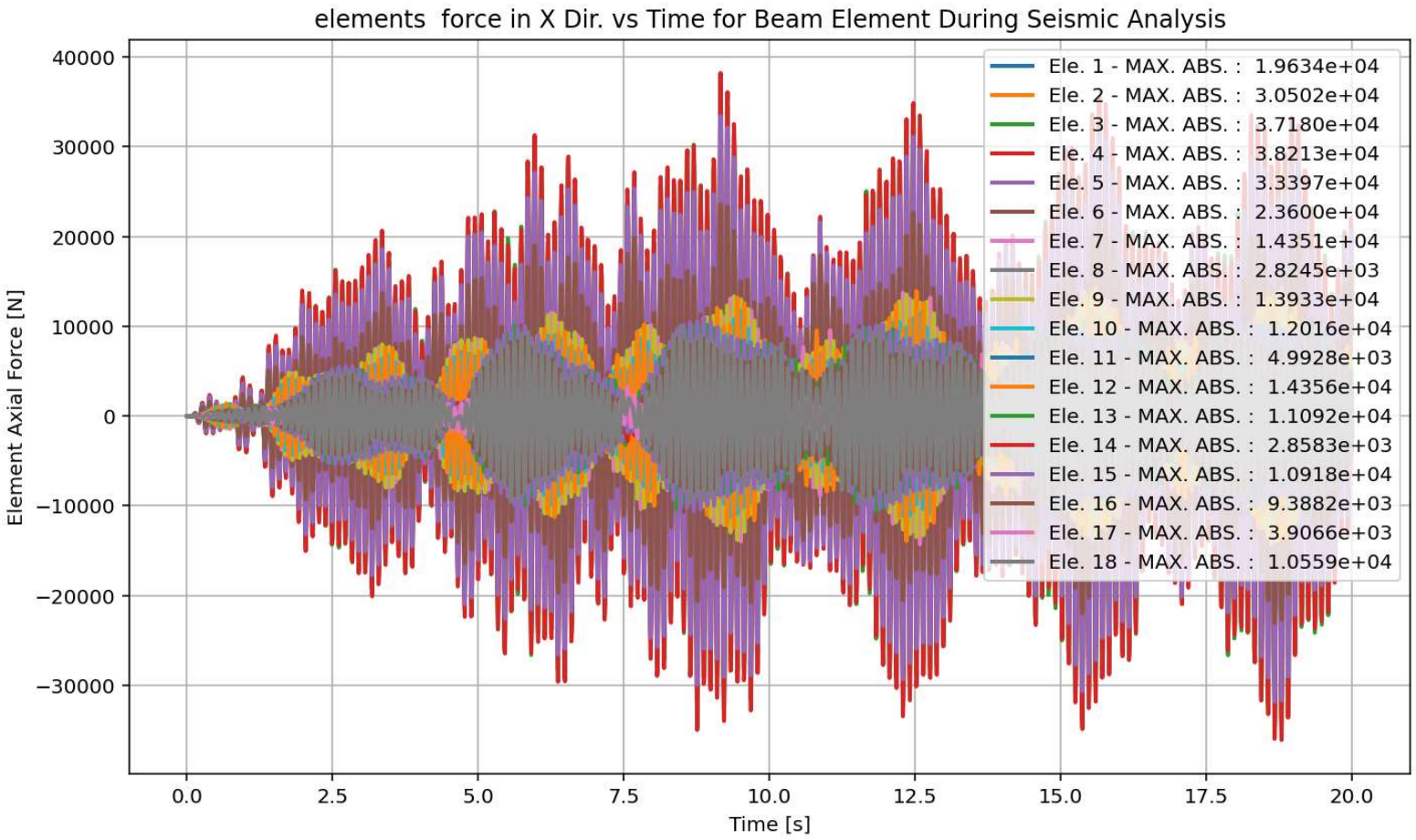


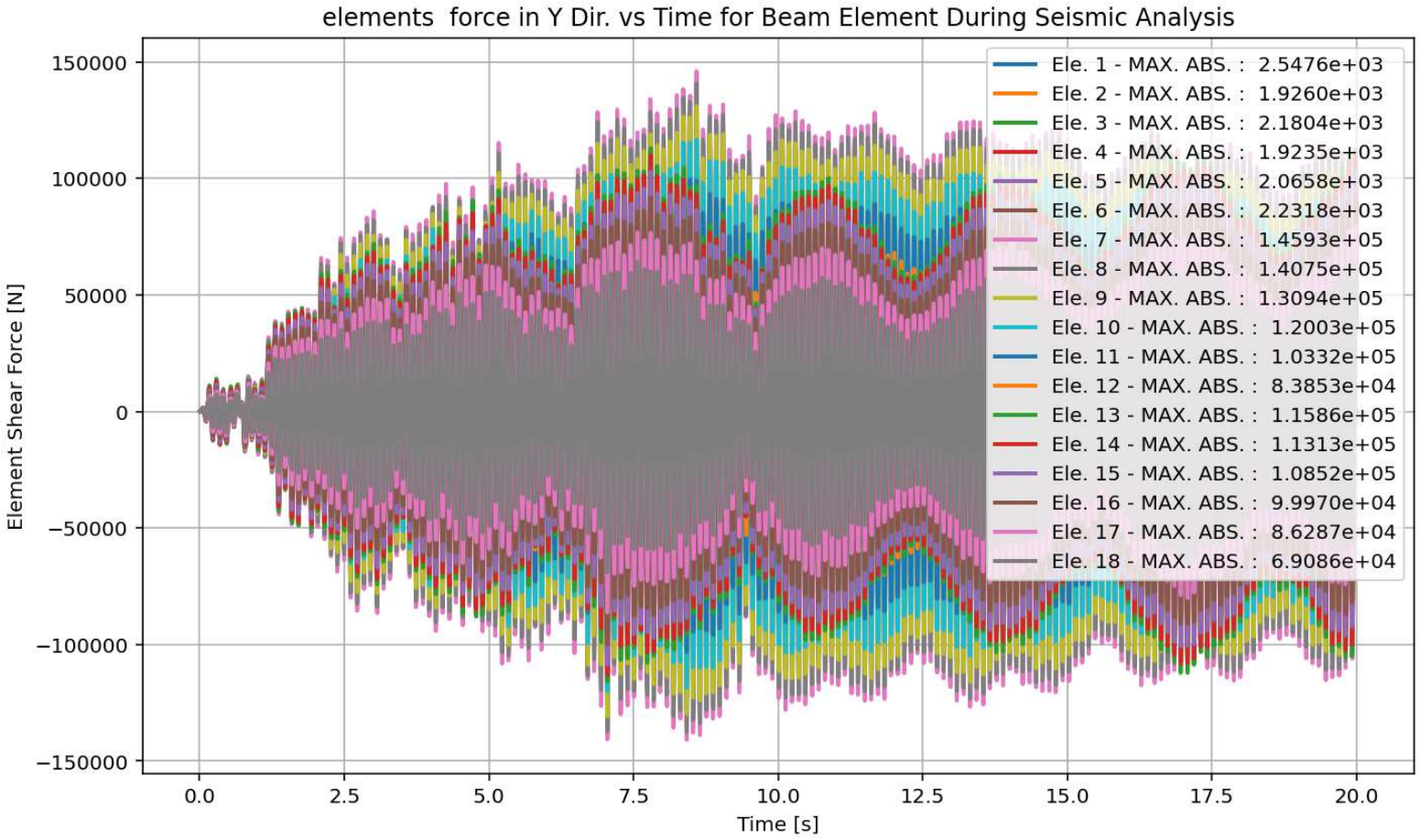
IPython Console | Files | Help | Variable Explorer | Debugger | Plots | History

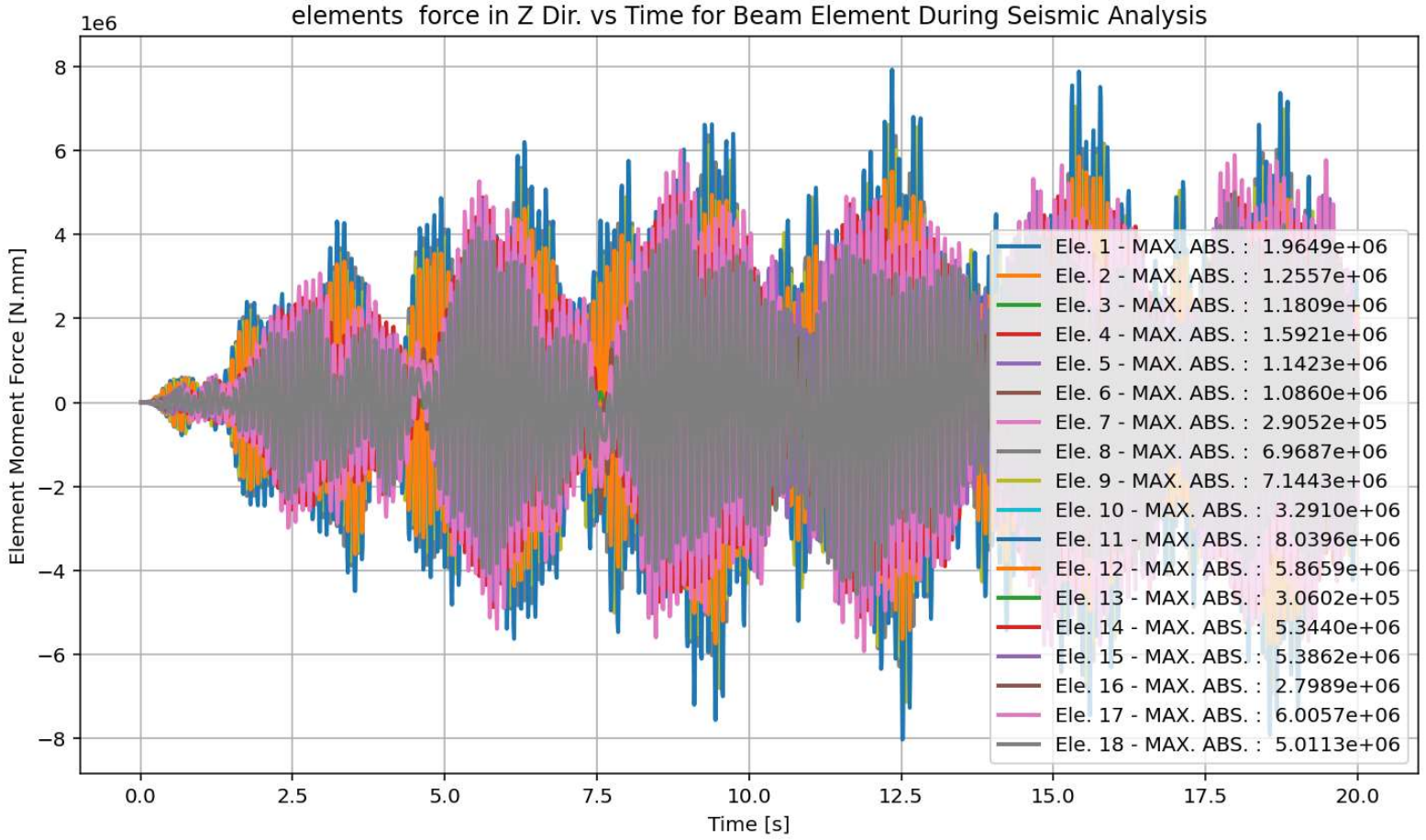
Inline Conda: anaconda3 (Python 3.12.7) LSP: Python Line 12, Col 105 UTF-8 CRLF RW Mem 43%

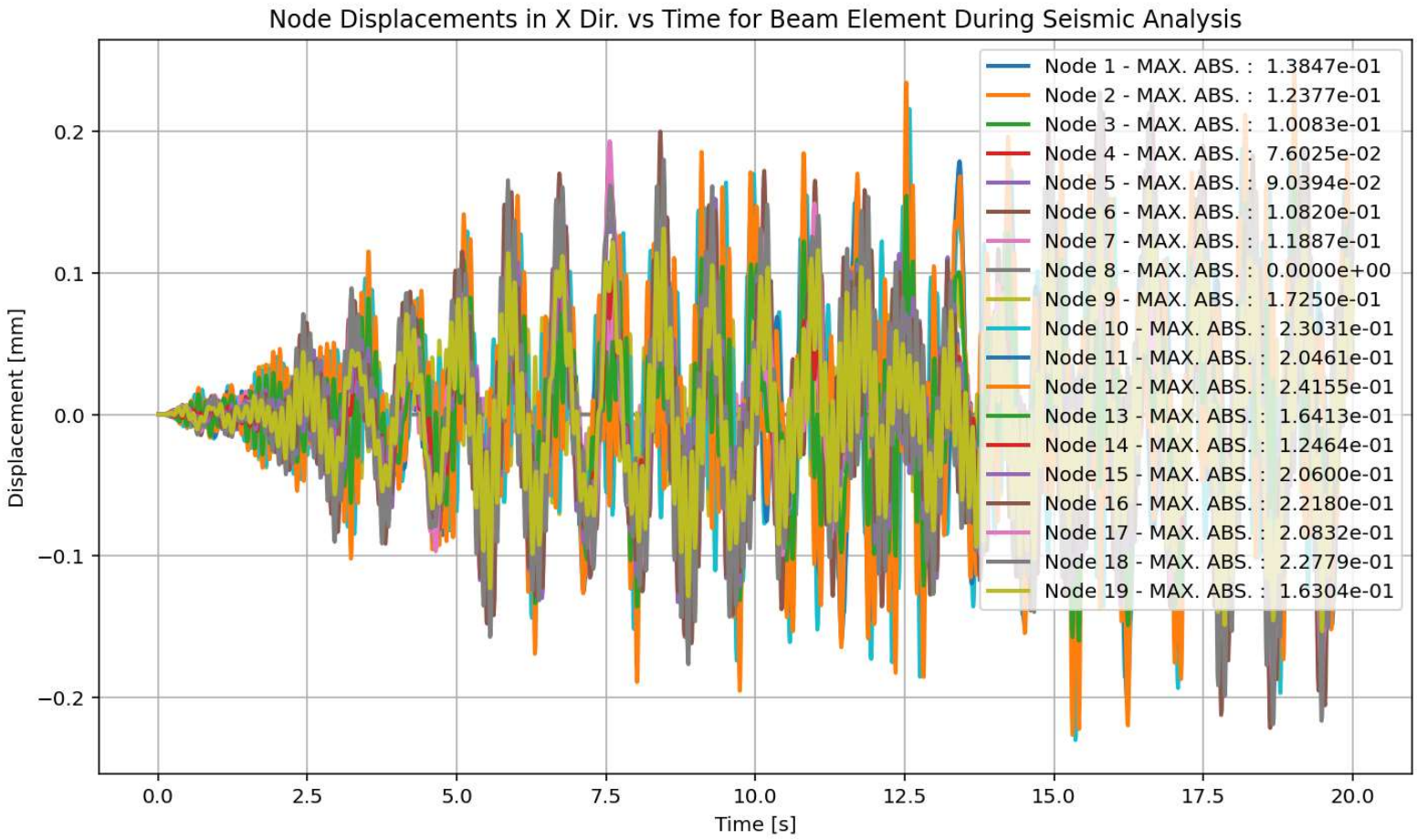

Base Reaction and Dispalcement of Structure During Seismic Analysis
Base Reaction [N]
200000
100000
0
−100000
−200000
−2
−1
0
1
2
Displacement in Middle Span [mm]

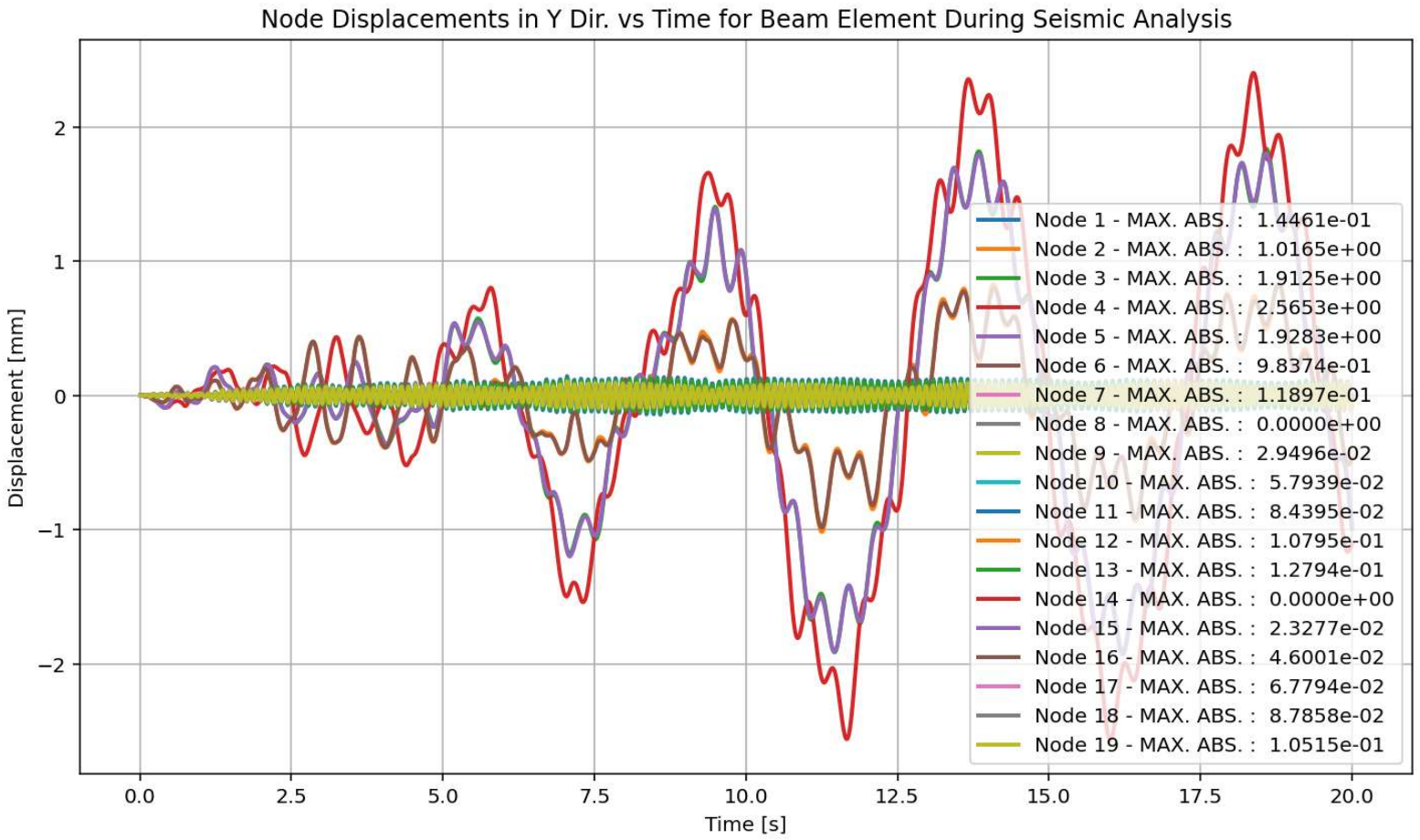

Reaction [N]
Time (s)
Mid Displacement [mm]
Time (s)
Disp X [mm]
Time (s)
Disp Y [mm]
Time (s)
Velocity X [mm/s]
Time (s)
Velocity Y [mm/s]
Time (s)
Acceleration X [mm/s²]
Time (s)
Acceleration Y [mm/s²]
Time (s)

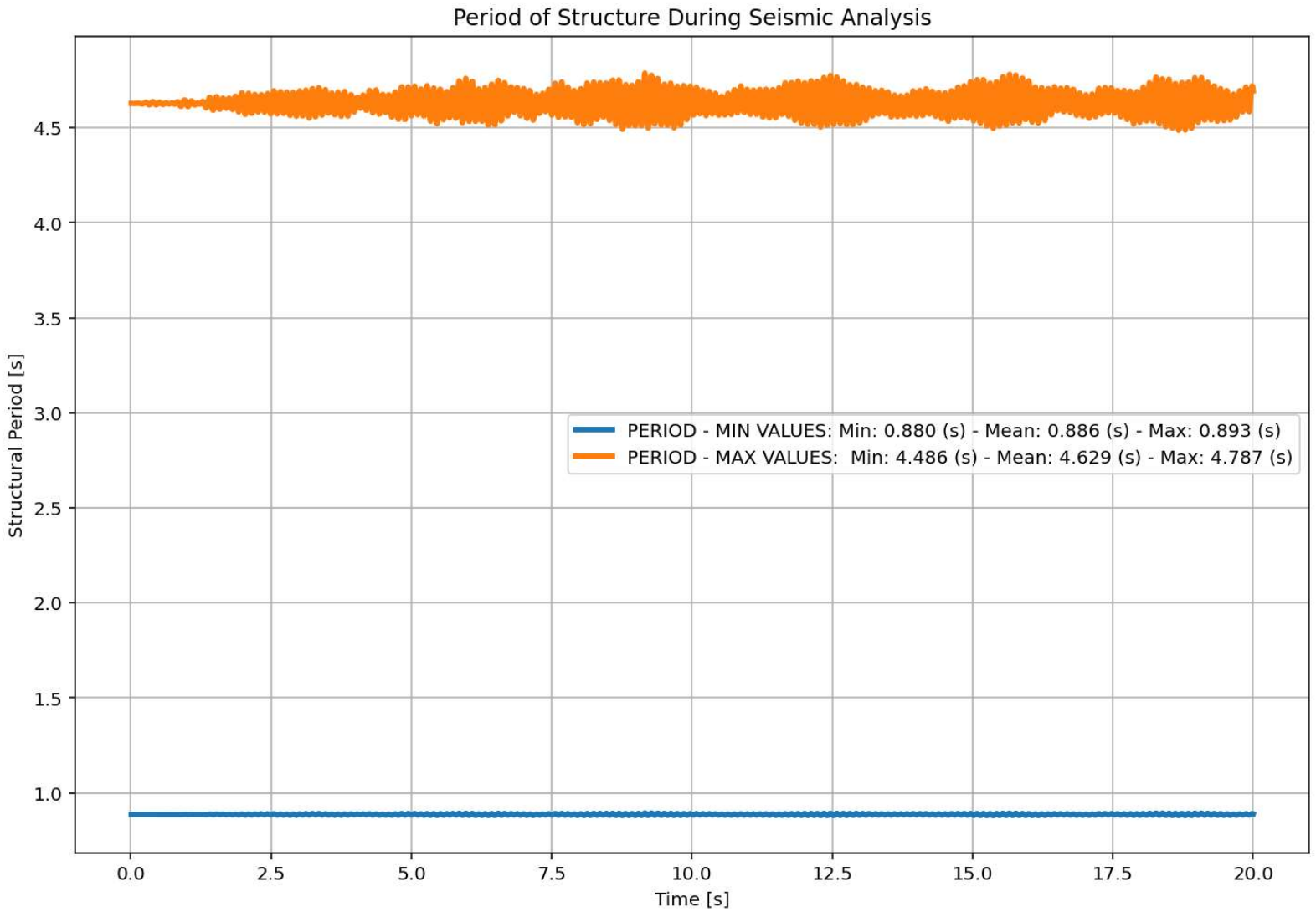

elements force in X Dir. vs Time for Beam Element During Seismic Analysis
Element Axial Force [N]
Time [s]
40000
30000
20000
10000
0
−10000
−20000
−30000
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Ele. 1 - MAX. ABS. : 1.9634e+04
Ele. 2 - MAX. ABS. : 3.0502e+04
Ele. 3 - MAX. ABS. : 3.7180e+04
Ele. 4 - MAX. ABS. : 3.8213e+04
Ele. 5 - MAX. ABS. : 3.3397e+04
Ele. 6 - MAX. ABS. : 2.3600e+04
Ele. 7 - MAX. ABS. : 1.4351e+04
Ele. 8 - MAX. ABS. : 2.8245e+03
Ele. 9 - MAX. ABS. : 1.3933e+04
Ele. 10 - MAX. ABS. : 1.2016e+04
Ele. 11 - MAX. ABS. : 4.9928e+03
Ele. 12 - MAX. ABS. : 1.4356e+04
Ele. 13 - MAX. ABS. : 1.1092e+04
Ele. 14 - MAX. ABS. : 2.8583e+03
Ele. 15 - MAX. ABS. : 1.0918e+04
Ele. 16 - MAX. ABS. : 9.3882e+03
Ele. 17 - MAX. ABS. : 3.9066e+03
Ele. 18 - MAX. ABS. : 1.0559e+04

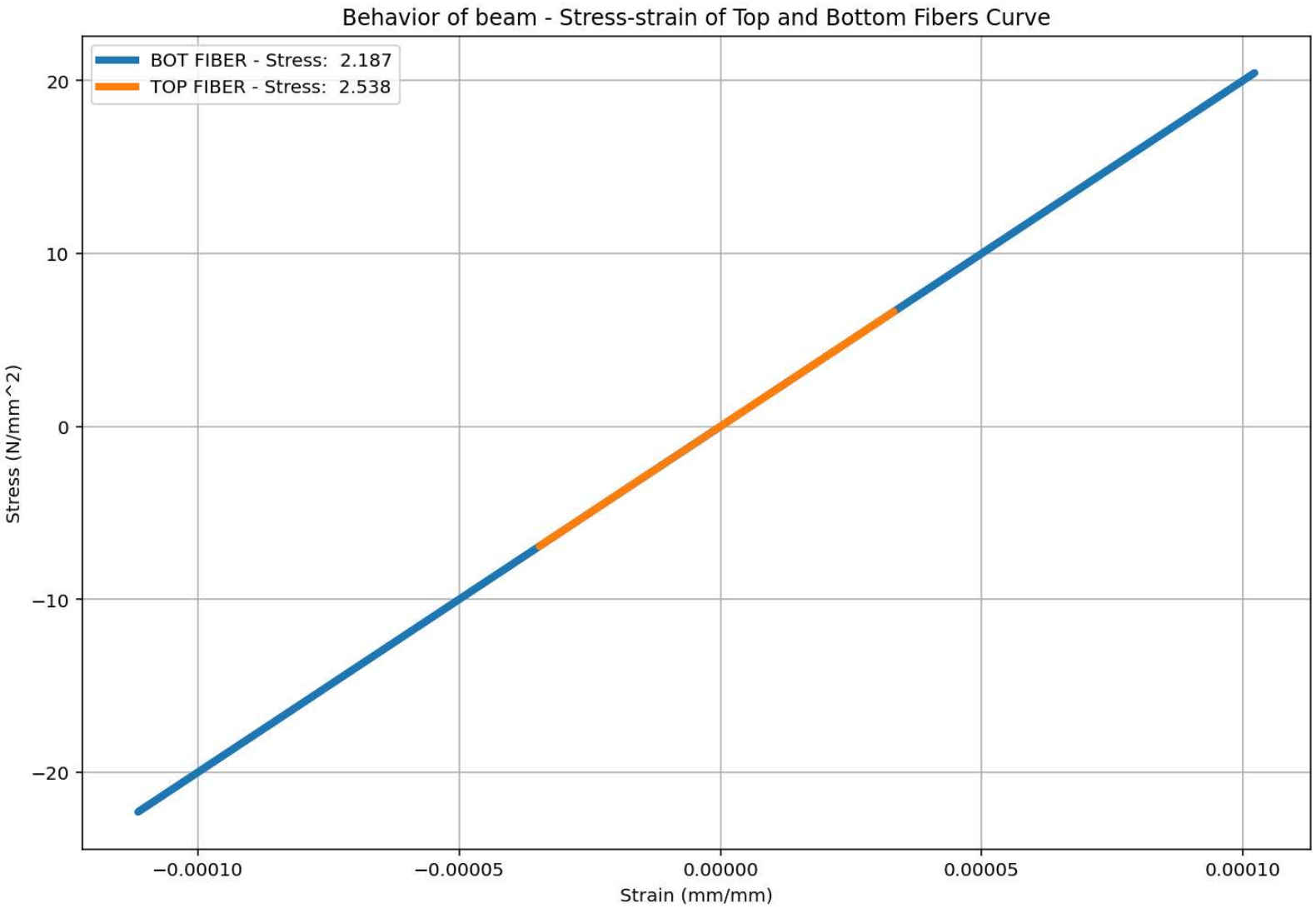

elements force in Y Dir. vs Time for Beam Element During Seismic Analysis
Element Shear Force [N]
Time [s]
150000
100000
50000
0
−50000
−100000
−150000
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Ele. 1 - MAX. ABS. : 2.5476e+03
Ele. 2 - MAX. ABS. : 1.9260e+03
Ele. 3 - MAX. ABS. : 2.1804e+03
Ele. 4 - MAX. ABS. : 1.9235e+03
Ele. 5 - MAX. ABS. : 2.0658e+03
Ele. 6 - MAX. ABS. : 2.2318e+03
Ele. 7 - MAX. ABS. : 1.4593e+05
Ele. 8 - MAX. ABS. : 1.4075e+05
Ele. 9 - MAX. ABS. : 1.3094e+05
Ele. 10 - MAX. ABS. : 1.2003e+05
Ele. 11 - MAX. ABS. : 1.0332e+05
Ele. 12 - MAX. ABS. : 8.3853e+04
Ele. 13 - MAX. ABS. : 1.1586e+05
Ele. 14 - MAX. ABS. : 1.1313e+05
Ele. 15 - MAX. ABS. : 1.0852e+05
Ele. 16 - MAX. ABS. : 9.9970e+04
Ele. 17 - MAX. ABS. : 8.6287e+04
Ele. 18 - MAX. ABS. : 6.9086e+04

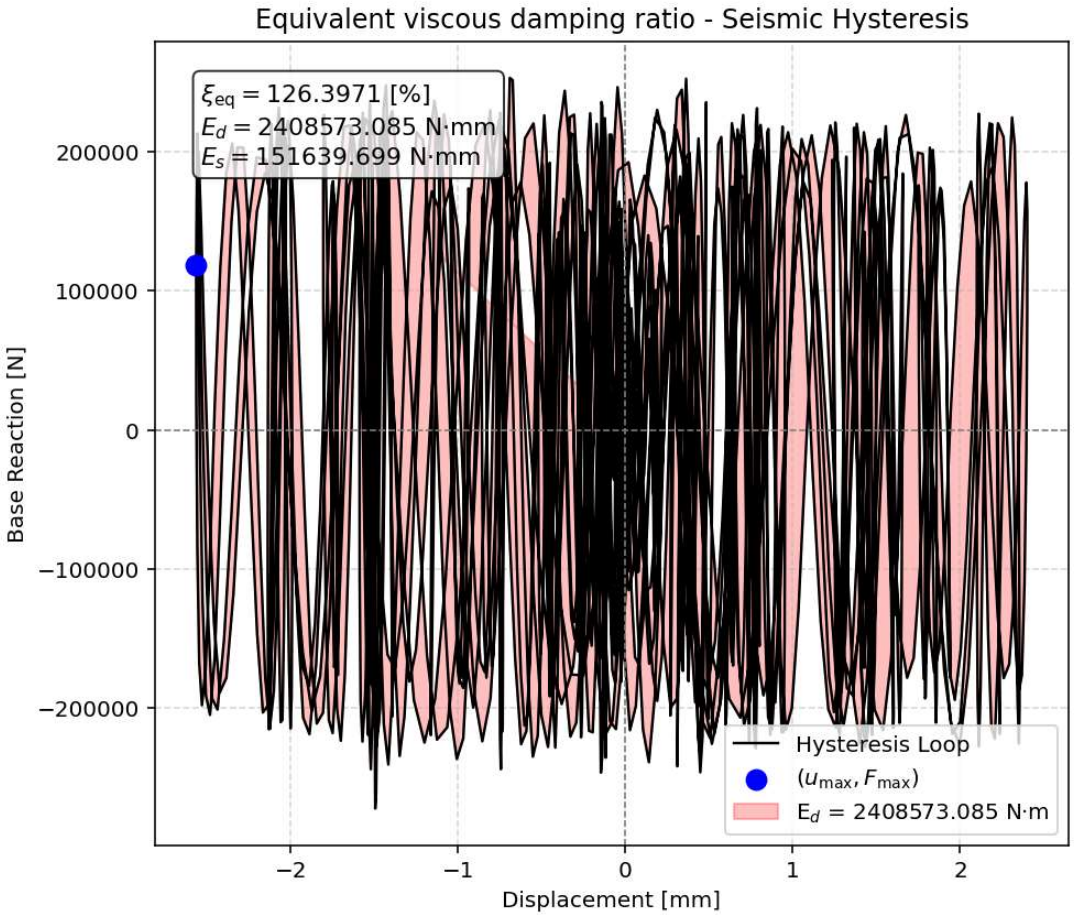

elements force in Z Dir. vs Time for Beam Element During Seismic Analysis
1e6
Element Moment Force [N.mm]
Time [s]
Ele. 1 - MAX. ABS. : 1.9649e+06
Ele. 2 - MAX. ABS. : 1.2557e+06
Ele. 3 - MAX. ABS. : 1.1809e+06
Ele. 4 - MAX. ABS. : 1.5921e+06
Ele. 5 - MAX. ABS. : 1.1423e+06
Ele. 6 - MAX. ABS. : 1.0860e+06
Ele. 7 - MAX. ABS. : 2.9052e+05
Ele. 8 - MAX. ABS. : 6.9687e+06
Ele. 9 - MAX. ABS. : 7.1443e+06
Ele. 10 - MAX. ABS. : 3.2910e+06
Ele. 11 - MAX. ABS. : 8.0396e+06
Ele. 12 - MAX. ABS. : 5.8659e+06
Ele. 13 - MAX. ABS. : 3.0602e+05
Ele. 14 - MAX. ABS. : 5.3440e+06
Ele. 15 - MAX. ABS. : 5.3862e+06
Ele. 16 - MAX. ABS. : 2.7989e+06
Ele. 17 - MAX. ABS. : 6.0057e+06
Ele. 18 - MAX. ABS. : 5.0113e+06

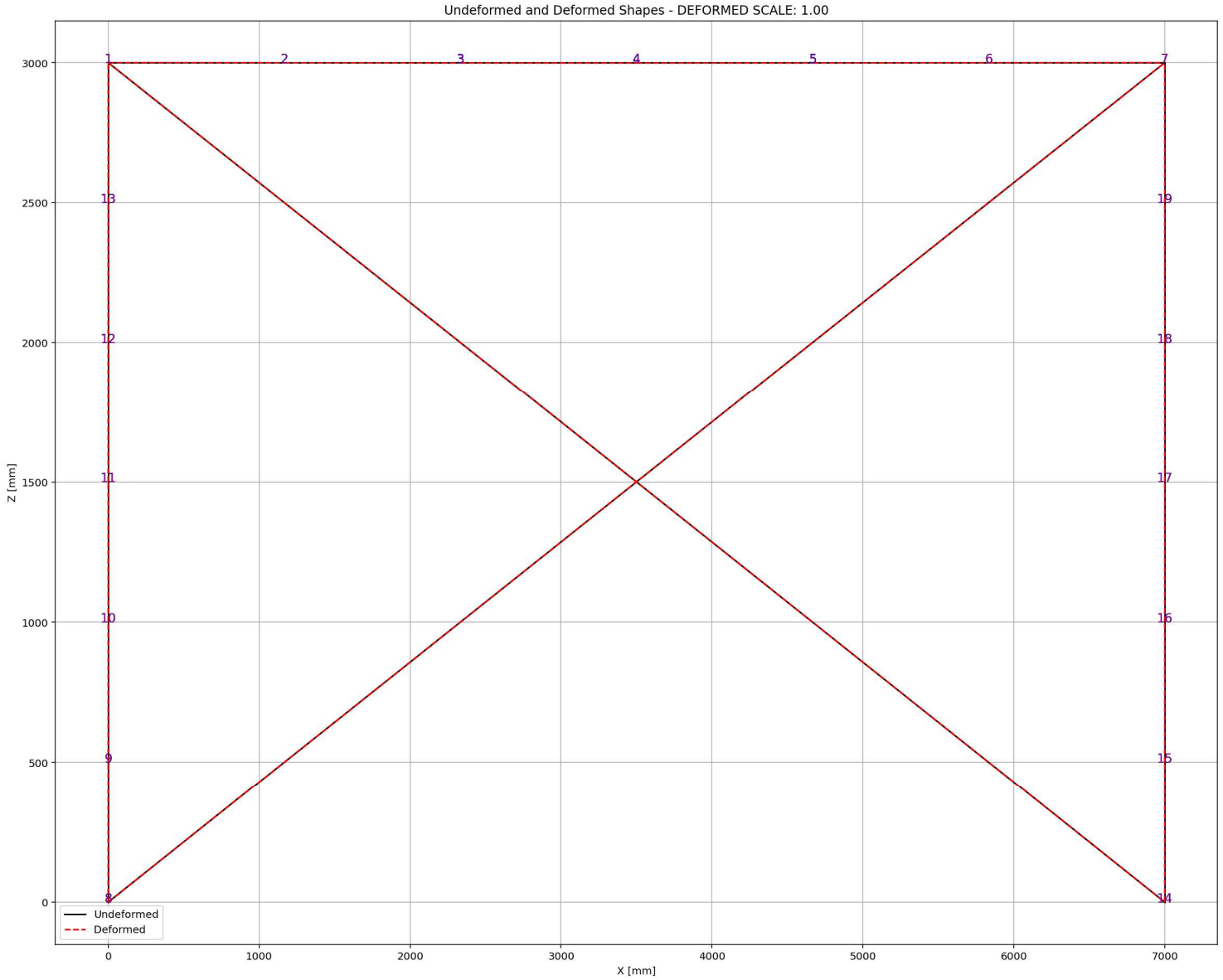

Node Displacements in X Dir. vs Time for Beam Element During Seismic Analysis
Displacement [mm]
Time [s]
0.2
0.1
0.0
−0.1
−0.2
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Node 1 - MAX. ABS. : 1.3847e-01
Node 2 - MAX. ABS. : 1.2377e-01
Node 3 - MAX. ABS. : 1.0083e-01
Node 4 - MAX. ABS. : 7.6025e-02
Node 5 - MAX. ABS. : 9.0394e-02
Node 6 - MAX. ABS. : 1.0820e-01
Node 7 - MAX. ABS. : 1.1887e-01
Node 8 - MAX. ABS. : 0.0000e+00
Node 9 - MAX. ABS. : 1.7250e-01
Node 10 - MAX. ABS. : 2.3031e-01
Node 11 - MAX. ABS. : 2.0461e-01
Node 12 - MAX. ABS. : 2.4155e-01
Node 13 - MAX. ABS. : 1.6413e-01
Node 14 - MAX. ABS. : 1.2464e-01
Node 15 - MAX. ABS. : 2.0600e-01
Node 16 - MAX. ABS. : 2.2180e-01
Node 17 - MAX. ABS. : 2.0832e-01
Node 18 - MAX. ABS. : 2.2779e-01
Node 19 - MAX. ABS. : 1.6304e-01

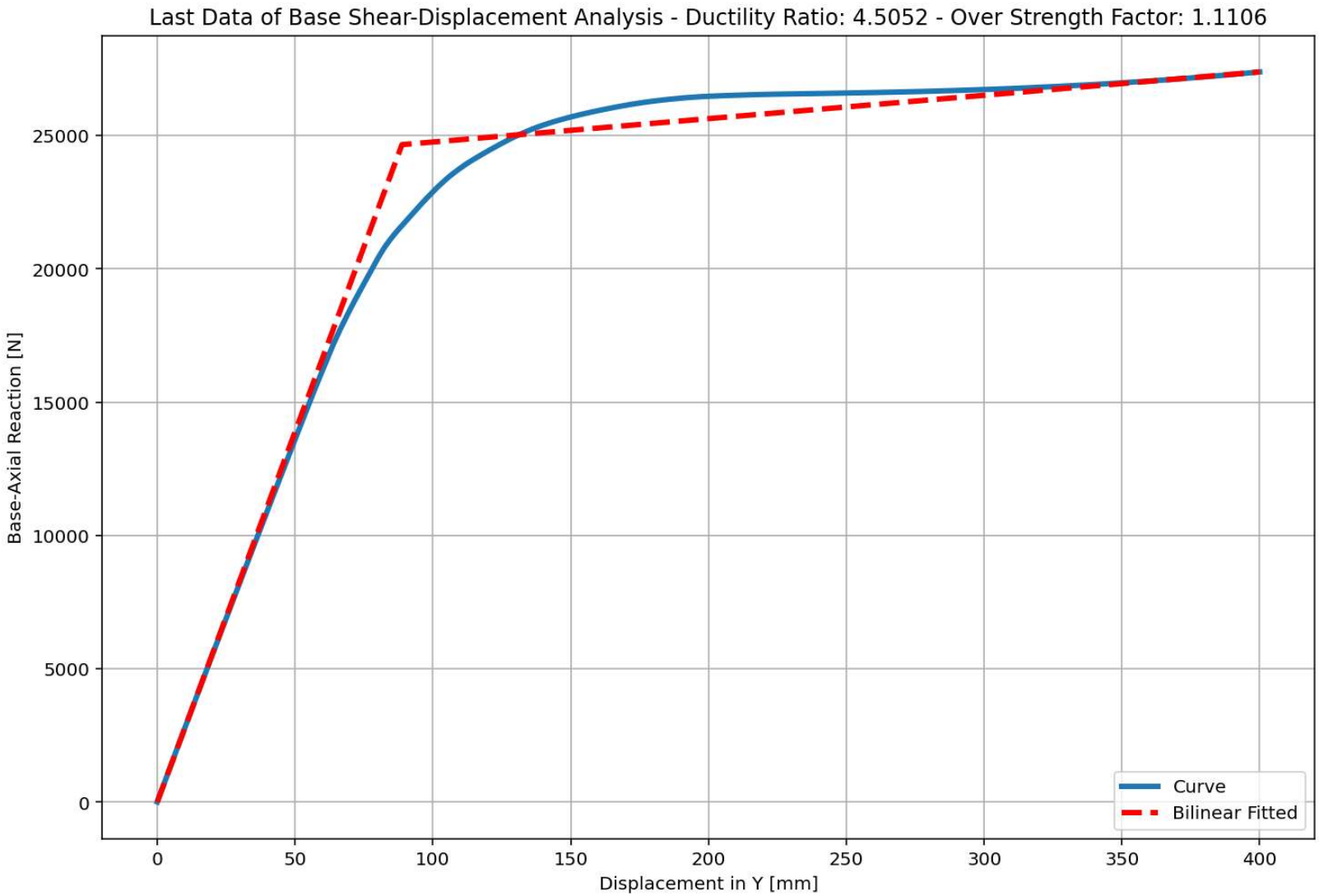

Node Displacements in Y Dir. vs Time for Beam Element During Seismic Analysis
Displacement [mm]
Time [s]
2
1
0
−1
−2
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Node 1 - MAX. ABS. : 1.4461e-01
Node 2 - MAX. ABS. : 1.0165e+00
Node 3 - MAX. ABS. : 1.9125e+00
Node 4 - MAX. ABS. : 2.5653e+00
Node 5 - MAX. ABS. : 1.9283e+00
Node 6 - MAX. ABS. : 9.8374e-01
Node 7 - MAX. ABS. : 1.1897e-01
Node 8 - MAX. ABS. : 0.0000e+00
Node 9 - MAX. ABS. : 2.9496e-02
Node 10 - MAX. ABS. : 5.7939e-02
Node 11 - MAX. ABS. : 8.4395e-02
Node 12 - MAX. ABS. : 1.0795e-01
Node 13 - MAX. ABS. : 1.2794e-01
Node 14 - MAX. ABS. : 0.0000e+00
Node 15 - MAX. ABS. : 2.3277e-02
Node 16 - MAX. ABS. : 4.6001e-02
Node 17 - MAX. ABS. : 6.7794e-02
Node 18 - MAX. ABS. : 8.7858e-02
Node 19 - MAX. ABS. : 1.0515e-01


Period of Structure During Seismic Analysis
Structural Period [s]
Time [s]
PERIOD - MIN VALUES: Min: 0.880 (s) - Mean: 0.886 (s) - Max: 0.893 (s)
PERIOD - MAX VALUES: Min: 4.486 (s) - Mean: 4.629 (s) - Max: 4.787 (s)
1.0
1.5
2.0
2.5
3.0
3.5
4.0
4.5
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0

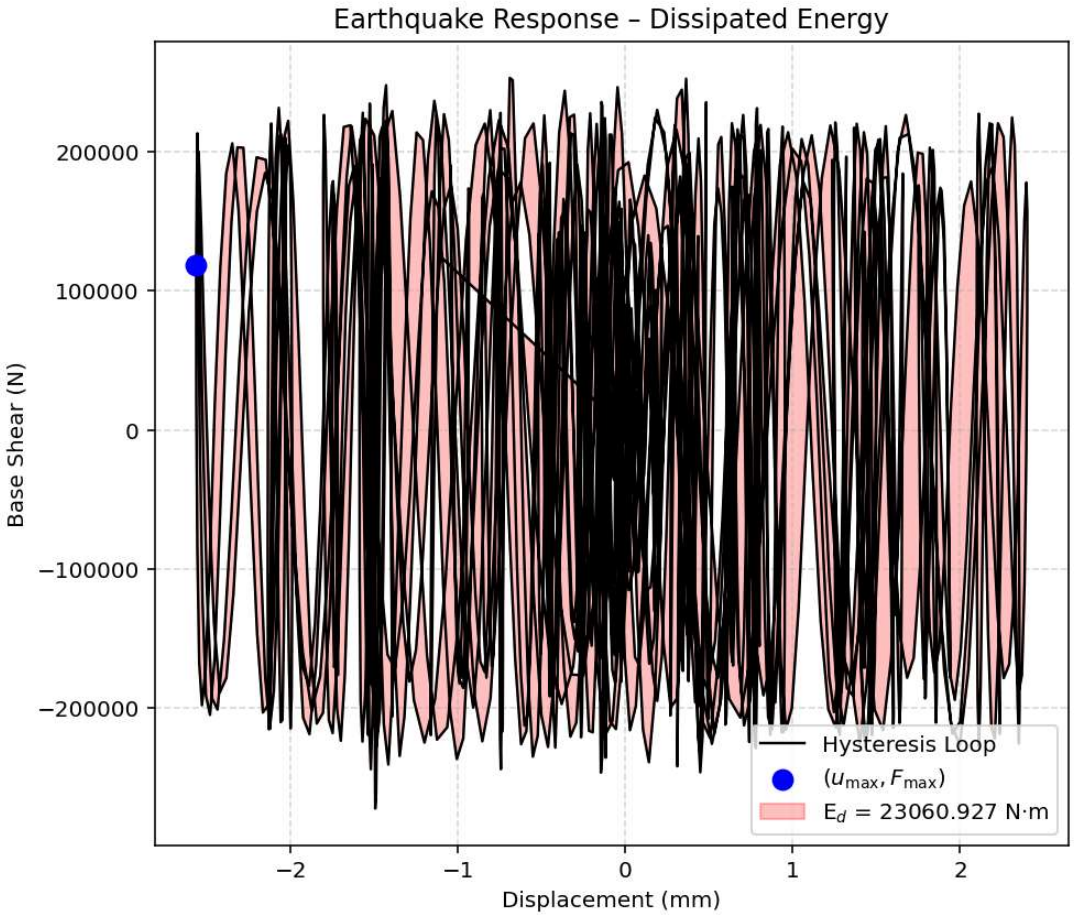

Behavior of beam - Stress-strain of Top and Bottom Fibers Curve
BOT FIBER - Stress: 2.187
TOP FIBER - Stress: 2.538
Stress (N/mm^2)
20
10
0
−10
−20
−0.00010
−0.00005
0.00000
0.00005
0.00010
Strain (mm/mm)

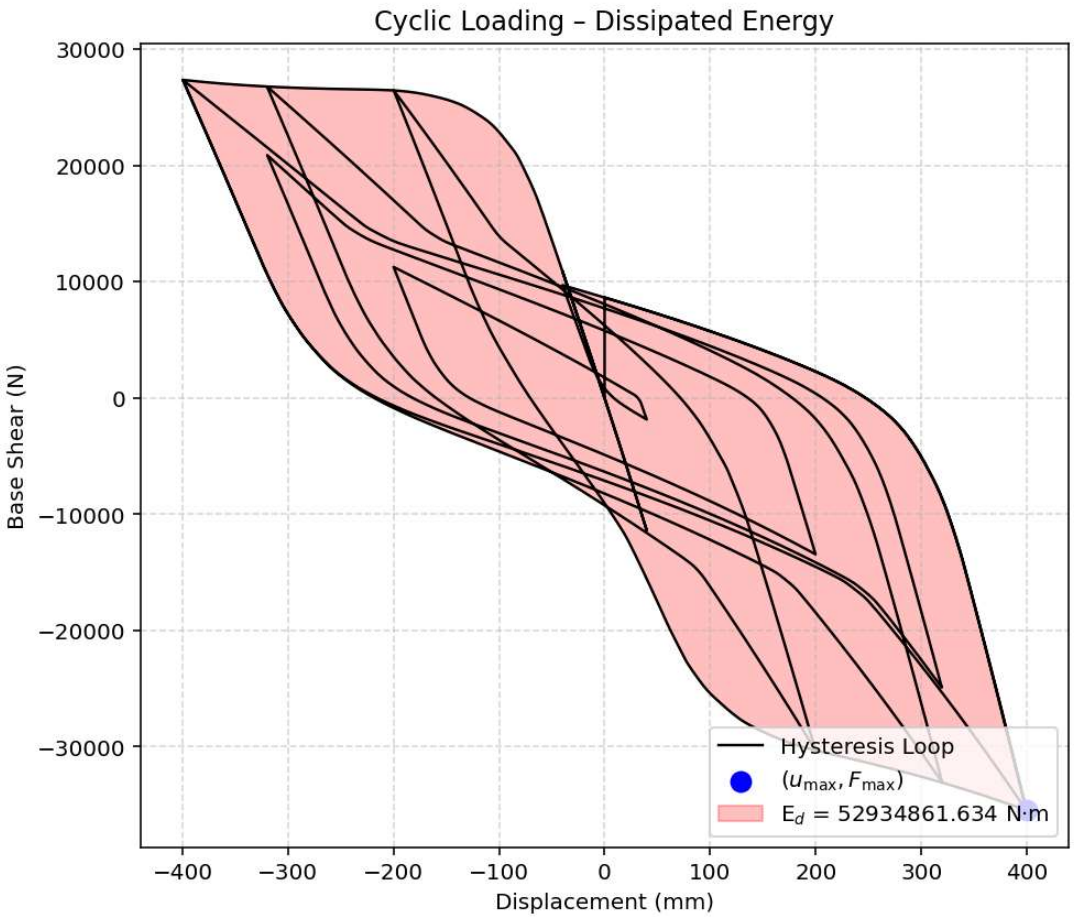

Equivalent viscous damping ratio - Seismic Hysteresis
$\xi_{eq} = 126.3971$ [%]
$E_d = 2408573.085$ N·mm
$E_s = 151639.699$ N·mm
200000
100000
0
−100000
−200000
Base Reaction [N]
−2
−1
0
1
2
Displacement [mm]
Hysteresis Loop
(u_{max}, F_{max})
E_d = 2408573.085 N·m


Undeformed and Deformed Shapes - DEFORMED SCALE: 1.00
Z [mm]
X [mm]
Undeformed
Deformed
1
2
3
4
5
6
7
8
9
10
11
12
13
14
15
16
17
18
19
0
500
1000
1500
2000
2500
3000
0
1000
2000
3000
4000
5000
6000
7000


Last Data of Base Shear-Displacement Analysis - Ductility Ratio: 4.5052 - Over Strength Factor: 1.1106
Base-Axial Reaction [N]
Displacement in Y [mm]
0
5000
10000
15000
20000
25000
0
50
100
150
200
250
300
350
400
Curve
Bilinear Fitted

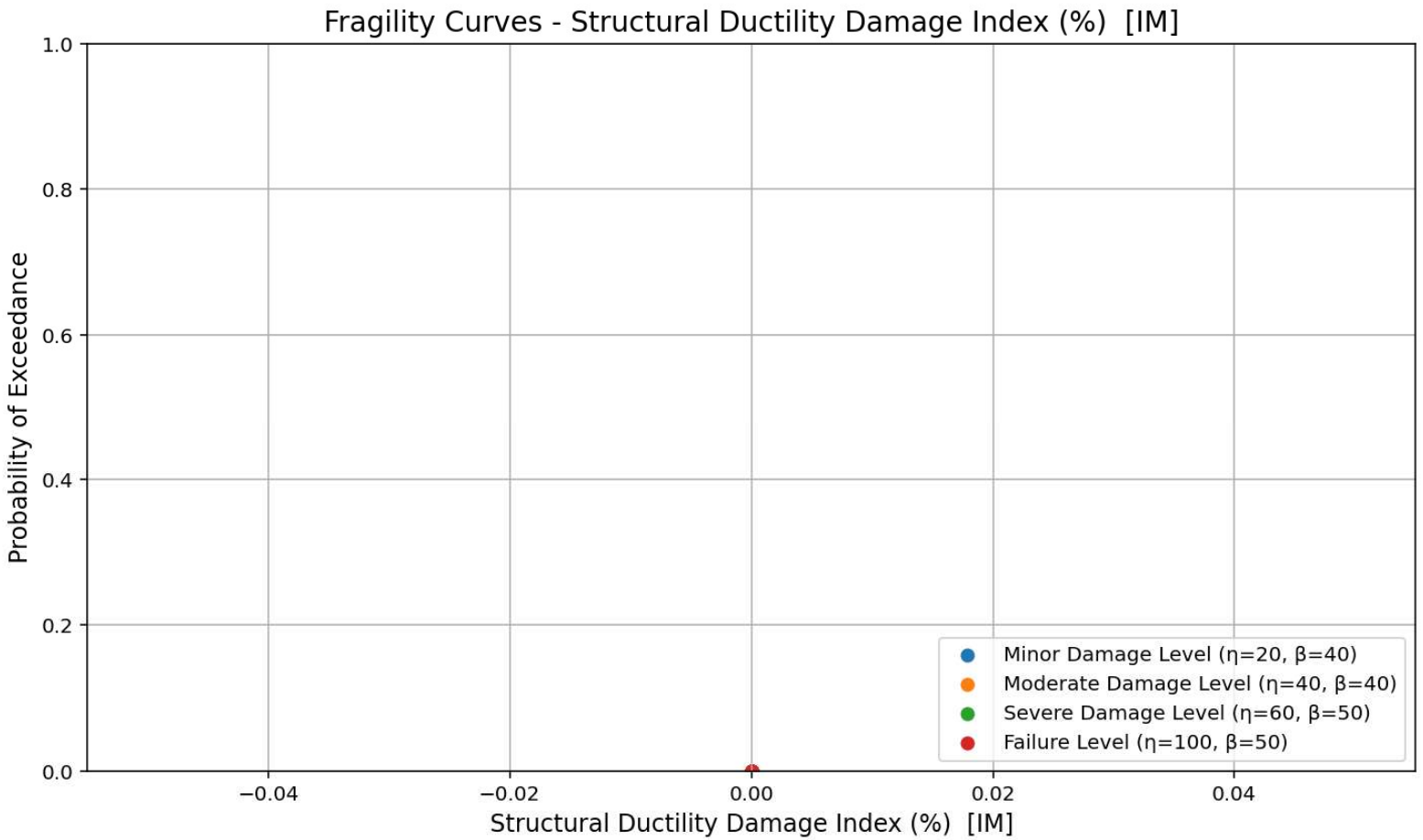
```
+==========================+
=  Analysis curve fitted =
    Disp       Base Shear
---------------------------
[[   0.          0.       ]
 [  88.78648283 24658.83282887]
 [ 400.       27386.67270937]]
+==========================+
+-------------------------------------------------+
 Structure Elastic Stiffness :    277.73
 Structure Plastic Stiffness :    68.47
 Structure Tangent Stiffness :    8.77
 Structure Ductility Ratio :      4.51
 Structure Over Strength Factor:  1.11
+-------------------------------------------------+
Over Strength Coefficient (Ω0):     1.1106
Displacement Ductility Ratio (μ):   4.5052
Ductility Coefficient (Rμ):         4.5052
Structural Behavior Coefficient (R): 5.0036
Structural Ductility Damage Index in Y Direction: -27.7048 (%)
```


Earthquake Response – Dissipated Energy
Base Shear (N)
Displacement (mm)
200000
100000
0
−100000
−200000
−2
−1
0
1
2
Hysteresis Loop
(u_{max}, F_{max})
E_d = 23060.927 N·m


Cyclic Loading – Dissipated Energy
Base Shear (N)
Displacement (mm)
30000
20000
10000
0
−10000
−20000
−30000
−400
−300
−200
−100
0
100
200
300
400
Hysteresis Loop
(u_{max}, F_{max})
E_d = 52934861.634 N·m

Dissipated Energy from Earthquake= 23060.93 N·mm
Dissipated Energy from Cyclic Displacement= 52934861.63 N·mm

DISSIPATED ENERGY CAPACITY INDEX: 0.044 [%]
ZONE 1: VERY MINOR DAMAGE


Fragility Curves - Structural Ductility Damage Index (%) [IM]
Probability of Exceedance
1.0
0.8
0.6
0.4
0.2
0.0
−0.04
−0.02
0.00
0.02
0.04
Structural Ductility Damage Index (%) [IM]
Minor Damage Level (η=20, β=40)
Moderate Damage Level (η=40, β=40)
Severe Damage Level (η=60, β=50)
Failure Level (η=100, β=50)